



Analisi e Progetto di Algoritmi

Laurea Triennale in Informatica
(codice: E3101Q113)

AA 2025/2026

Algoritmo di Prim
(Problema del Minimum Spanning Tree)

Docente: Raffaella Rizzi

[Algoritmo generico]

GENERIC-MST (G, W)

$A \leftarrow \emptyset$

while $|V| - |A| > 1$ **do**

trova $(V', V - V')$ che rispetti A

$(u, v) \leftarrow$ arco leggero per $(V', V - V')$

$A = A \cup \{(u, v)\}$

return (V, A) $// (V, A)$ è MST

[Algoritmo generico]

Alla generica iterazione,

$A \subset T \Rightarrow G_A = (V, A)$ è una foresta di $|V| - |A|$ alberi (componenti connesse)

Considero uno qualsiasi degli alberi di $G_A \rightarrow C = (V_C, A_C)$

$\Rightarrow$ il taglio $(V_C, V - V_C)$ rispetta A

COROLLARIO (DEL TEOREMA DELL'ARCO SICURO)

Sia:

- ✓ $G = (V, E)$, grafo non orientato, connesso e pesato
- ✓ $A \subset T$, sottoinsieme degli archi di MST
- ✓ una componente connessa $C = (V_C, A_C)$ della foresta $G_A = (V, A)$

Un arco (u, v) leggero rispetto al taglio $(V_C, V - V_C)$ è sicuro per A

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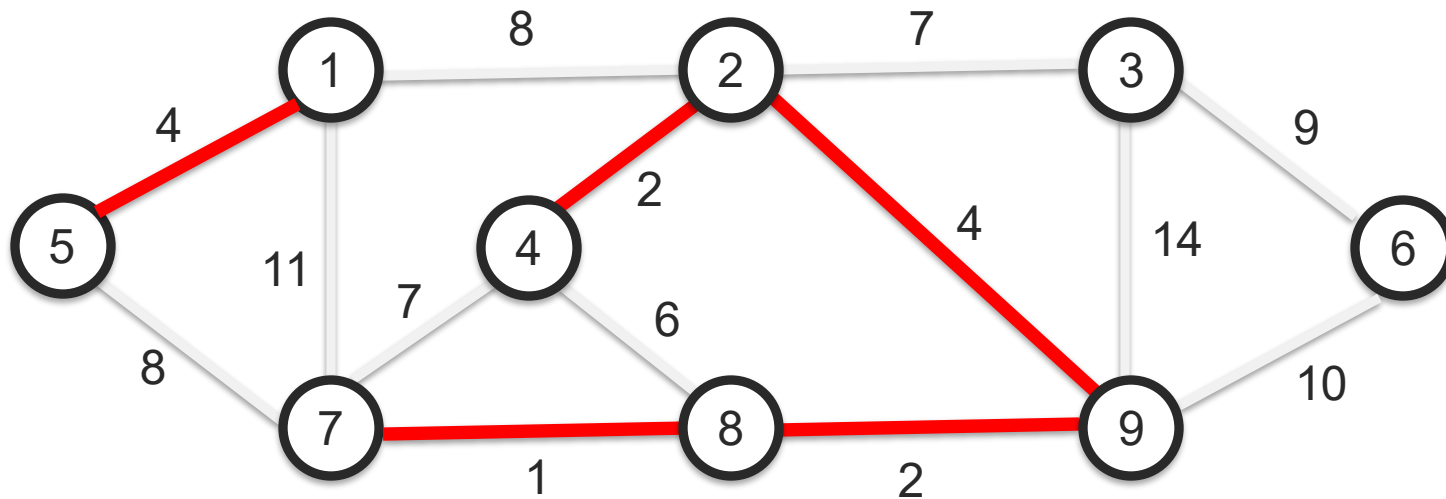
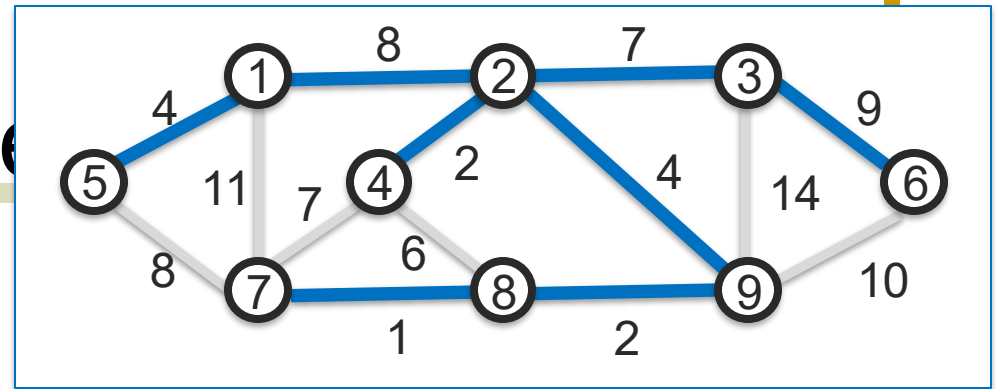
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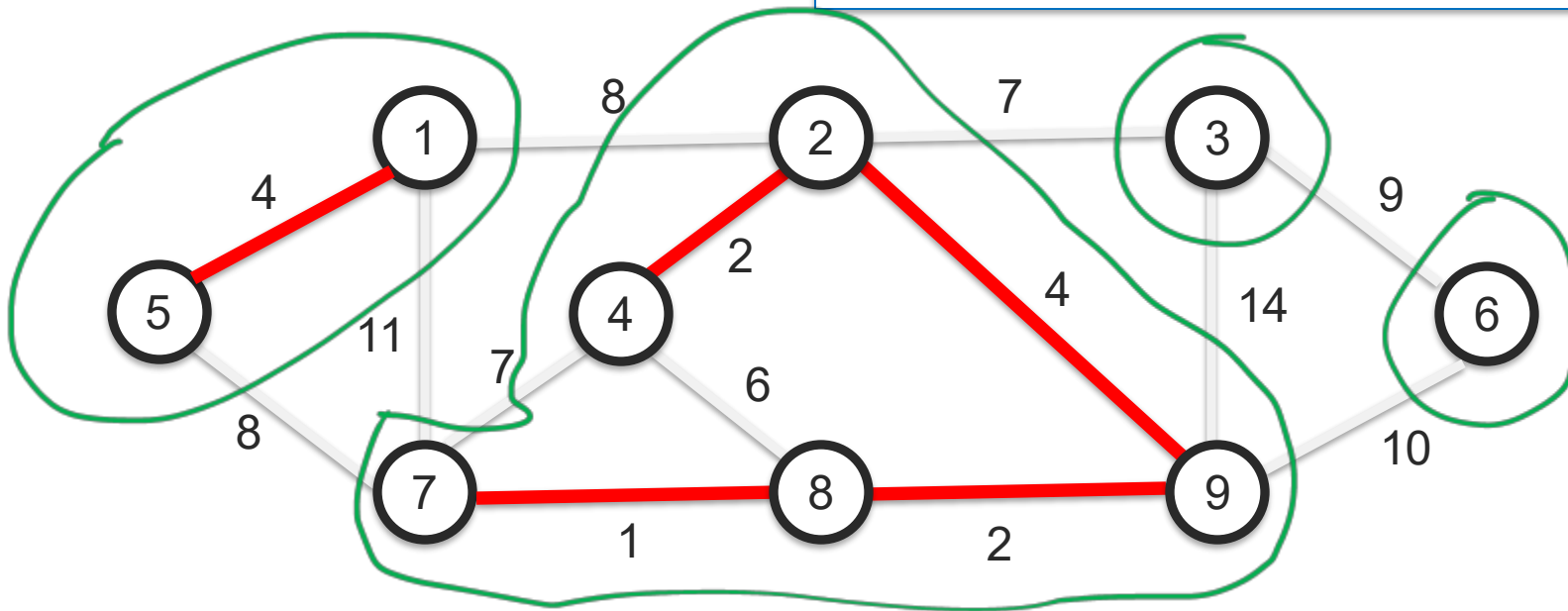
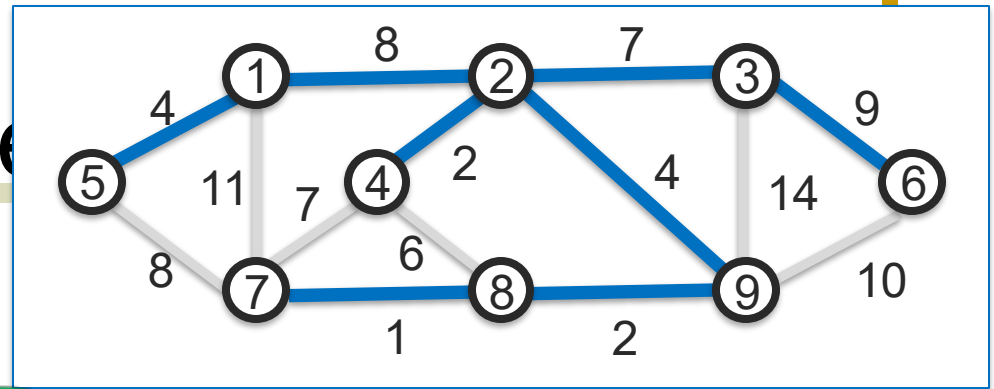
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$$A = \{(7,8), (2,4), (8,9), (2,9), (1,5)\}$$

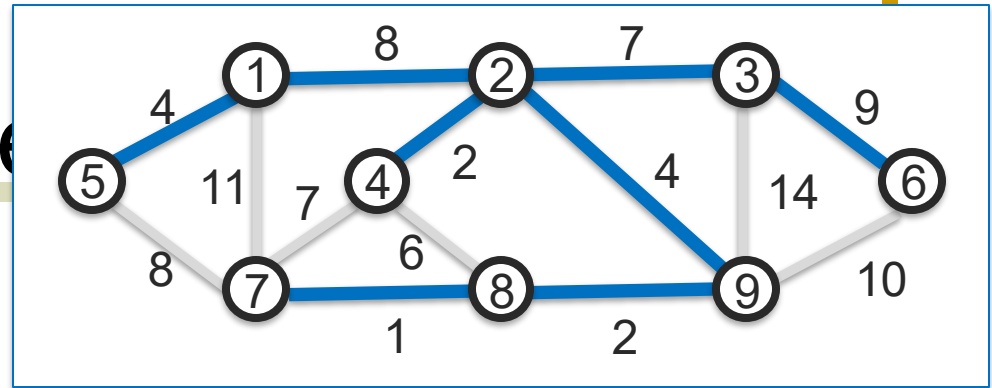
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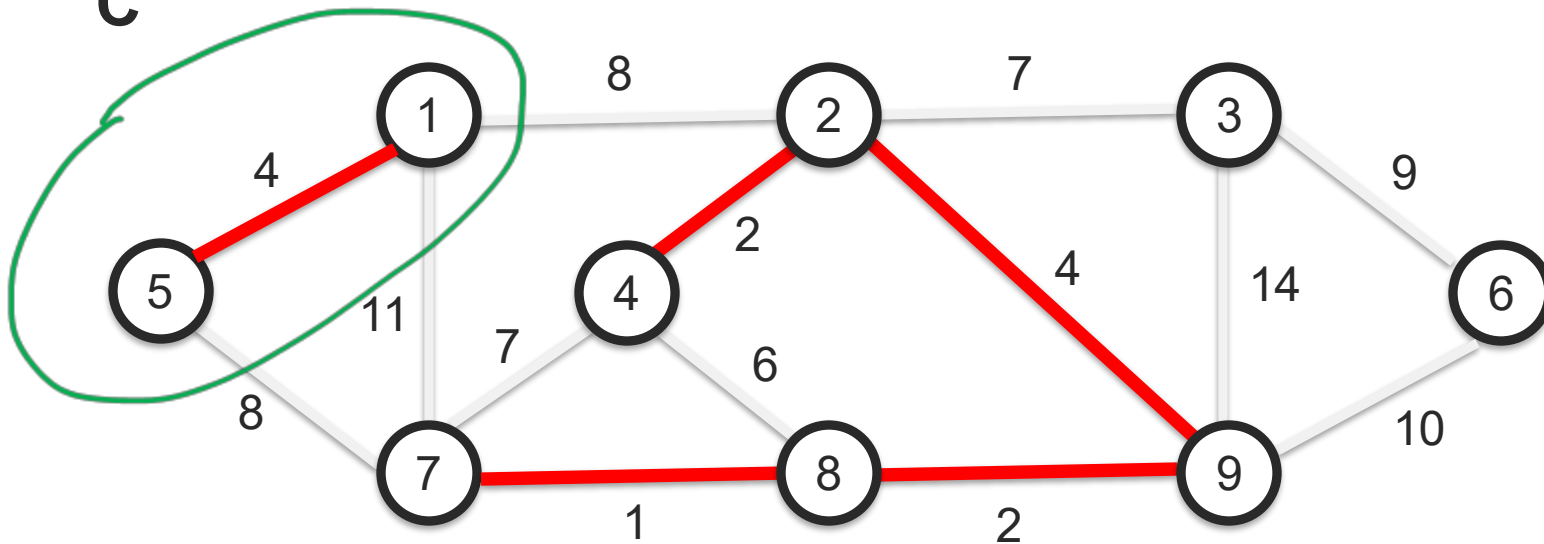
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G_A ha 4 componenti connesse

[Algoritmo gene



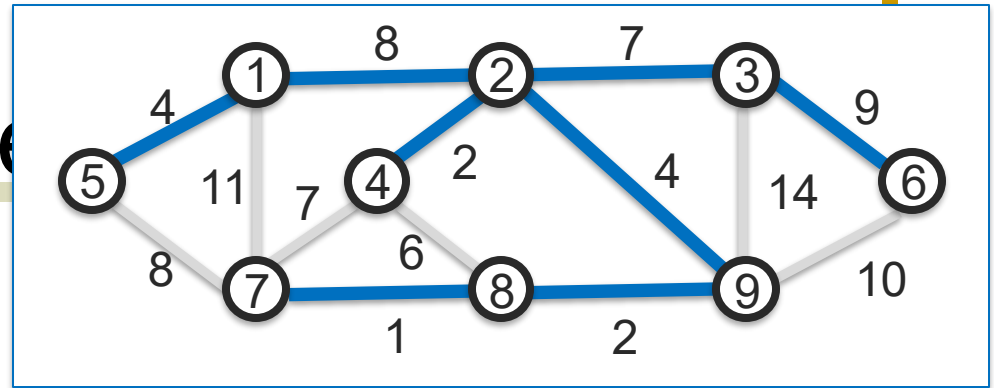
c



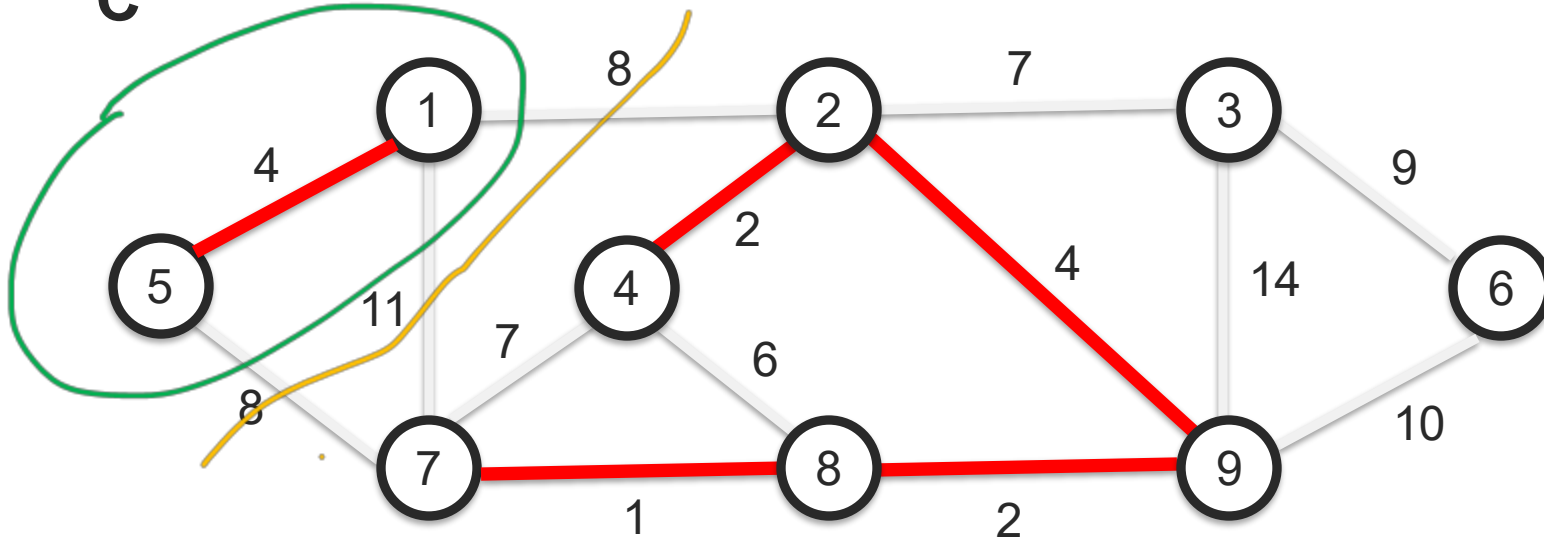
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$$V_C = \{1,5\}$$

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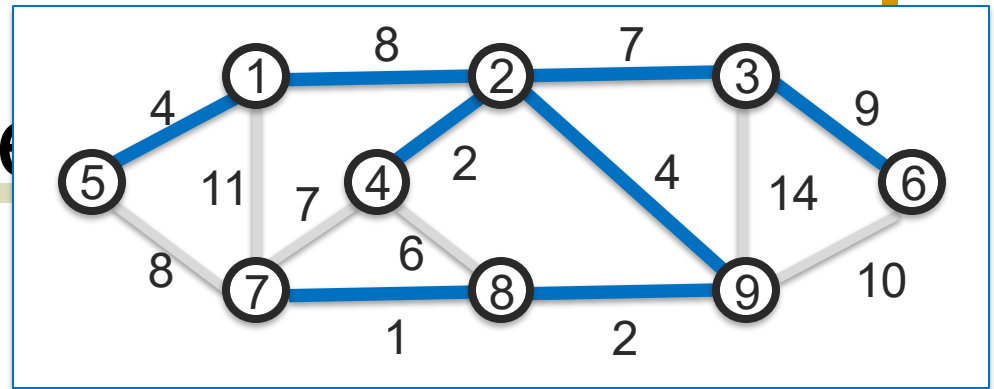
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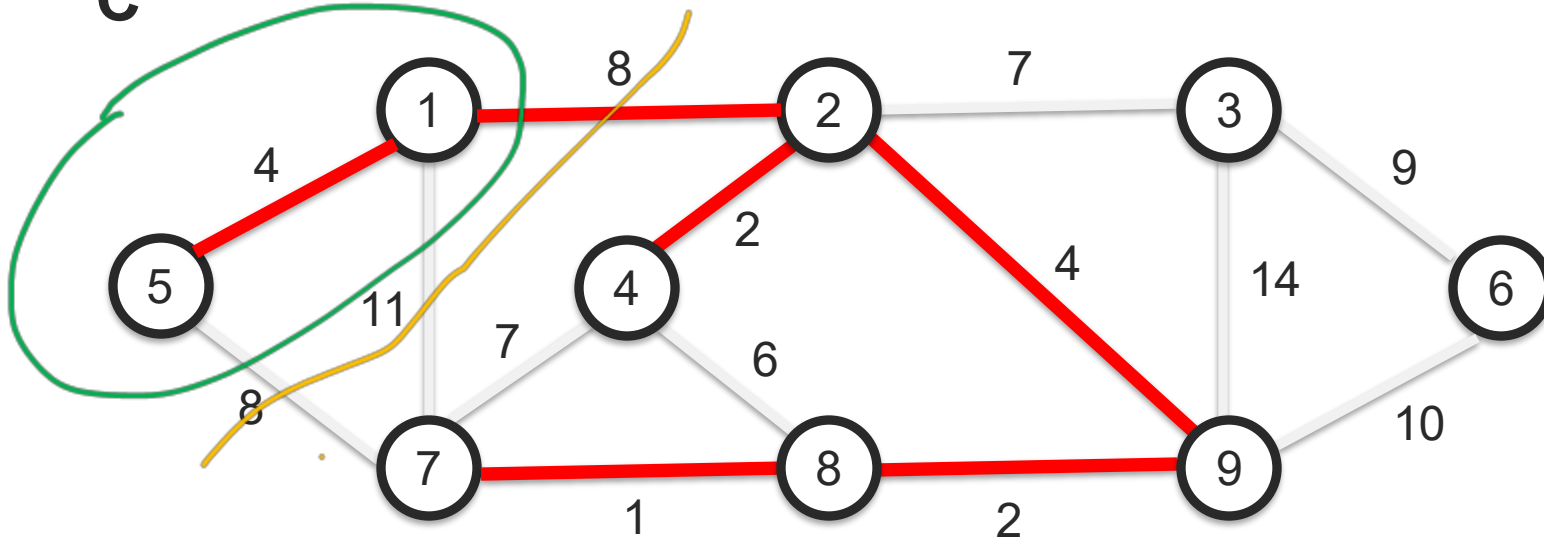
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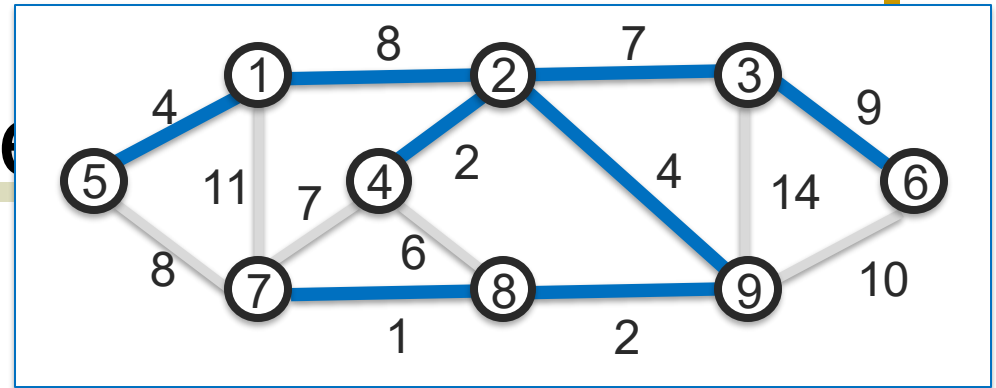


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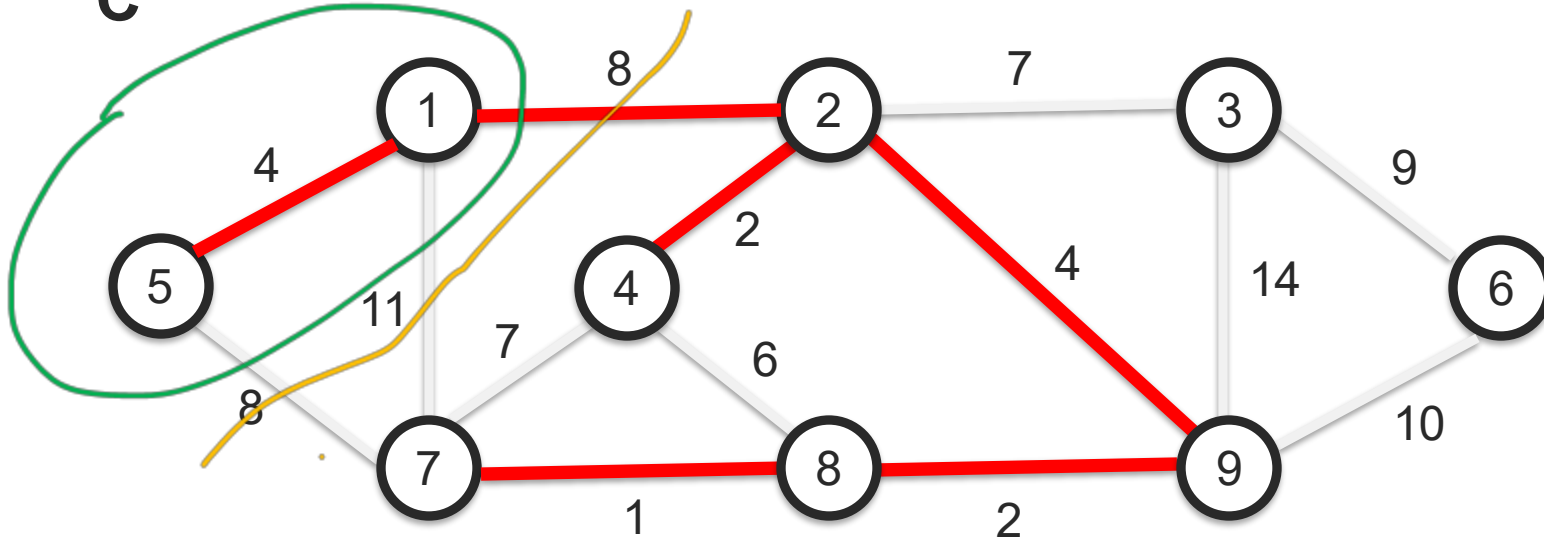
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$(1,2) \rightarrow$ arco sicuro

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$r \leftarrow$ **vertice arbitrario**

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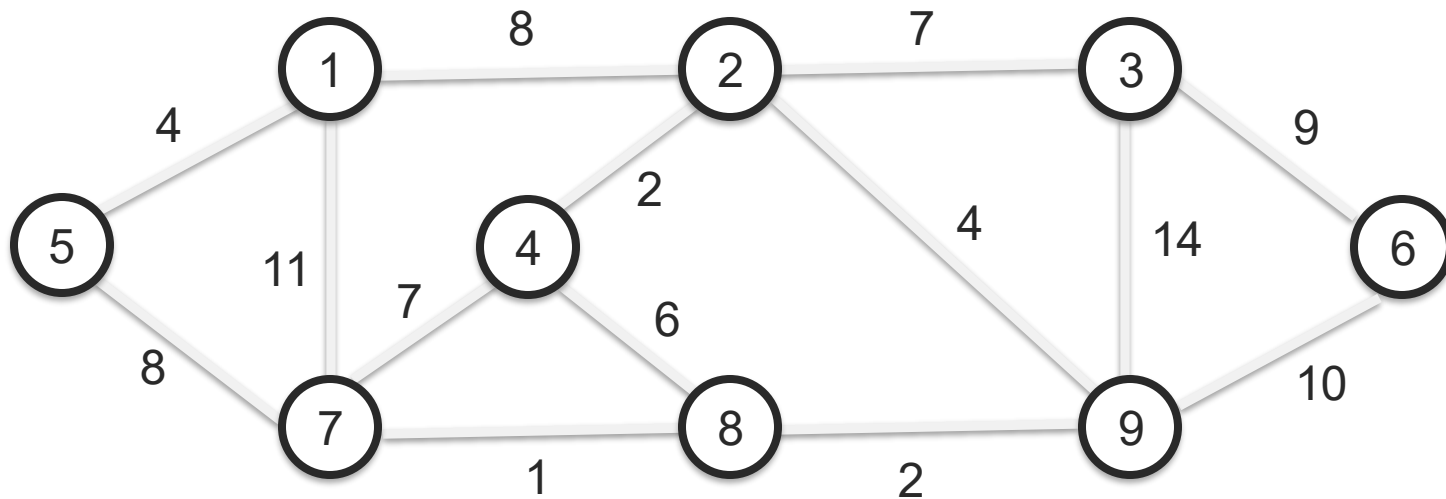
scegli la componente $C = (V_C, E_C)$ **contenente** r

$(u, v) \leftarrow$ **arco di peso minimo che collega un**
 vertex in C **con un vertex esterno a** C

$A = A \cup \{ (u, v) \}$

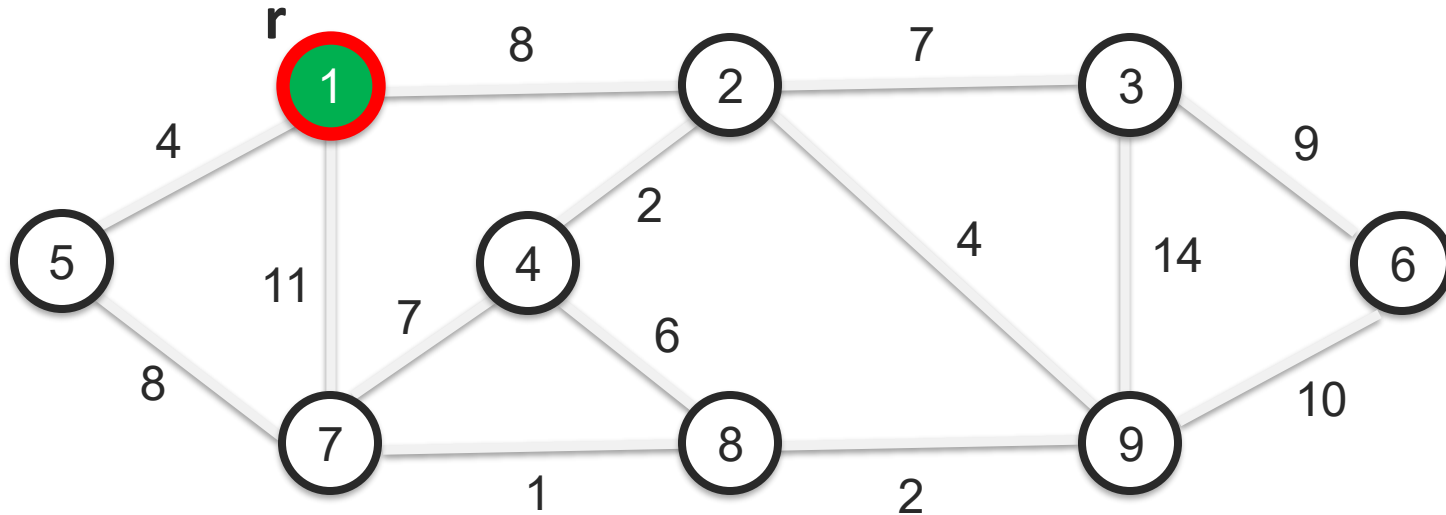
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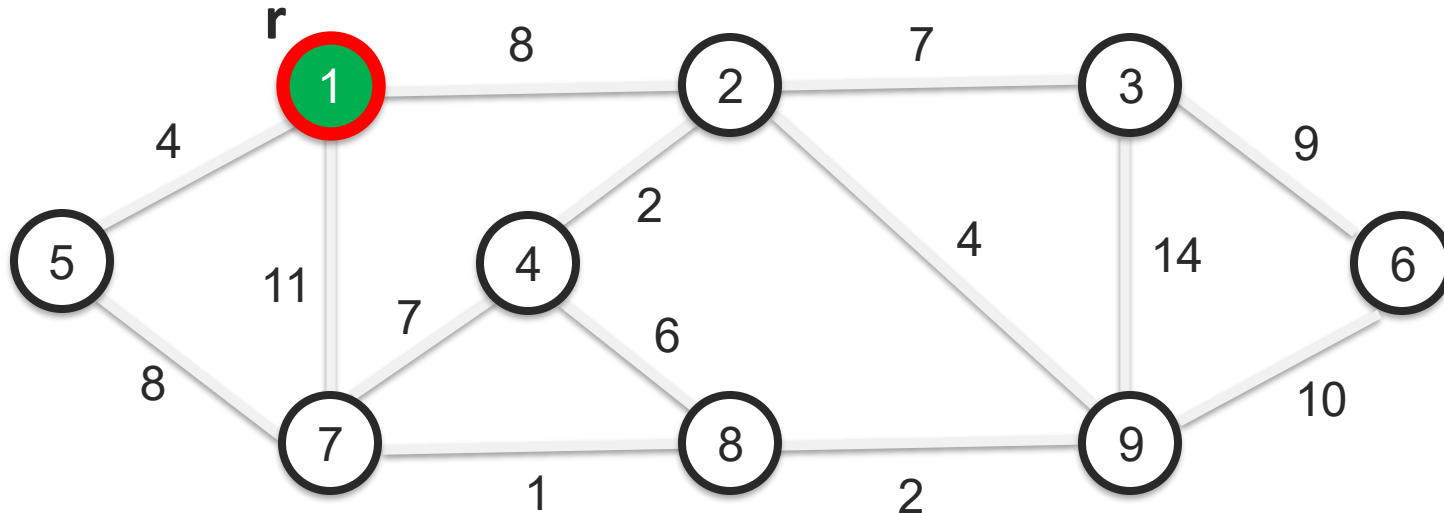
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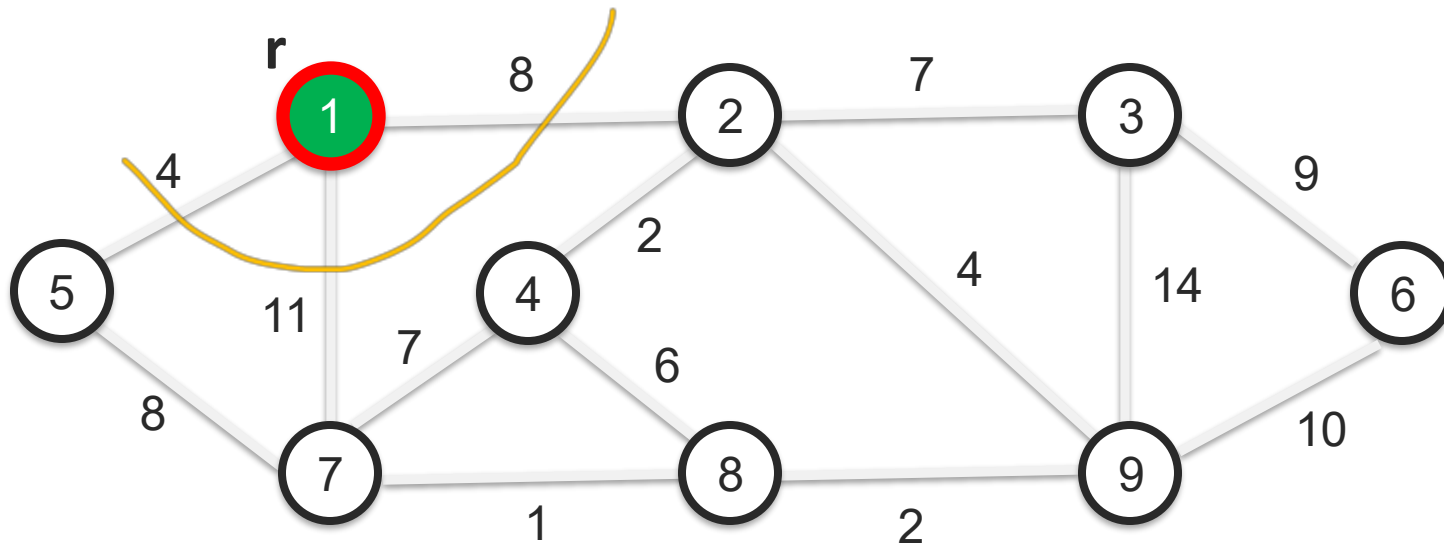
$$V_C = \{1\}$$



$$A = \emptyset$$

Idea dell'algoritmo di Prim

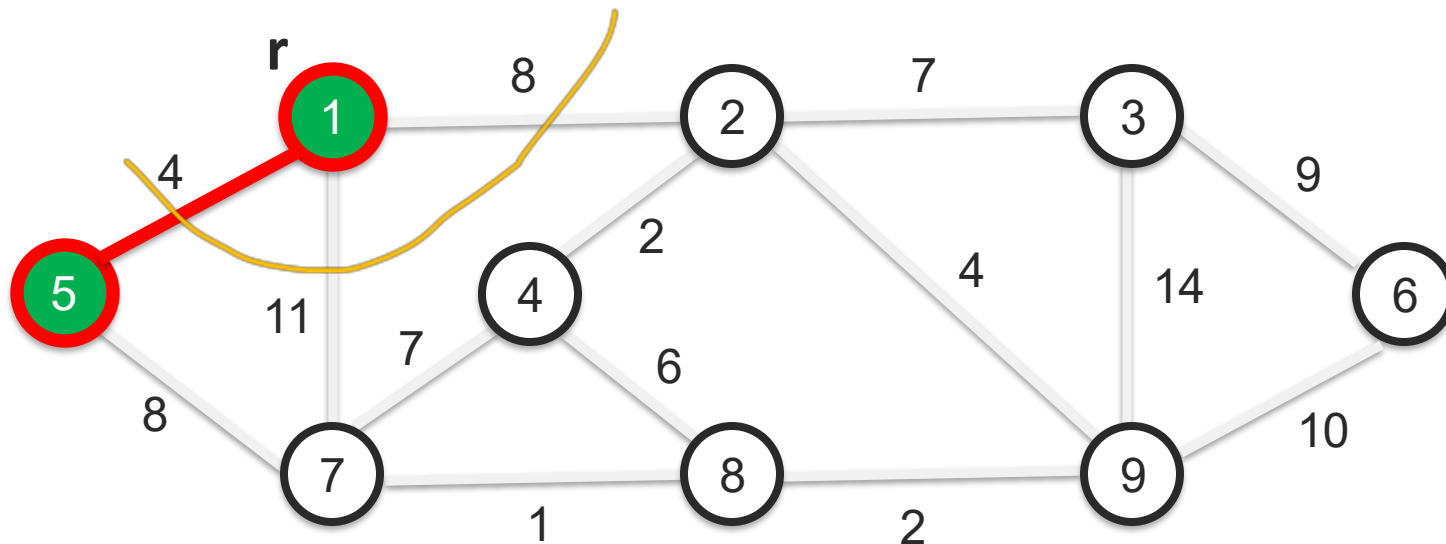
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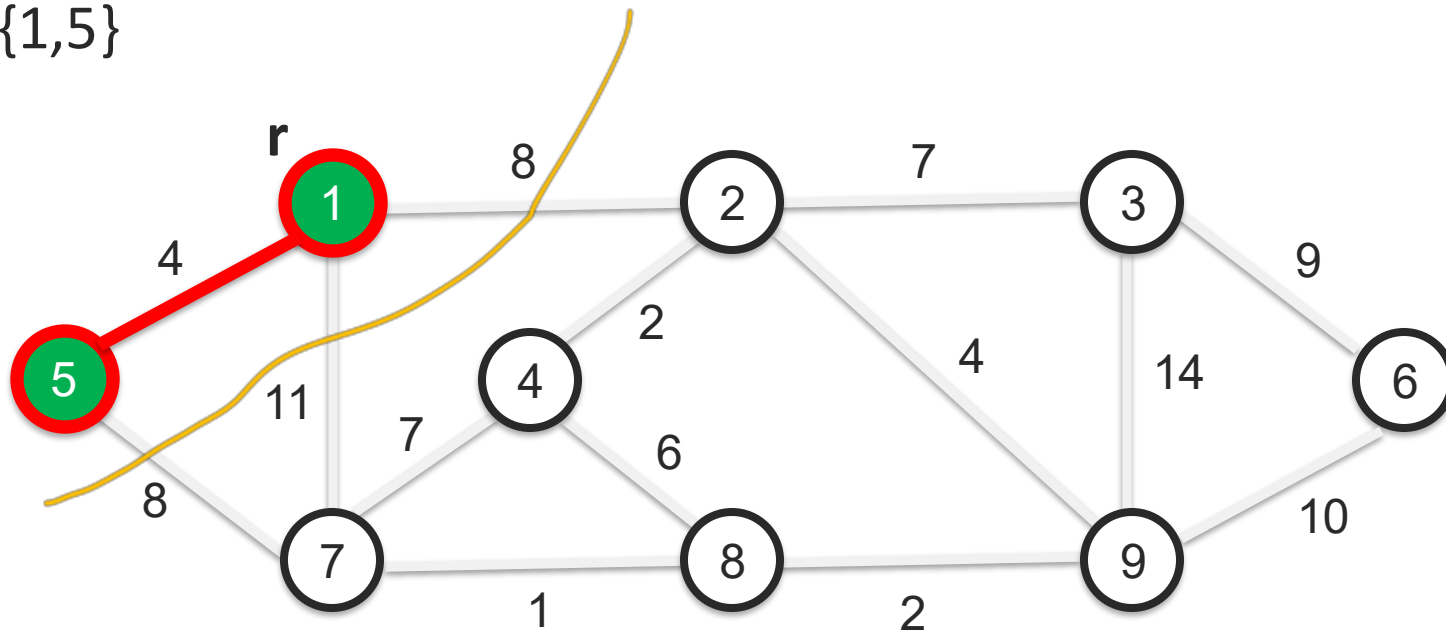
$V_C = \{1, 5\}$



$A = \{(1,5)\}$

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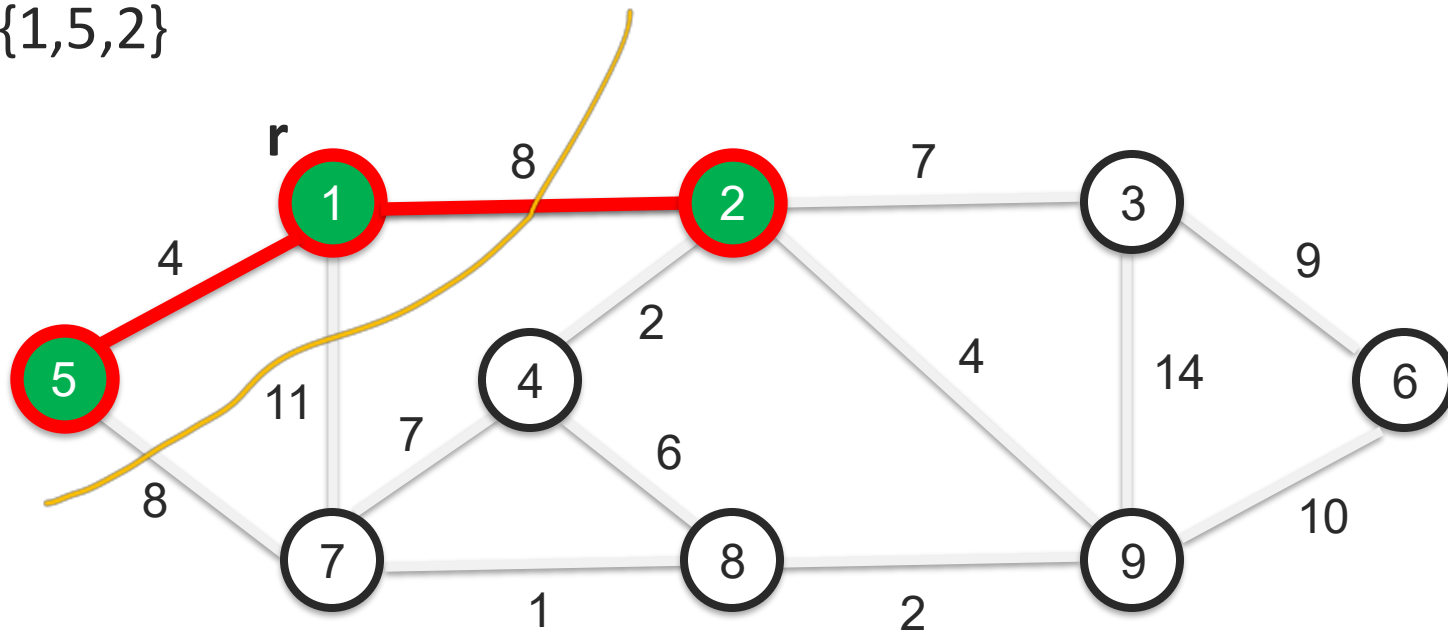
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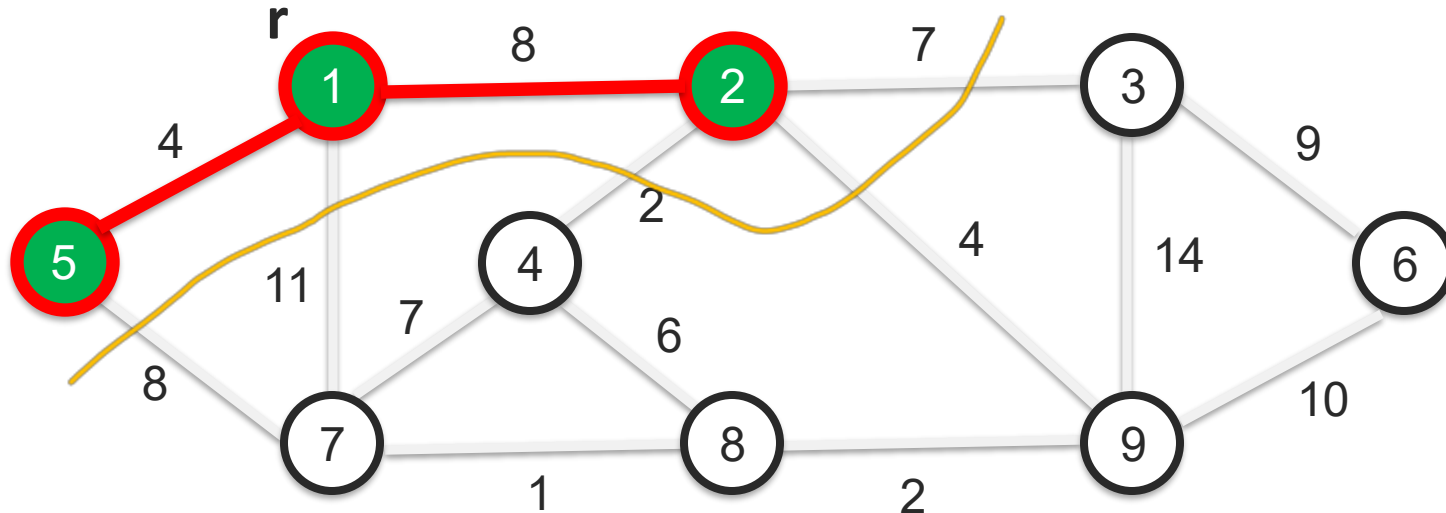
$V_C = \{1, 5, 2\}$



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Idea dell'algoritmo di Prim

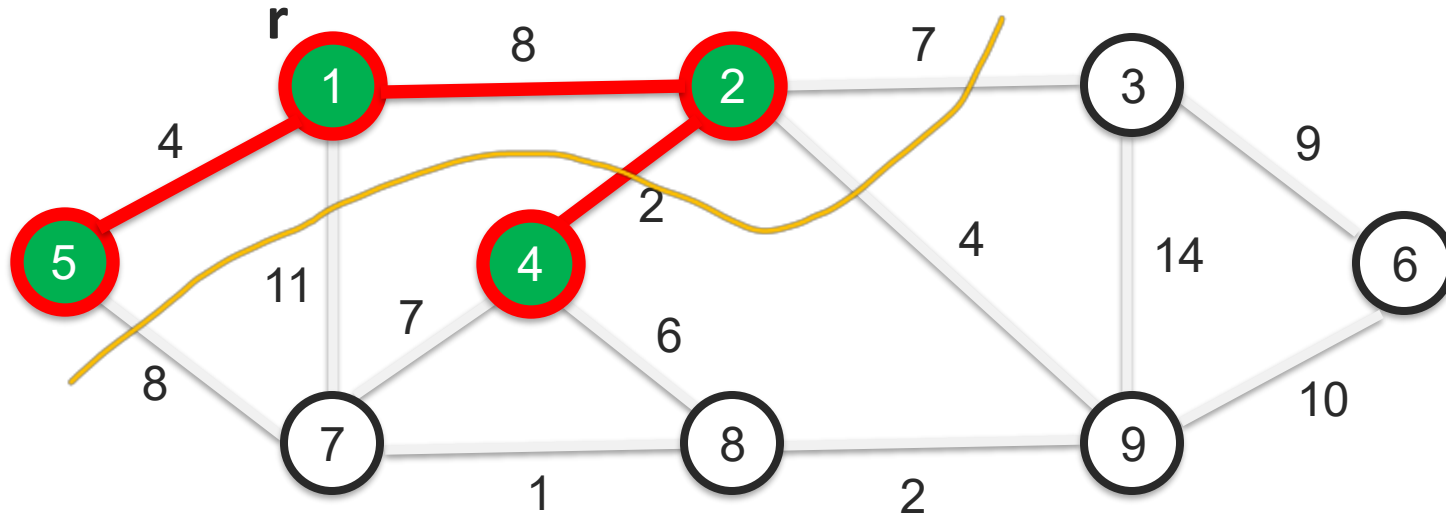
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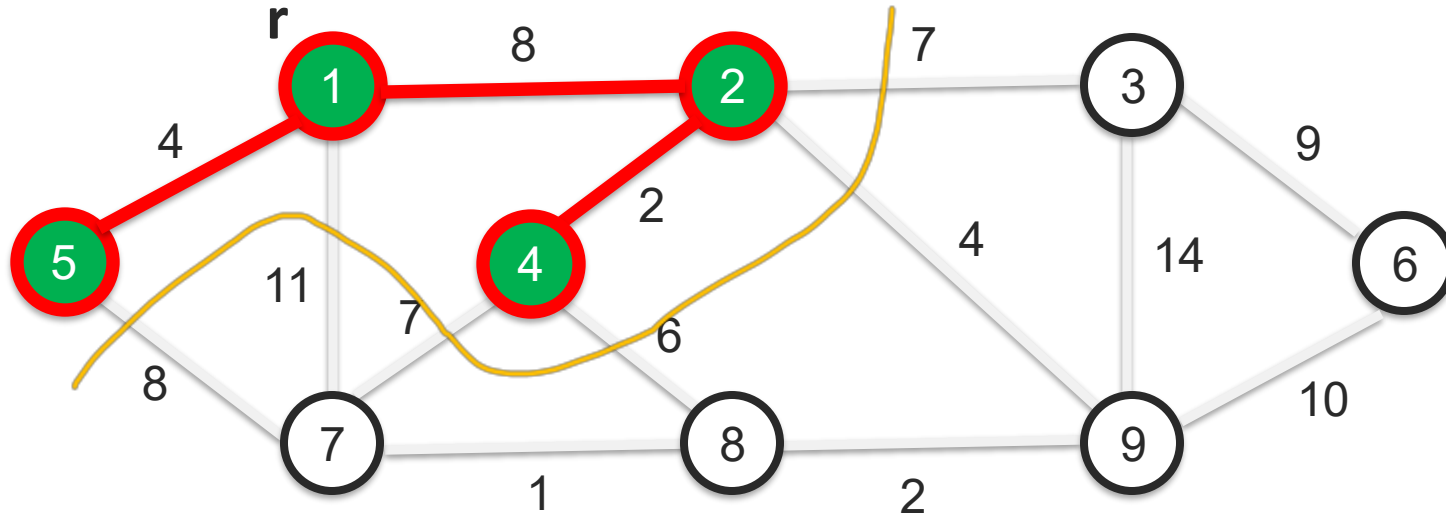
$V_C = \{1, 5, 2, 4\}$



$A = \{(1,5), (1,2), (2,4)\}$

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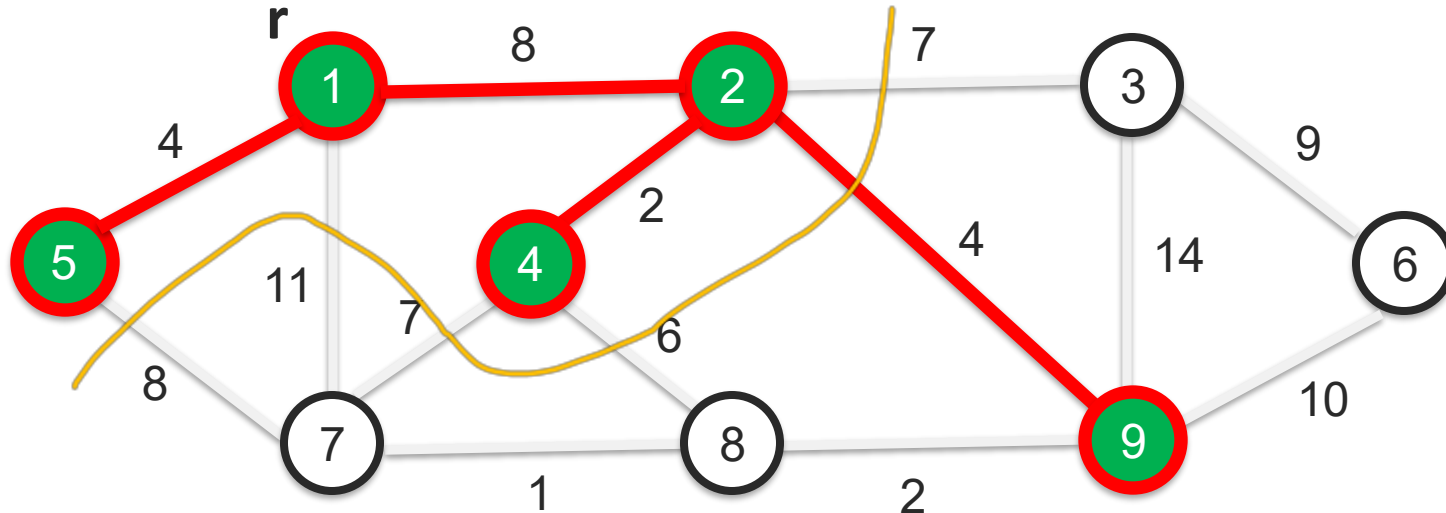
$V_C = \{1, 5, 2, 4\}$



$A = \{(1,5), (1,2), (2,4)\}$

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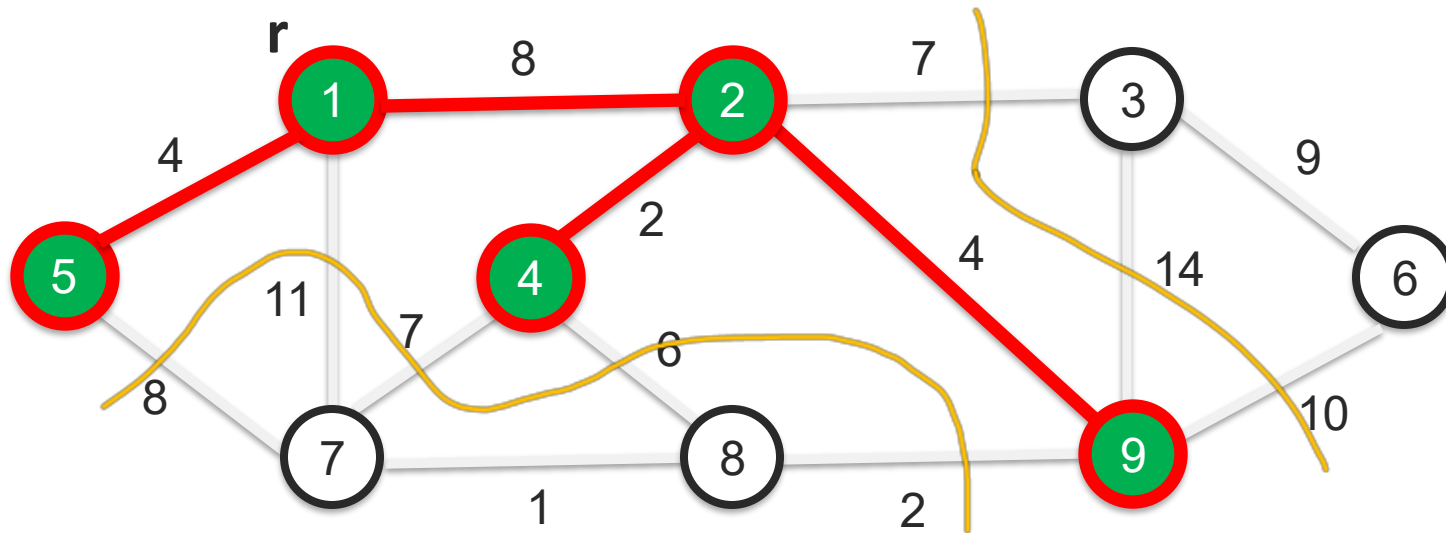
$V_C = \{1, 5, 2, 4, 9\}$



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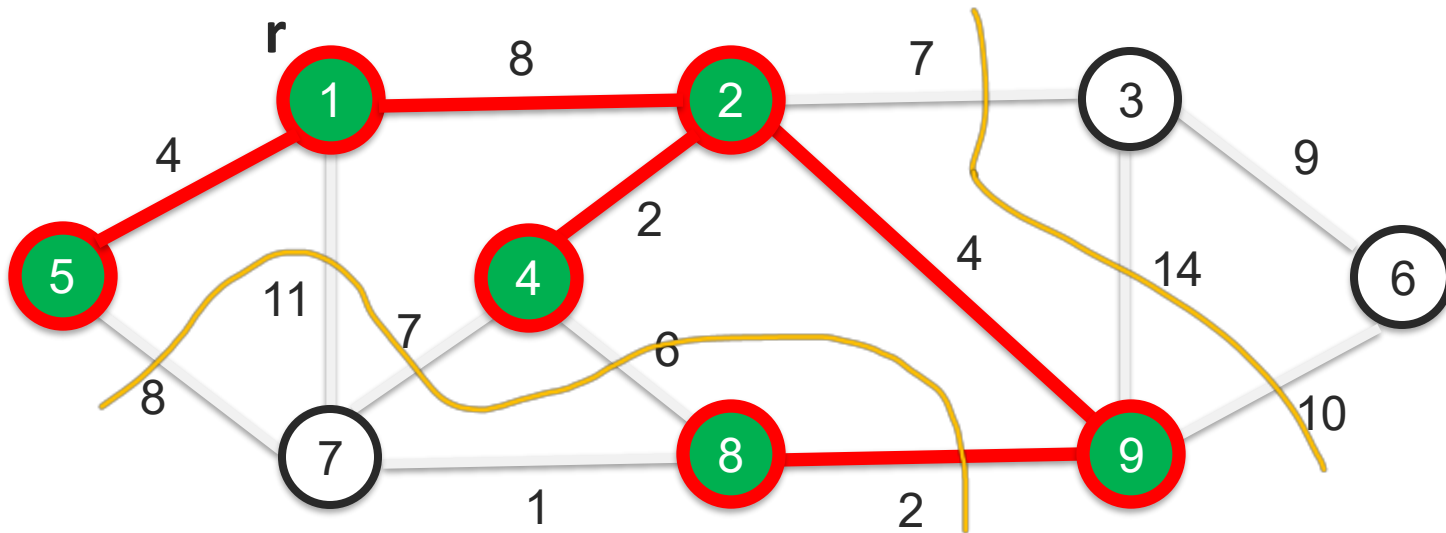
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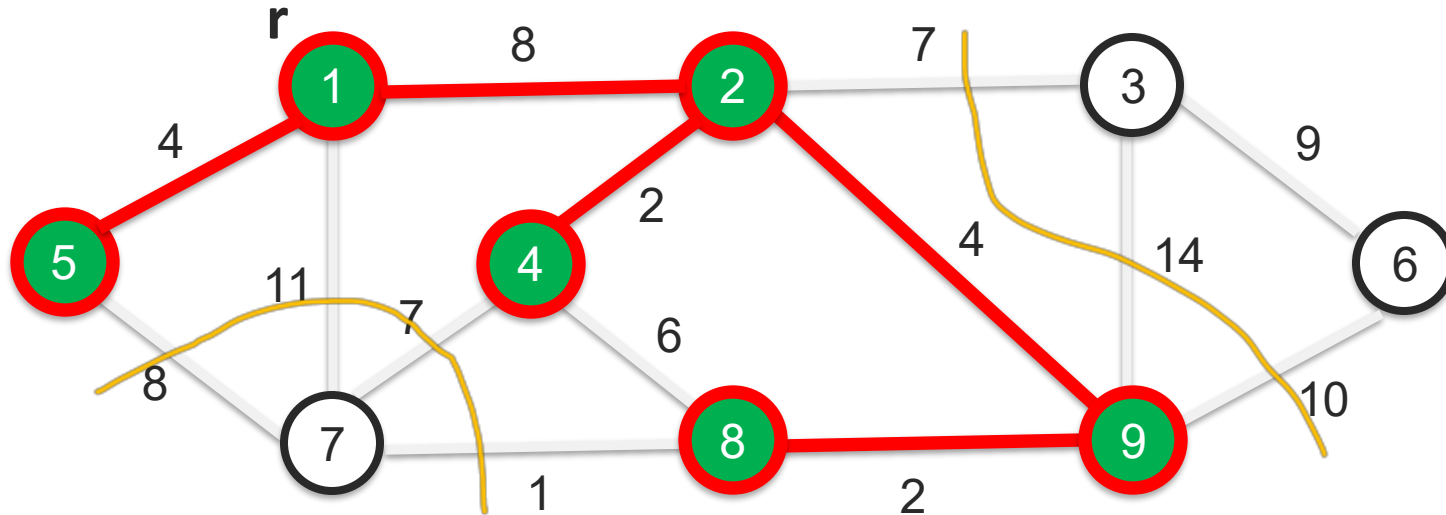
$V_C = \{1, 5, 2, 4, 9, 8\}$



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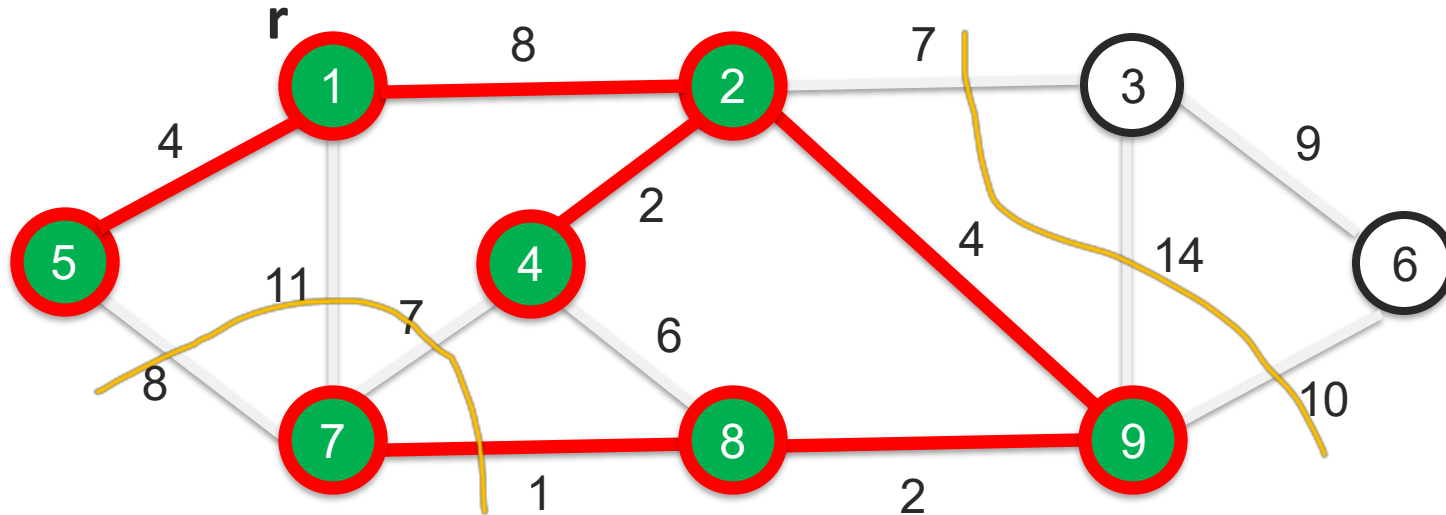
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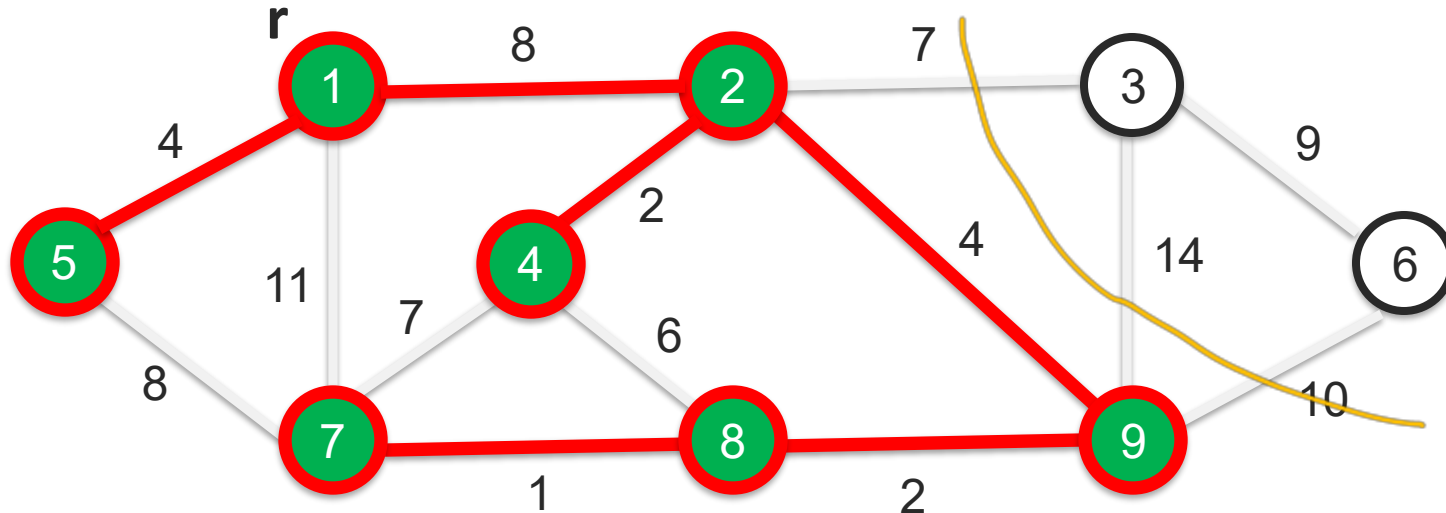
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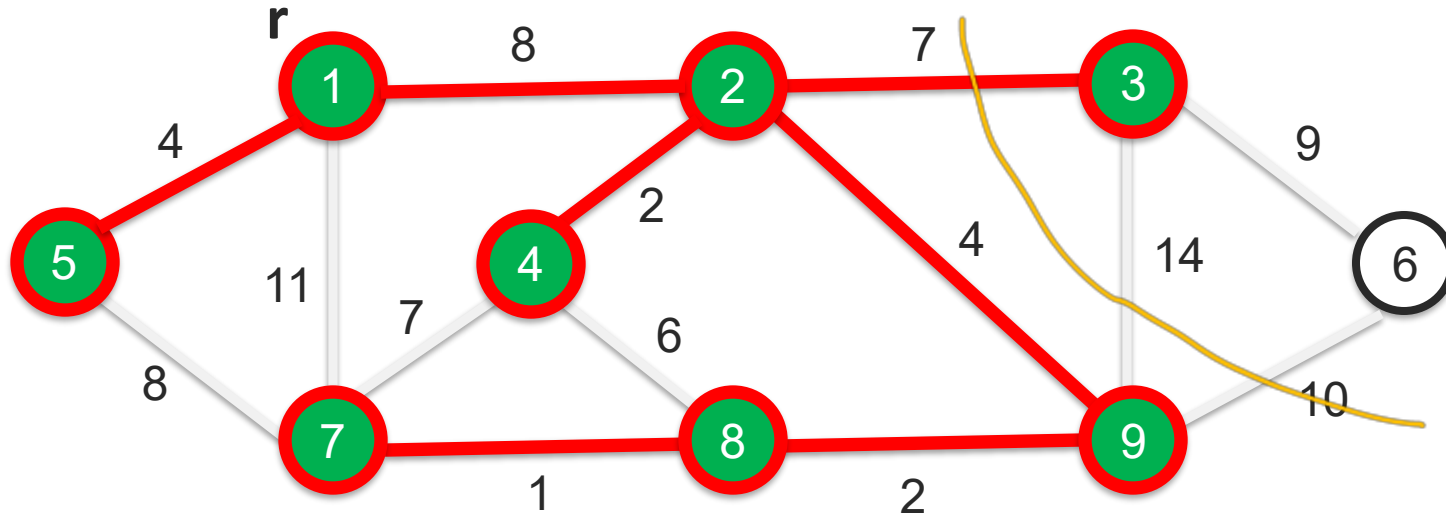
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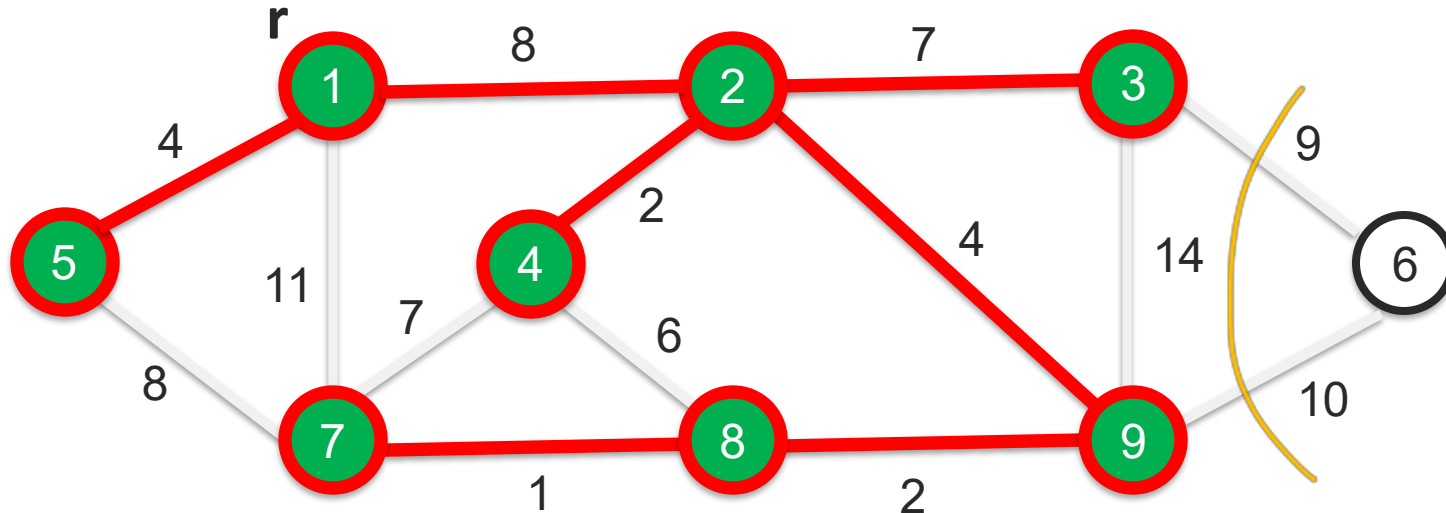
$V_C = \{1, 5, 2, 4, 9, 8, 7, 3\}$



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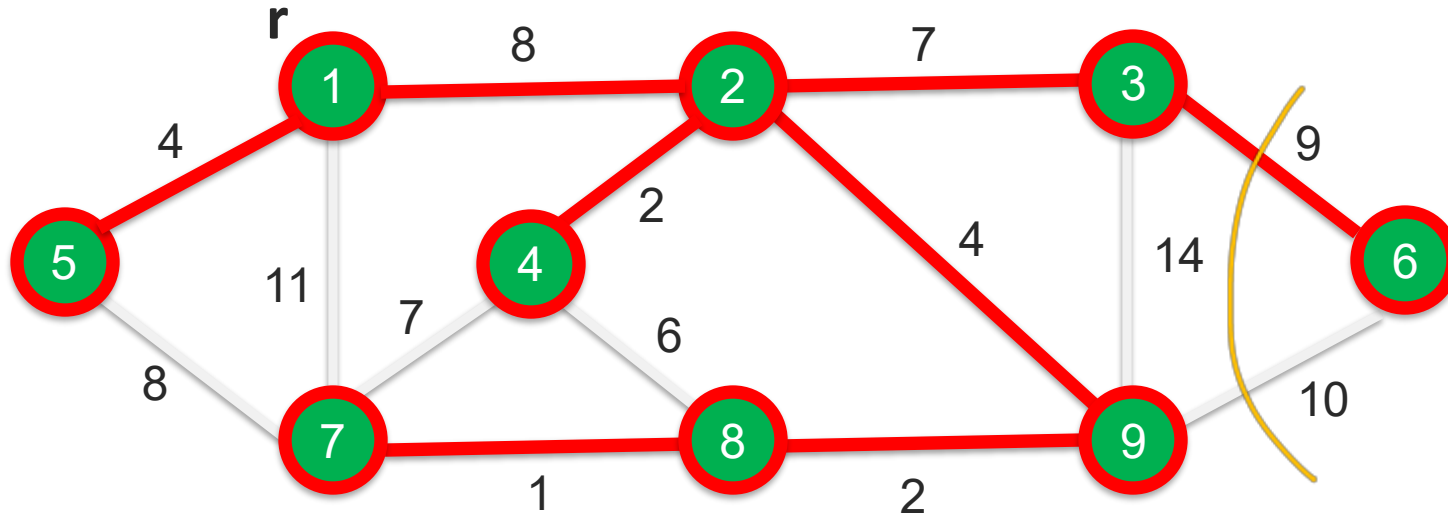
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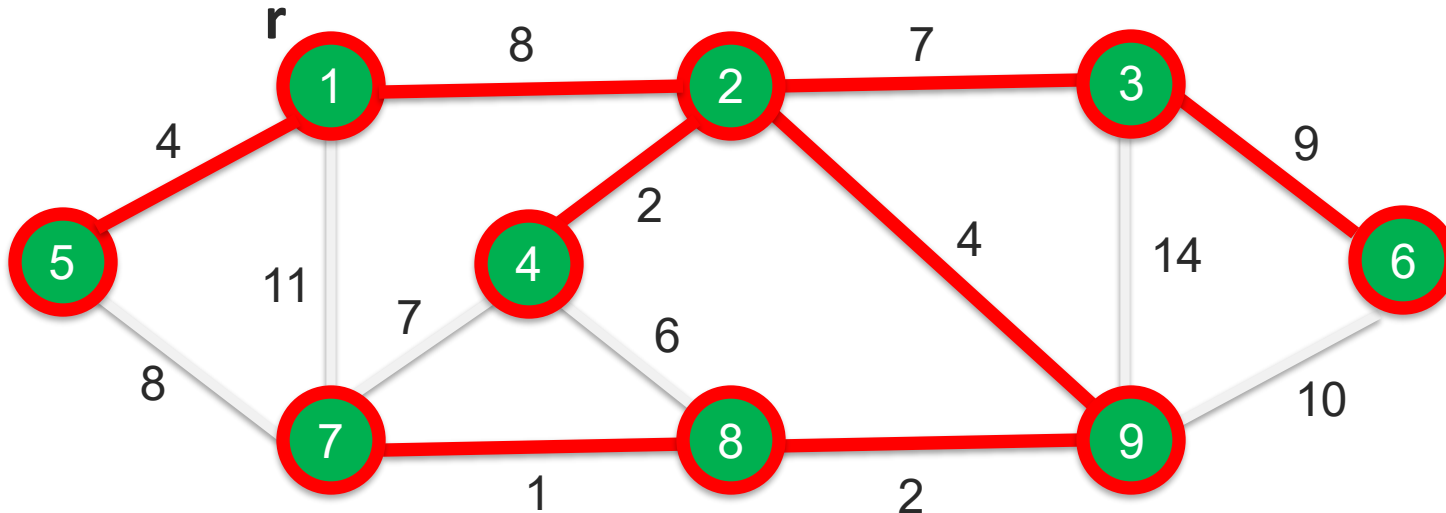
$V_C = \{1, 5, 2, 4, 9, 8, 7, 3, 6\}$



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Idea dell'algoritmo di Prim

$V_C = \{1, 5, 2, 4, 9, 8, 7, 3, 6\}$



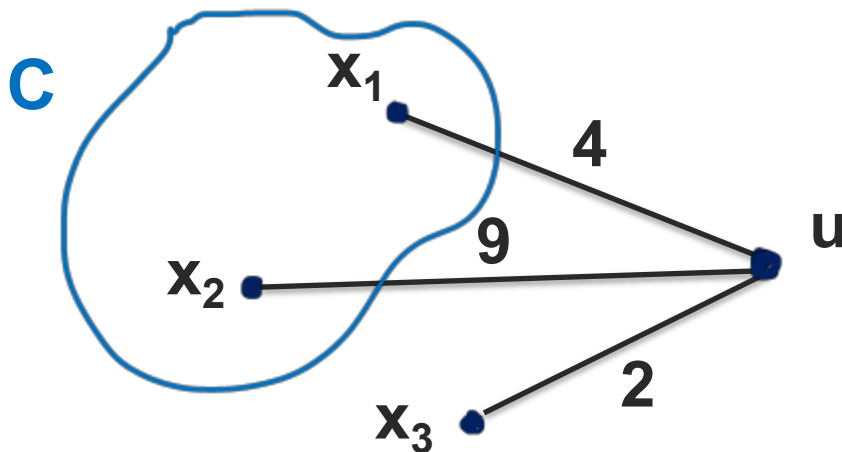
$A = \{(1,5), (1,2), (2,4), (2,9), (8,9), (7,8), (2,3), (3,6)\}$

$\Rightarrow G_A = (V, A) \rightarrow \text{Minimum Spanning Tree}$

[Algoritmo di Prim]

Due attributi per ogni vertice u del grafo:

- $u.key$ = minimo tra i pesi degli archi (x,u) tale che $x \in C$
- $u.\pi$ = vertice che ha fornito il valore $u.key$



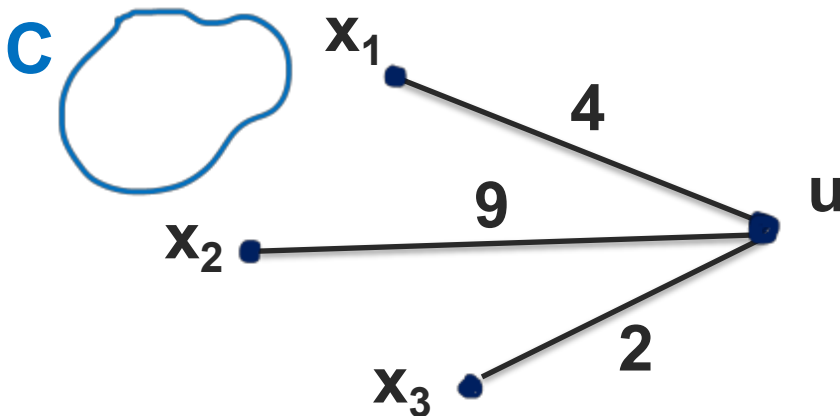
$u.key = 4$
 $u.\pi = x_1$

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$\nexists (x,u) \in E$ t.c. $x \in C \Rightarrow u.key = +\infty, u.\pi = NIL$



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INIZIALIZZAZIONE

per ogni vertice $u \in V$

$u.key \leftarrow +\infty$

$u.\pi \leftarrow NIL$

$r.key \leftarrow 0$

[Algoritmo di Prim]

Due attributi per ogni vertice u del grafo:

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- $u.\pi$ = vertice che ha fornito il valore $u.key$

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$p = \min\{u'.key \mid u' \notin C\}$ è il peso di un arco sicuro
 $(u.\pi, u)$ tale che $u.key = p$ è un arco sicuro

[Algoritmo di Prim]

Prima di un'iterazione:

- ✓ V_C , insieme dei vertici nella componente C
- ✓ $V - V_C$, insieme dei vertici che stanno fuori da C

Generica iterazione:

- viene estratto da $V - V_C$ il vertice u con il minimo valore di **key**
 - $\Rightarrow u.key$ è il peso di un arco sicuro
 - $\Rightarrow (u.\pi, u)$ è un arco sicuro
- per ogni $v \in \text{adj}(u)$ che sta in $V - V_C$ e tale per cui $W(u,v) < v.key$:
 - $v.key$ viene aggiornato a $W(u,v)$
 - $v.\pi$ viene aggiornato a u

$$V - V_C = \emptyset$$

$$\Rightarrow T = \{(u.\pi, u) \text{ tale che } u \in V, u \neq r\}$$

[Algoritmo di Prim]

PRIM-MST(V, E, W, r)

foreach $v \in V$ **do**

$v.\text{key} \leftarrow +\infty$

$v.\pi \leftarrow \text{NIL}$

$r.\text{key} \leftarrow 0$

$Q \leftarrow$ empty min_priority queue // $V - V_c$

foreach $v \in V$ **do**

 INSERT(Q, v)

while $Q \neq \emptyset$ **do**

$u \leftarrow$ EXTRACT-MIN(Q)

foreach $v \in \text{adj}(u)$ **do**

if $v \in Q$ and $W(u, v) < v.\text{key}$ **then**

$v.\text{key} \leftarrow W(u, v)$

$v.\pi \leftarrow u$

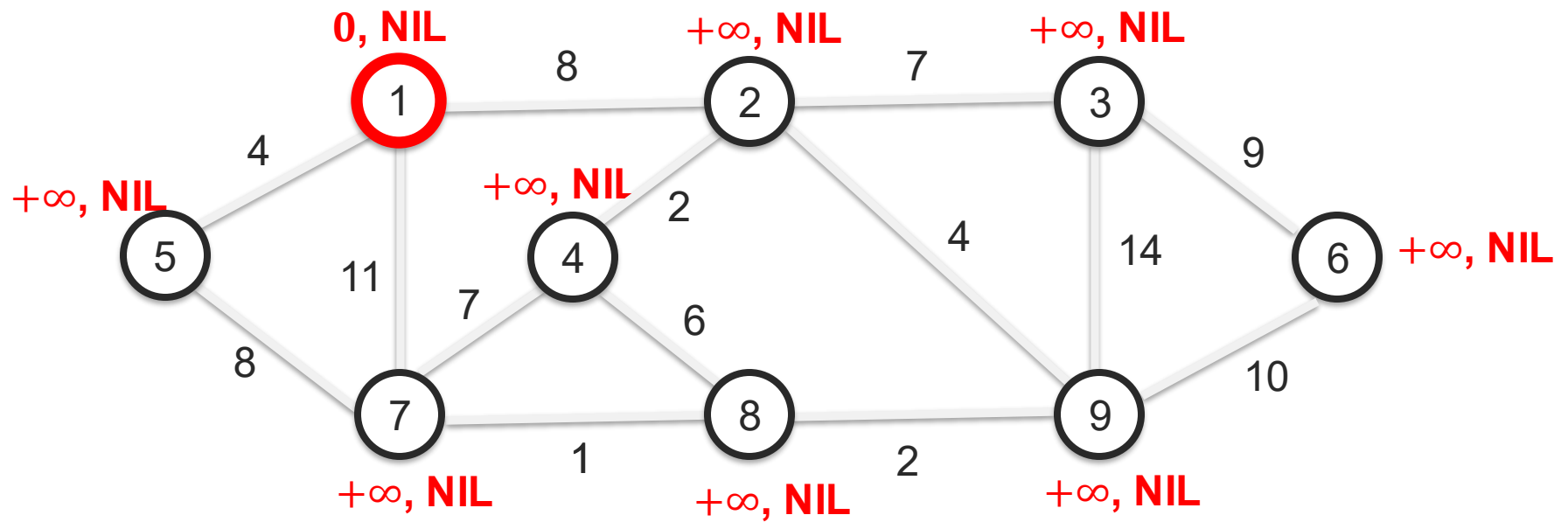
 DECREASE-KEY($Q, v, v.\text{key}$)

[Algoritmo di Prim]

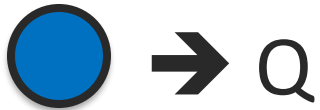
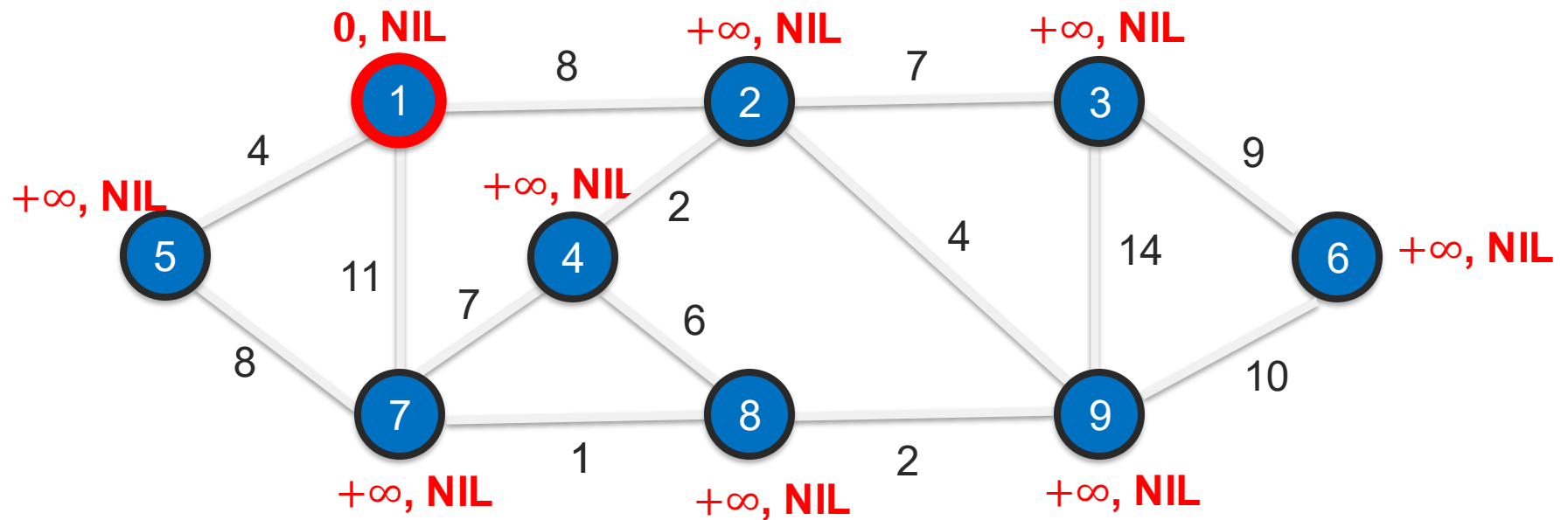
```
PRIM-MST( $V, E, W, r$ )  
  foreach  $v \in V$  do  
     $v.\text{key} \leftarrow +\infty$   
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   $r.\text{key} \leftarrow 0$   
   $Q \leftarrow$  empty min_priority queue //  $V - V_c$   
  foreach  $v \in V$  do  
    INSERT( $Q, v$ )  
  while  $Q \neq \emptyset$  do  
     $u \leftarrow$  EXTRACT-MIN( $Q$ )  
    foreach  $v \in \text{adj}(u)$  do  
      if  $v \in Q$  and  $W(u, v) <$   
         $v.\text{key} \leftarrow W(u, v)$   
         $v.\pi \leftarrow u$   
    DECREASE-KEY( $Q, v, v.\text{key}$ )
```

Complessità in tempo
 $O(|E|\log|E|)$

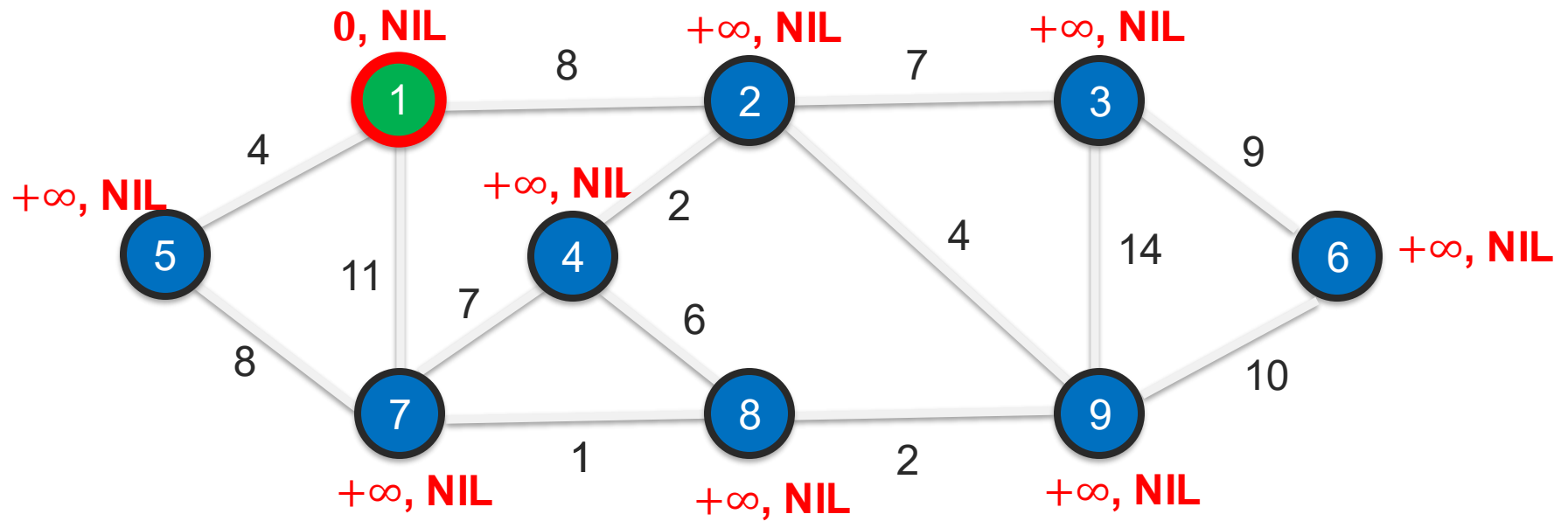
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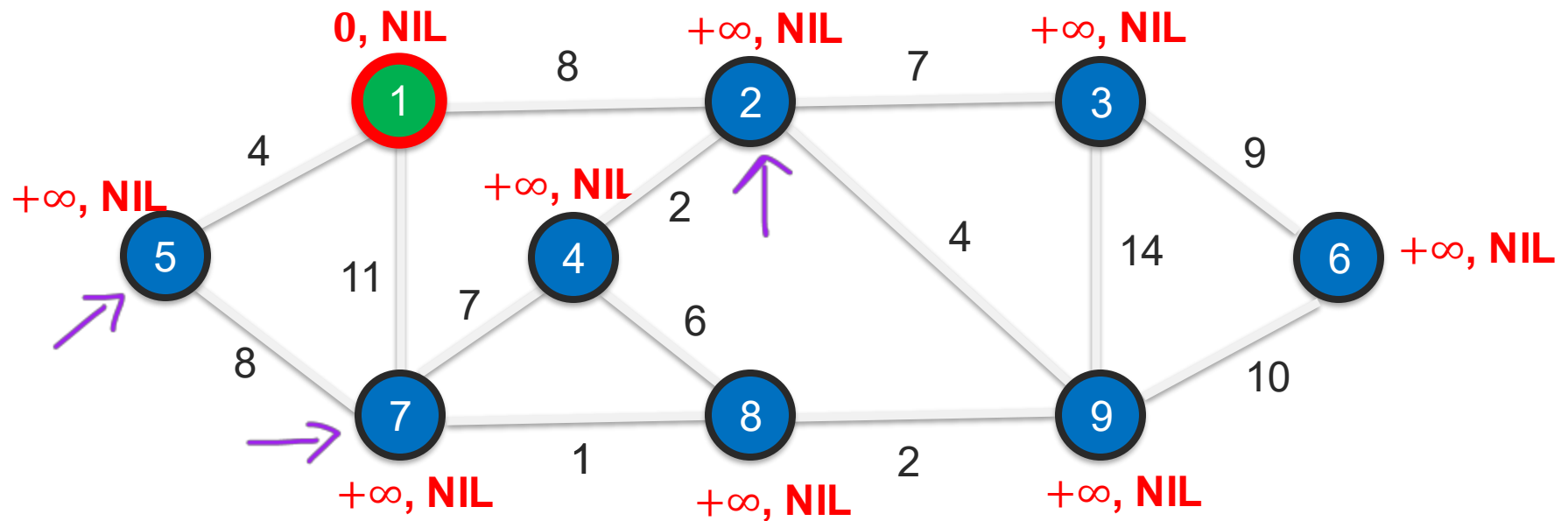


[Algoritmo di Prim]



Estrazione da Q del vertice 1

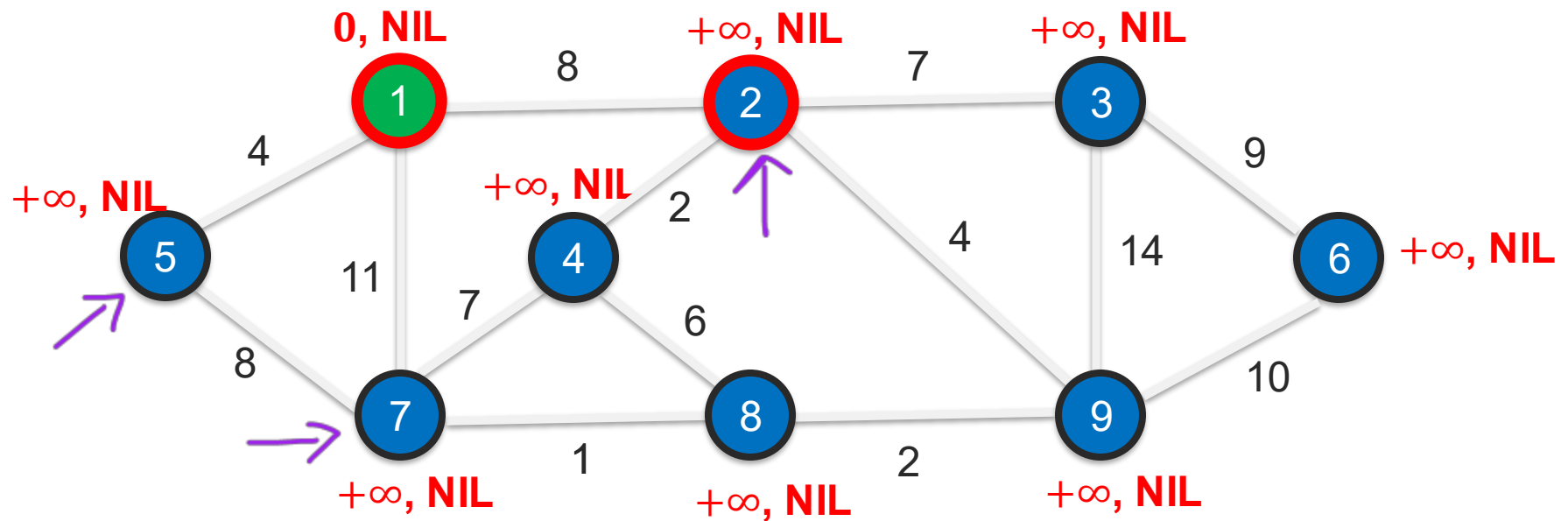
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Estrazione da Q del vertice 1

Adiacenti del vertice 1 $\rightarrow$ 2, 7, 5

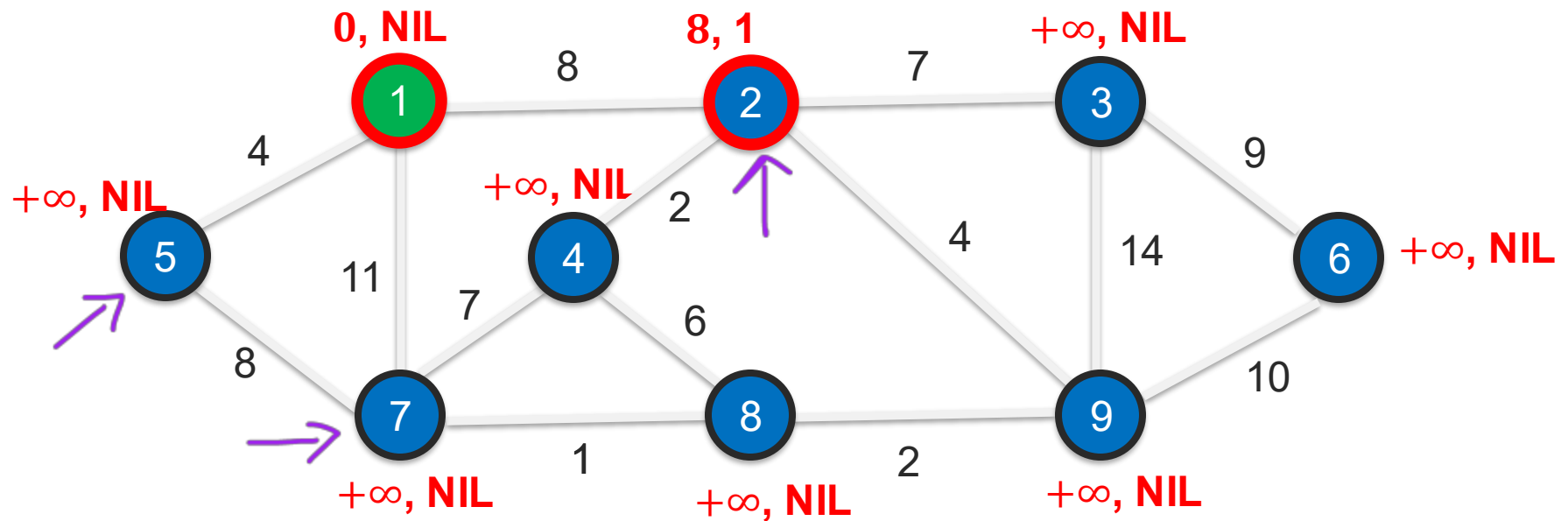
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Estrazione da Q del vertice 1

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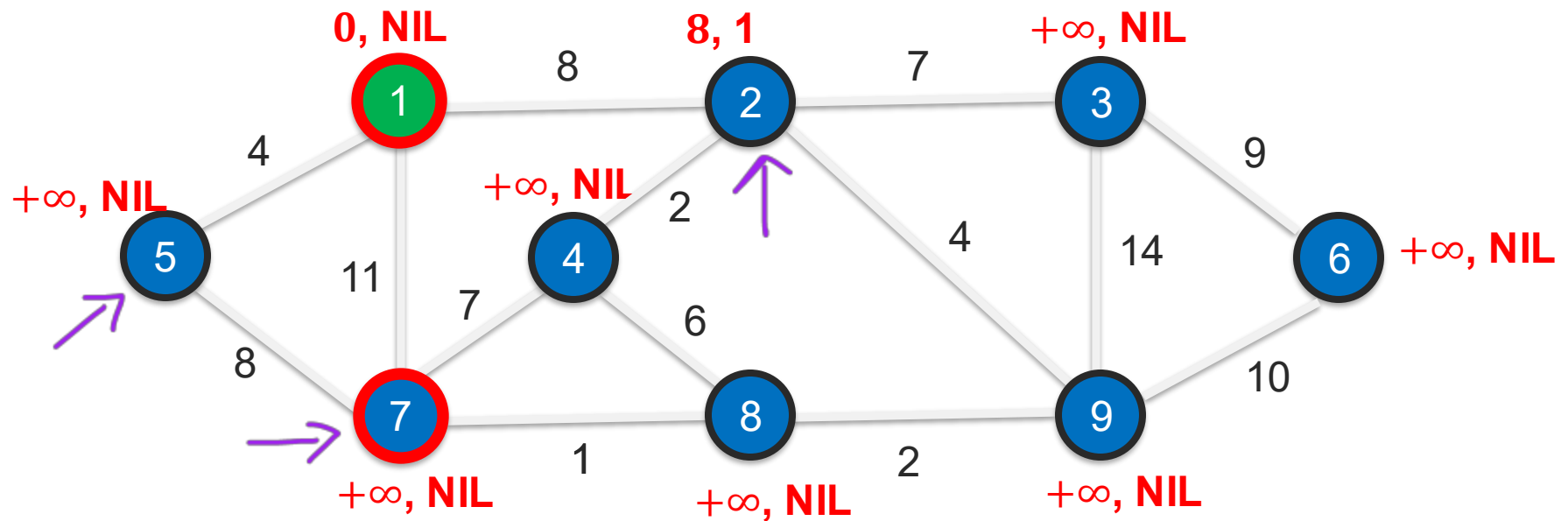
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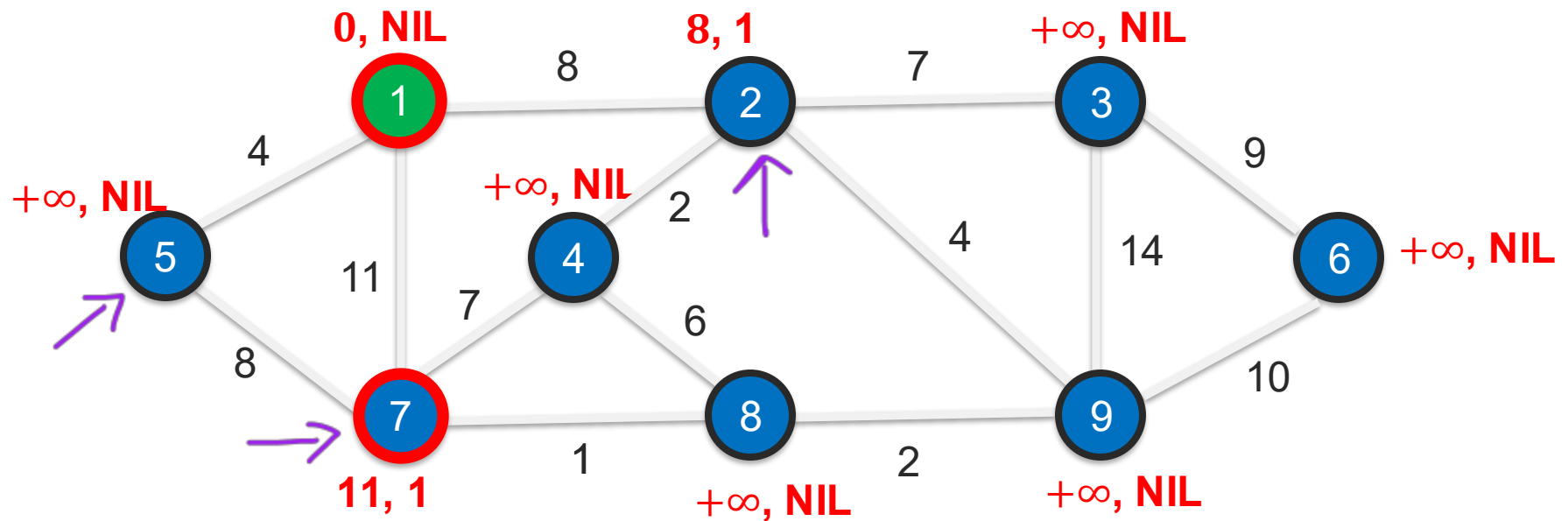
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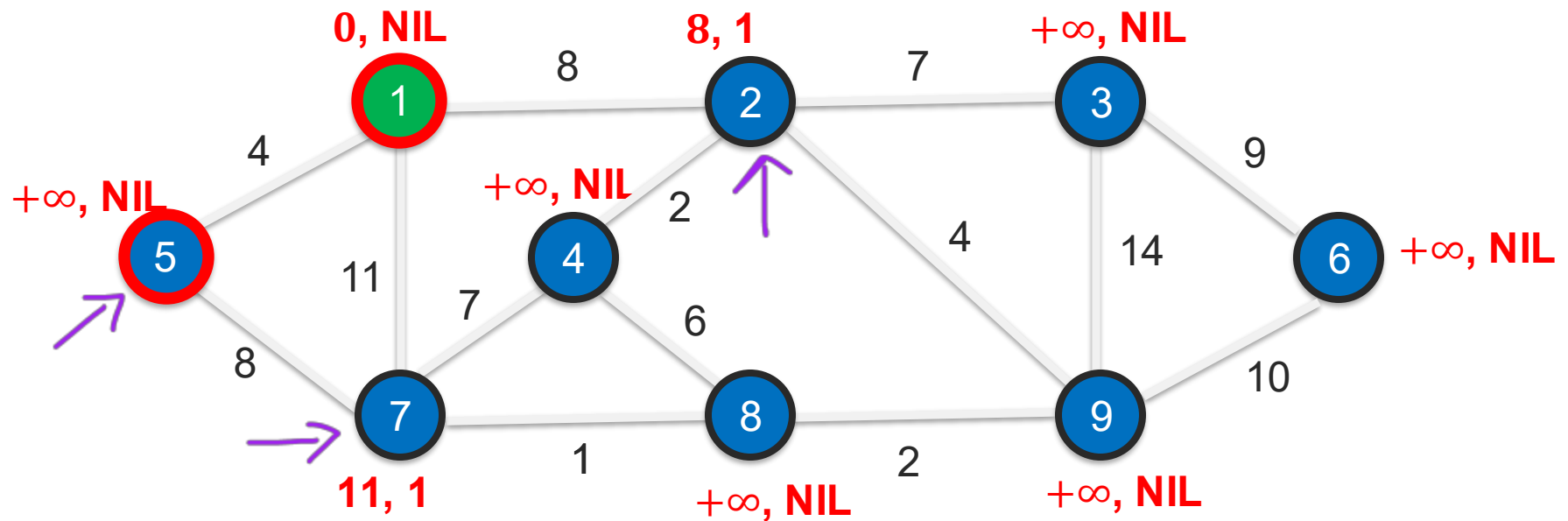
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Estrazione da Q del vertice 1

Adiacenti del vertice 1 $\rightarrow$ 2, 7, 5

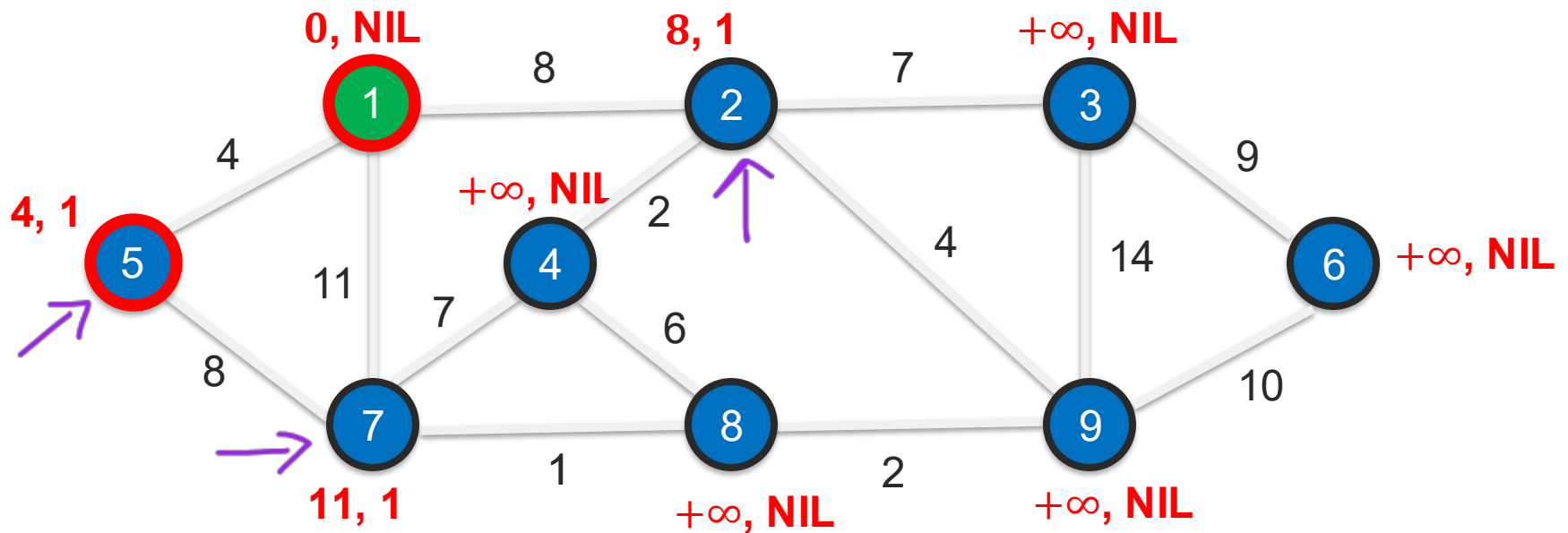
[Algoritmo di Prim]



Estrazione da Q del vertice 1

Adiacenti del vertice 1 $\rightarrow$ 2, 7, 5

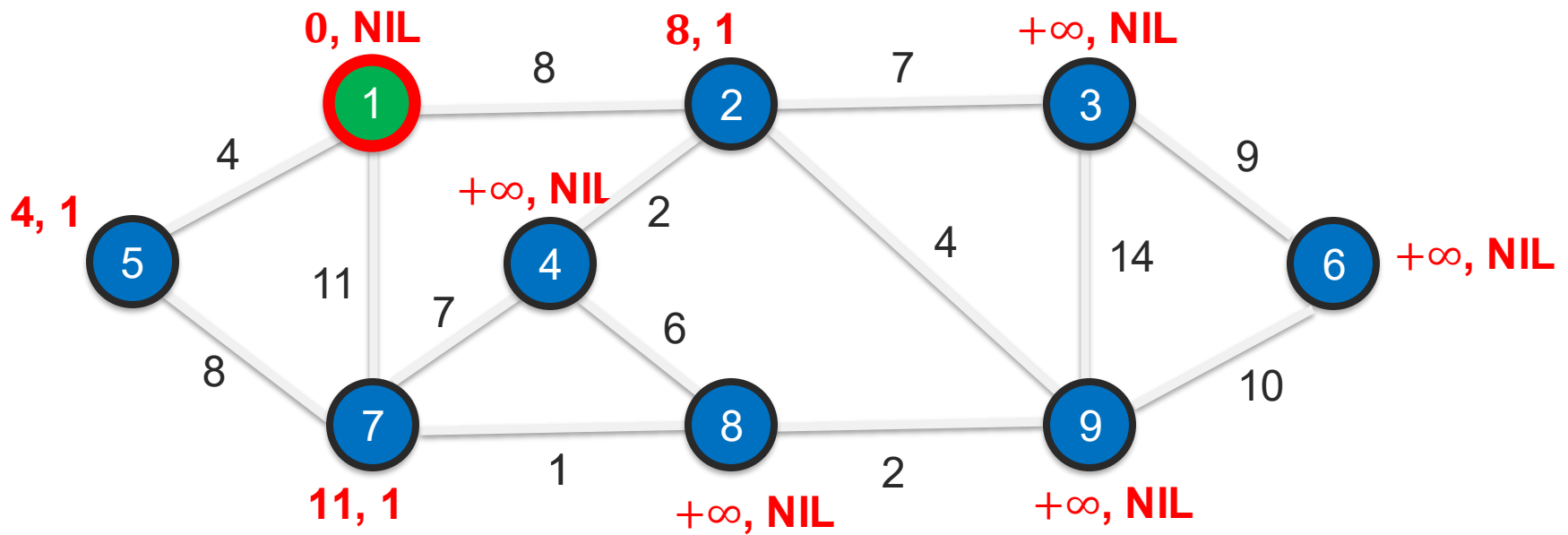
[Algoritmo di Prim]



Estrazione da Q del vertice 1

Adiacenti del vertice 1 $\rightarrow$ 2, 7, 5

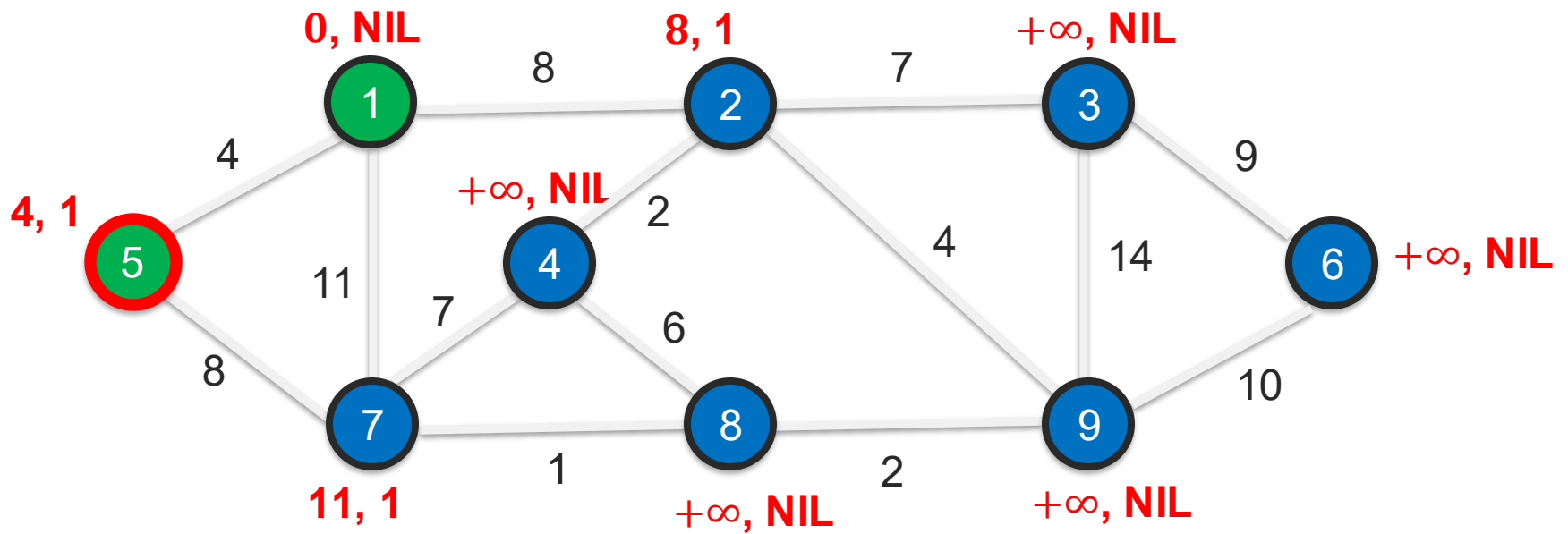
[Algoritmo di Prim]



Estrazione da Q del vertice 1

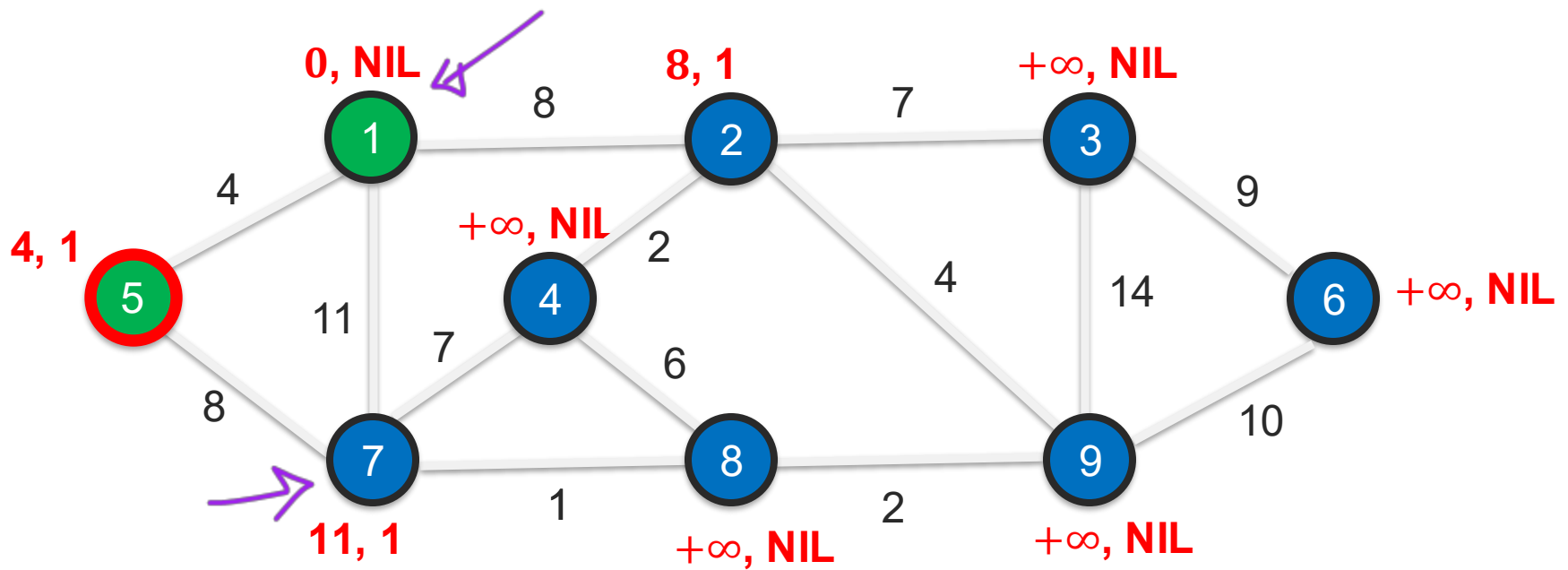
Adiacenti del vertice 1 $\rightarrow$ 2, 7, 5

[Algoritmo di Prim]



Estrazione da Q del vertice 5

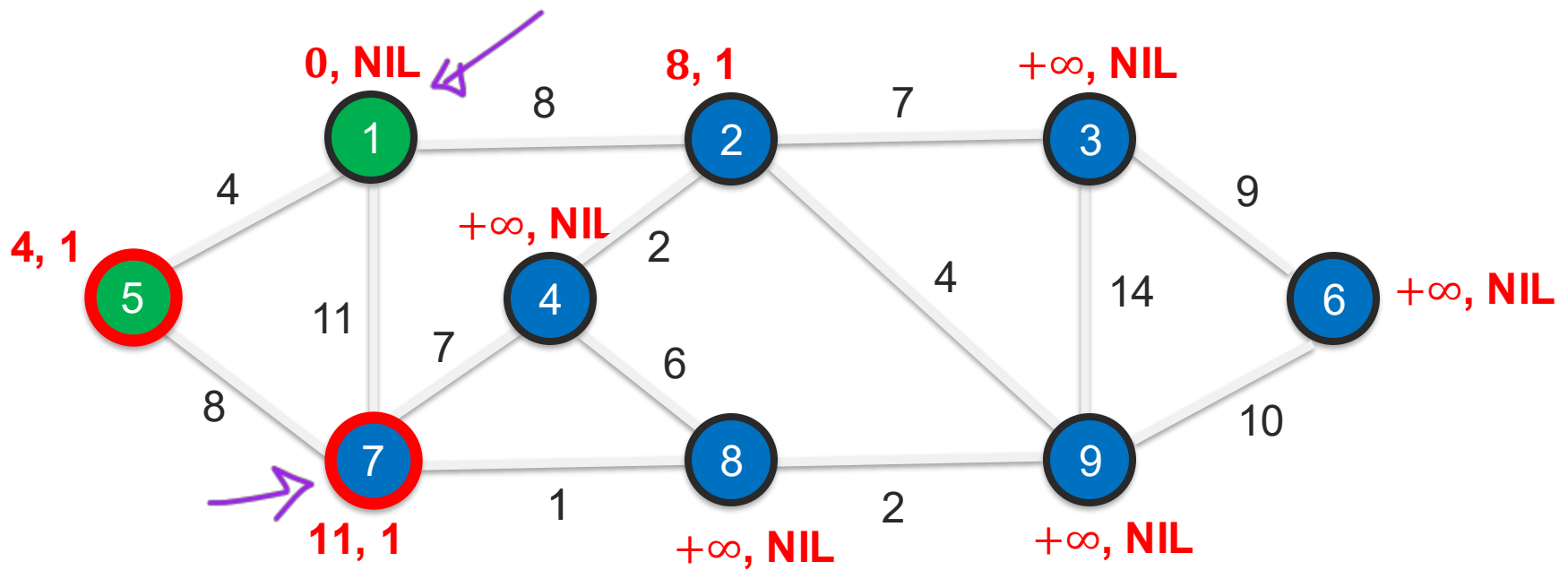
[Algoritmo di Prim]



Estrazione da Q del vertice 5

Adiacenti del vertice 5 $\rightarrow$ 1, 7

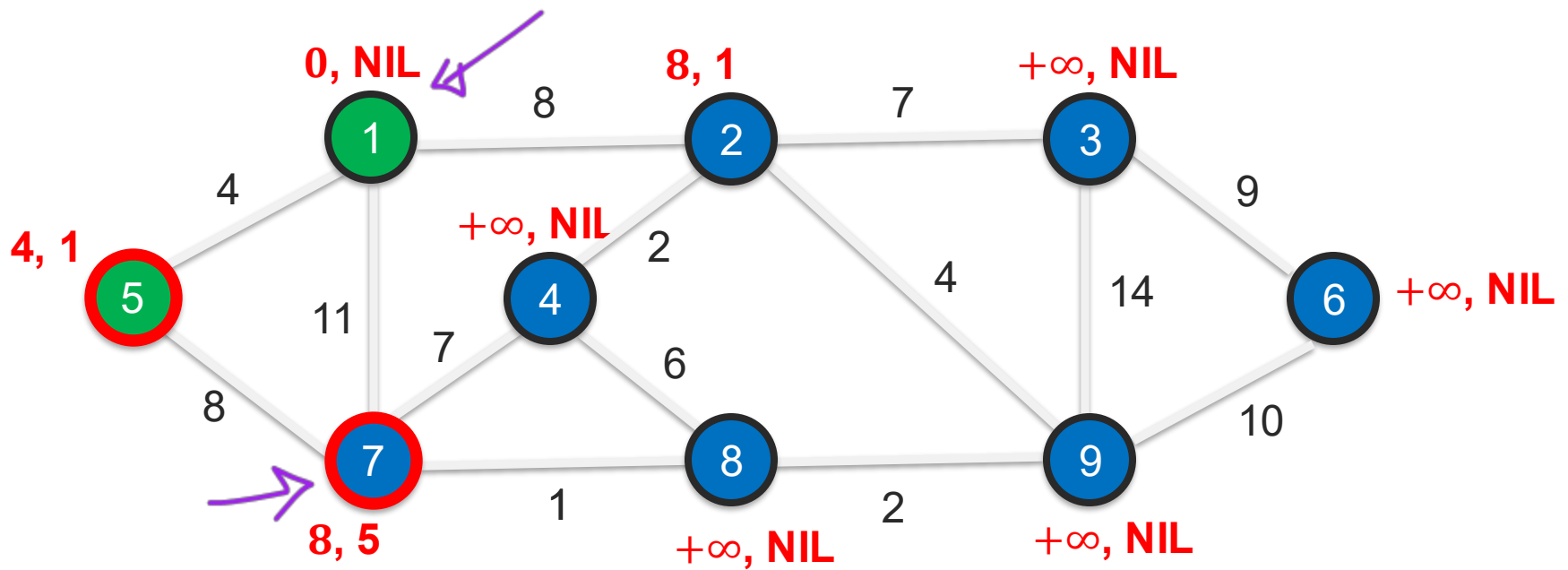
[Algoritmo di Prim]



Estrazione da Q del vertice 5

Adiacenti del vertice 5 $\rightarrow$ 1, 7

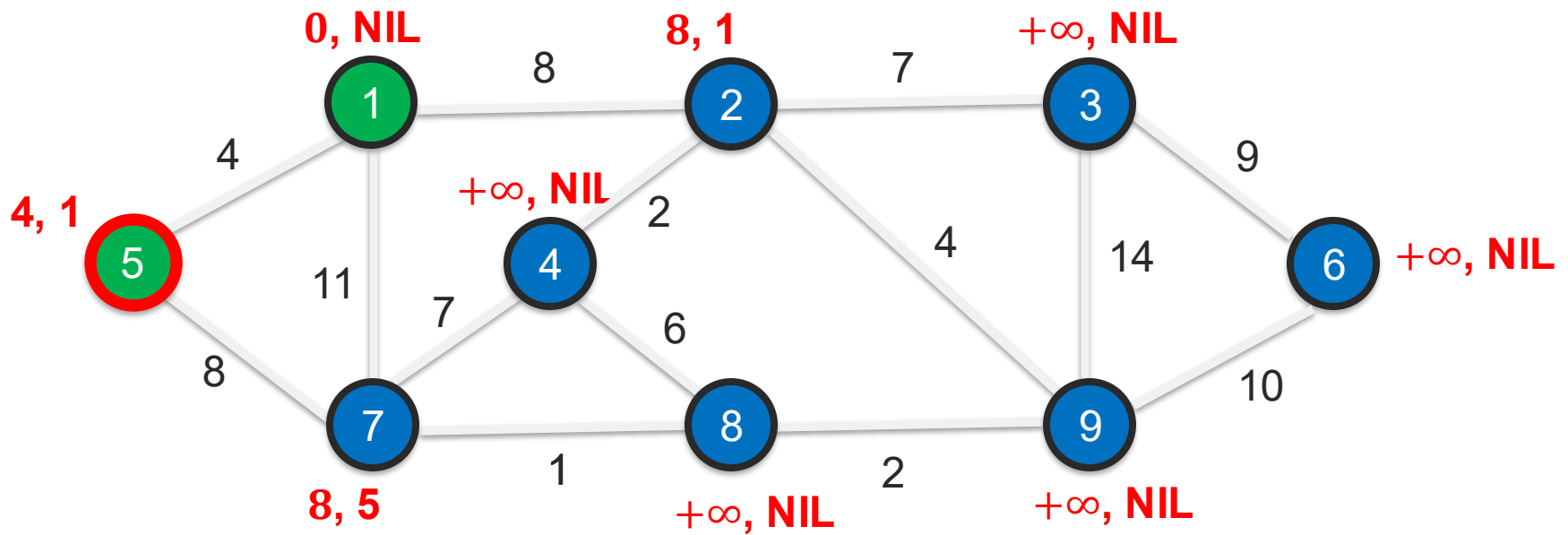
[Algoritmo di Prim]



Estrazione da Q del vertice 5

Adiacenti del vertice 5 $\rightarrow$ 1, 7

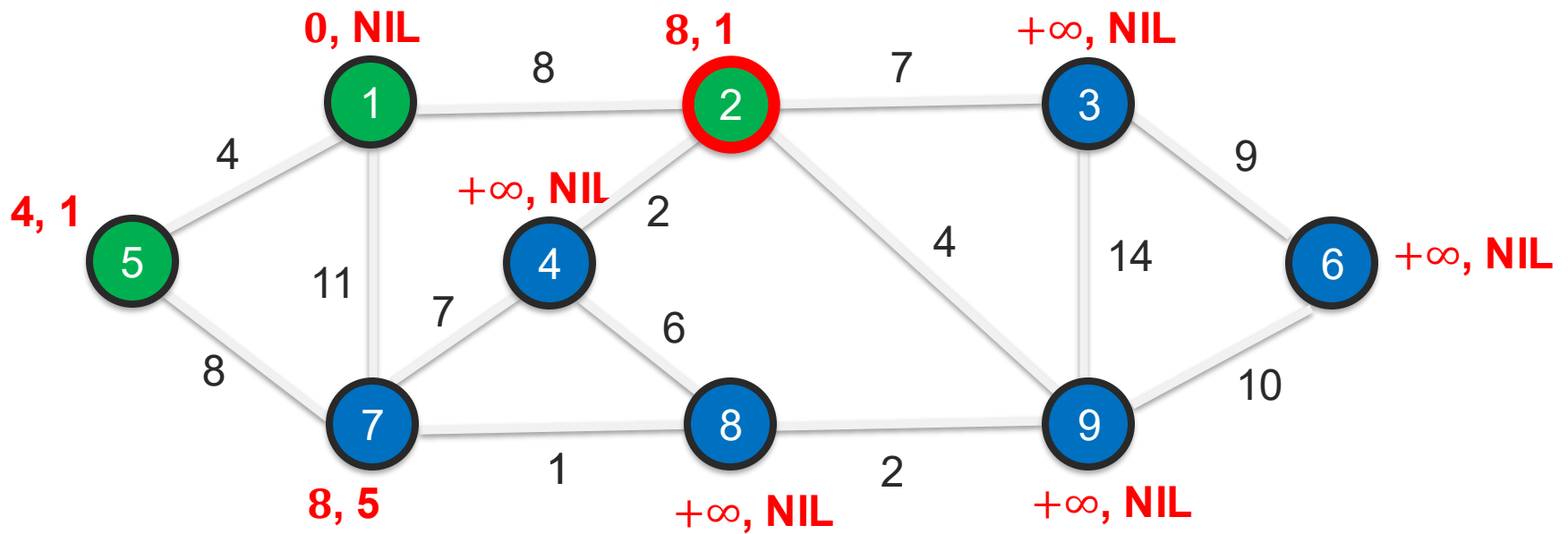
[Algoritmo di Prim]



Estrazione da Q del vertice 5

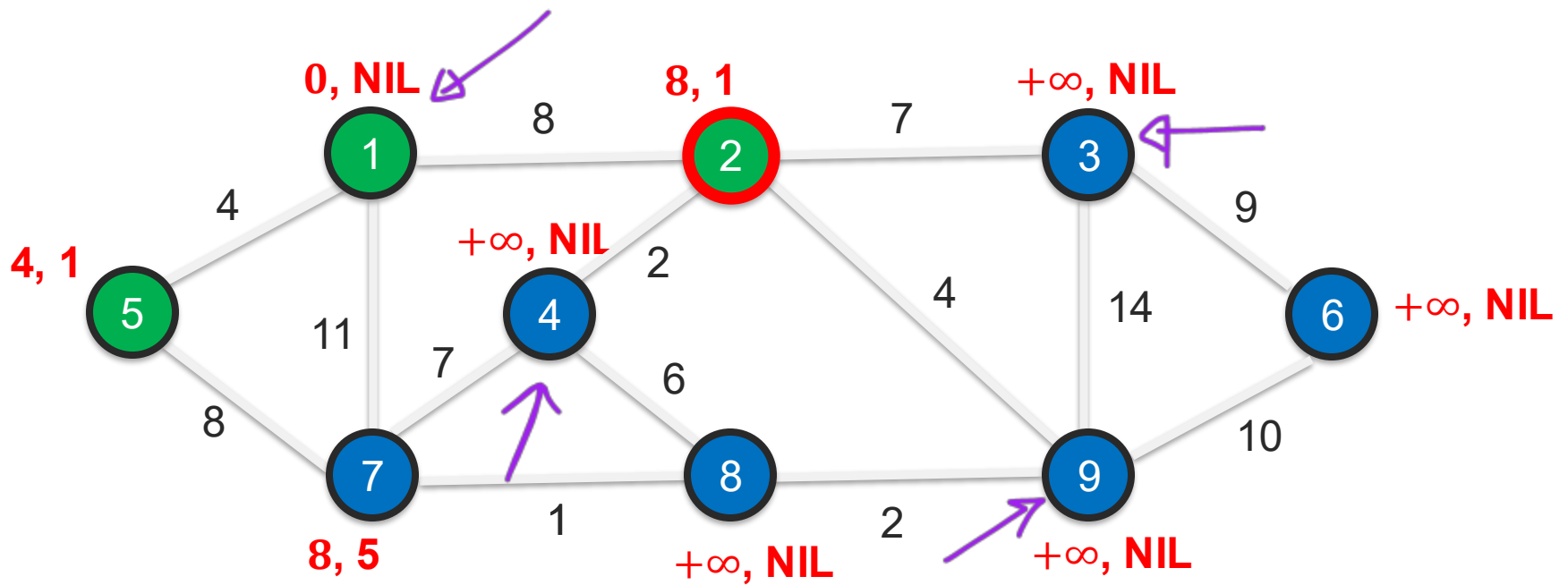
Adiacenti del vertice 5 $\rightarrow$ 1, 7

[Algoritmo di Prim]



Estrazione da Q del vertice 2

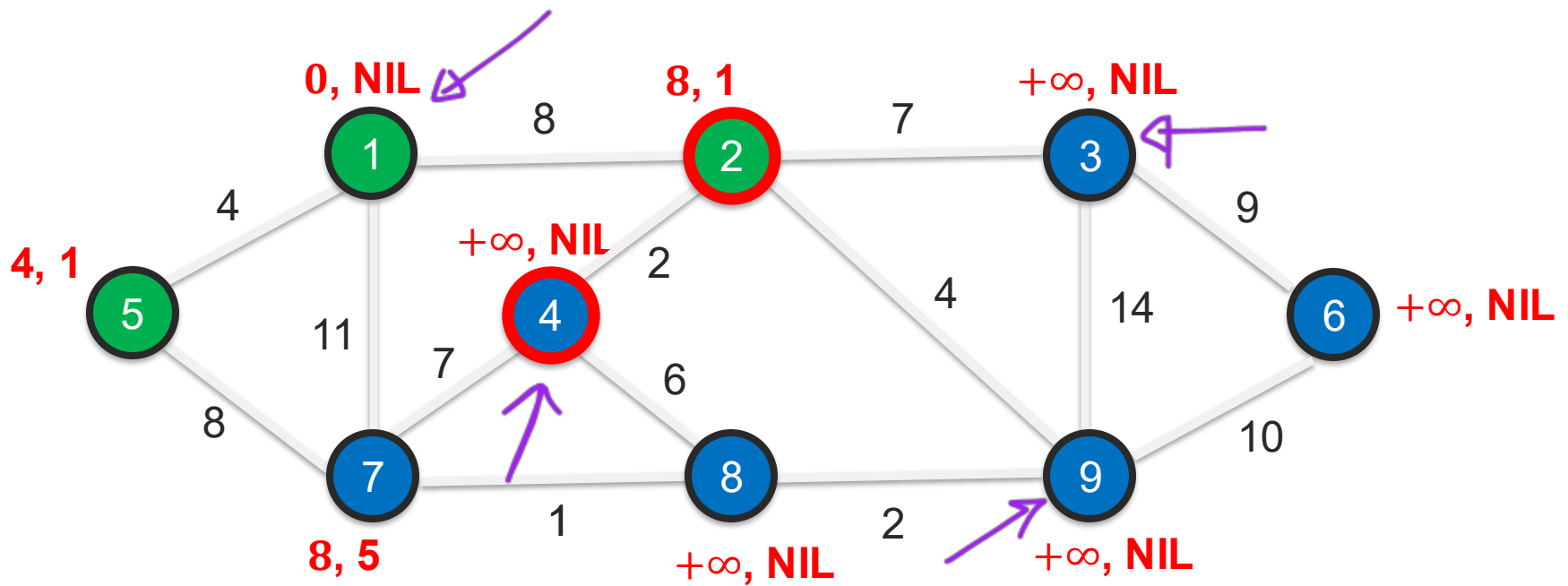
[Algoritmo di Prim]



Estrazione da Q del vertice 2

Adiacenti del vertice 2 $\rightarrow$ 1, 4, 9, 3

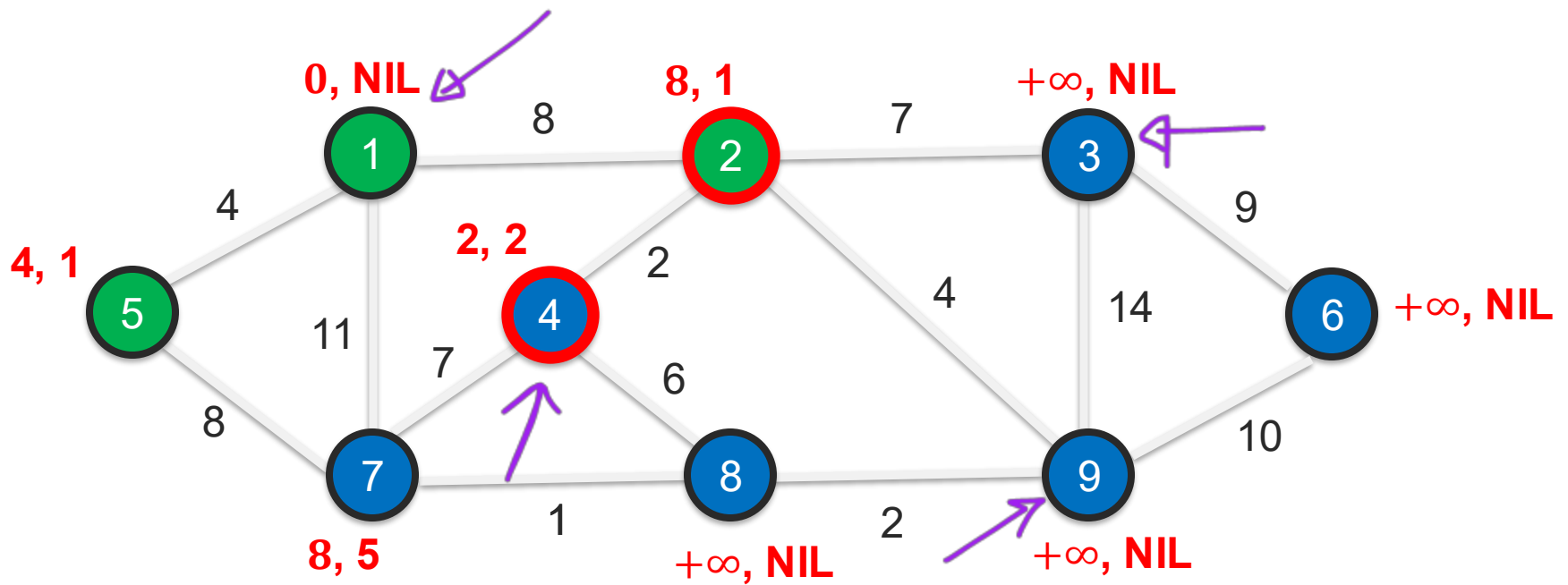
[Algoritmo di Prim]



Estrazione da Q del vertice 2

Adiacenti del vertice 2 $\rightarrow$ 1, 4, 9, 3

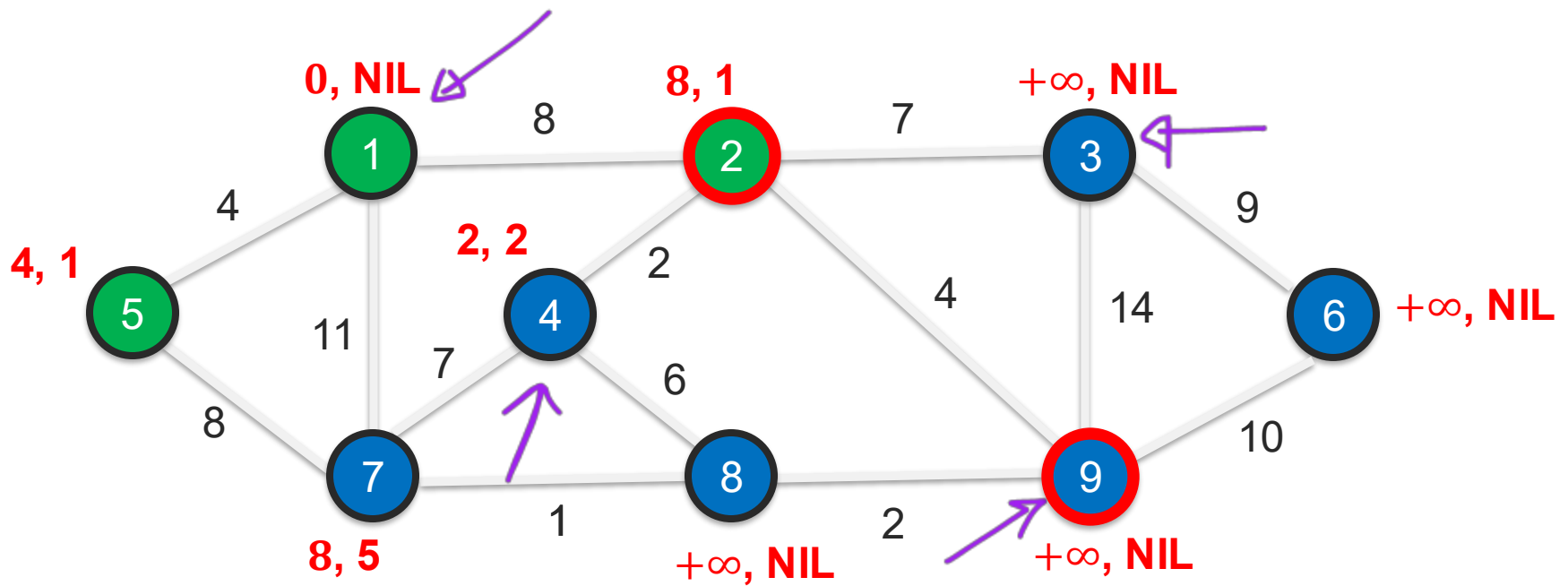
[Algoritmo di Prim]



Estrazione da Q del vertice 2

Adiacenti del vertice 2 $\rightarrow$ 1, 4, 9, 3

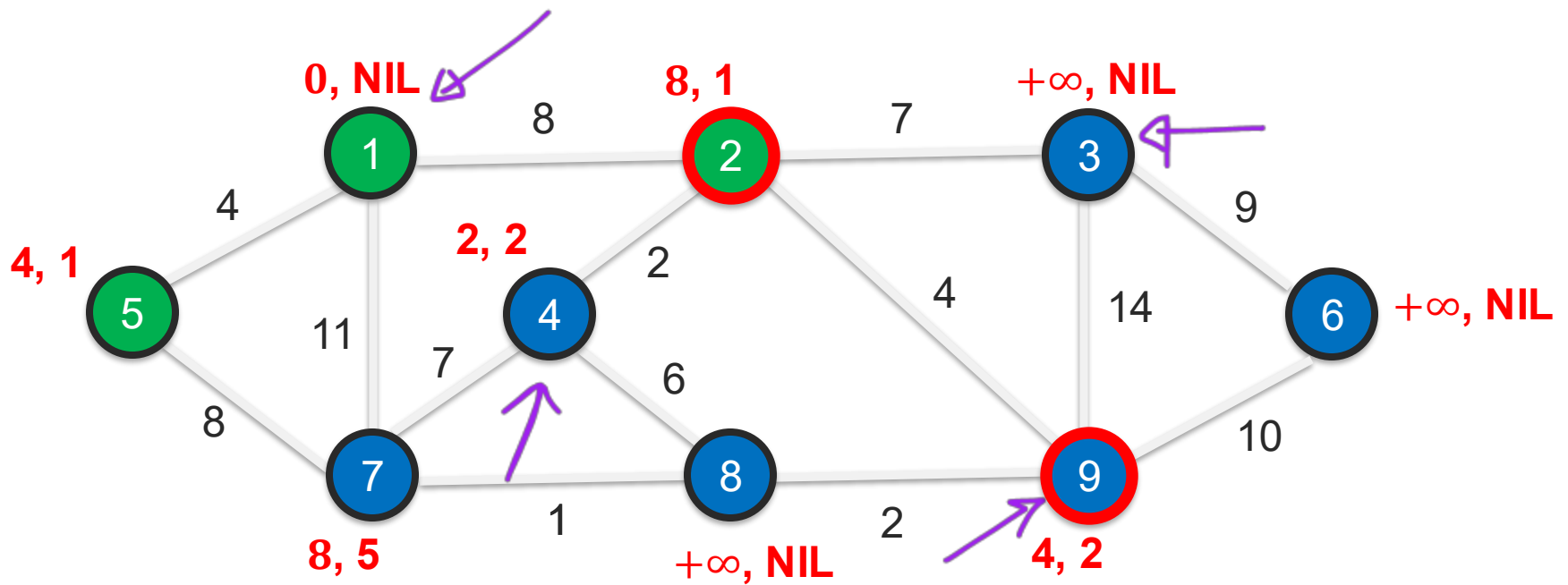
[Algoritmo di Prim]



Estrazione da Q del vertice 2

Adiacenti del vertice 2 → 1, 4, 9, 3

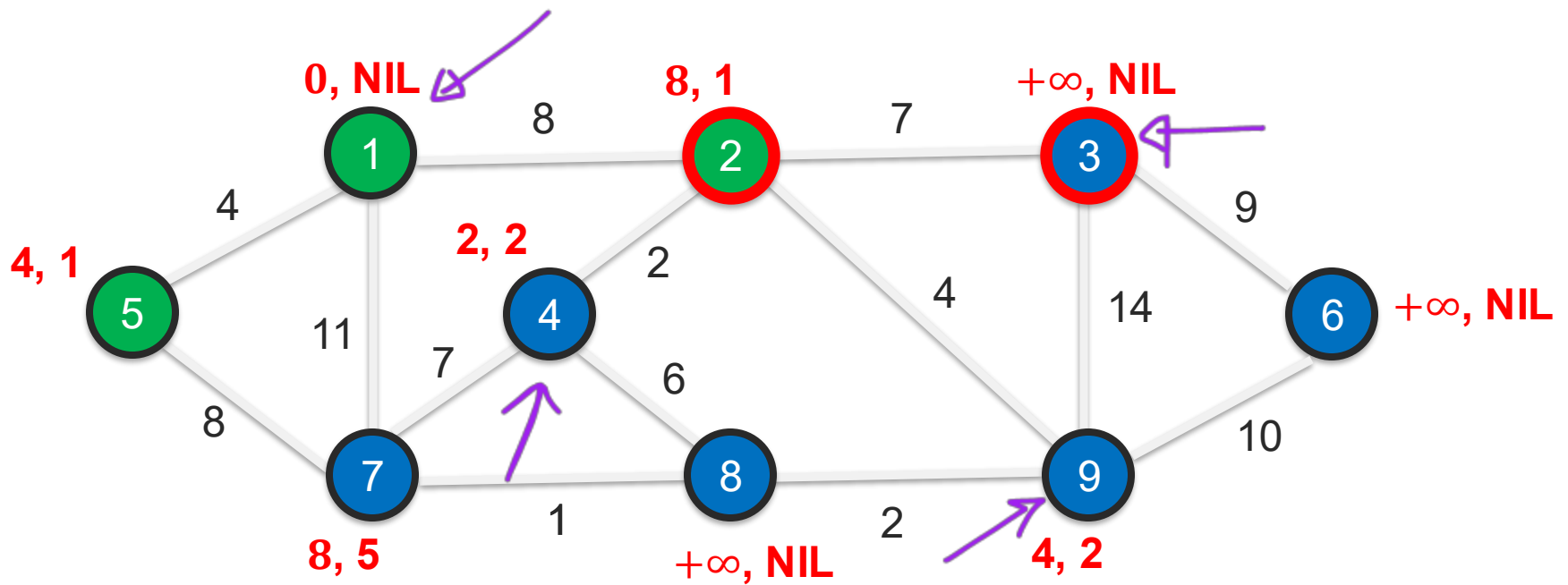
[Algoritmo di Prim]



Estrazione da Q del vertice 2

Adiacenti del vertice 2 $\rightarrow$ 1, 4, 9, 3

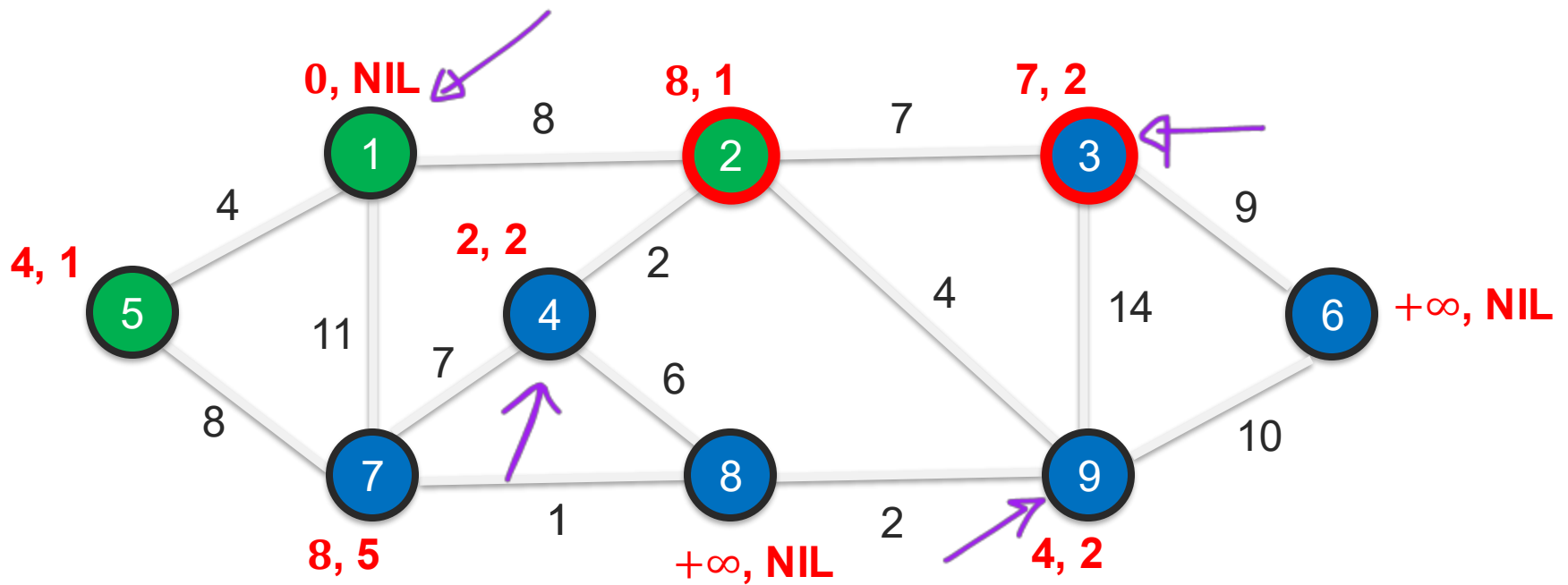
[Algoritmo di Prim]



Estrazione da Q del vertice 2

Adiacenti del vertice 2 $\rightarrow$ 1, 4, 9, 3

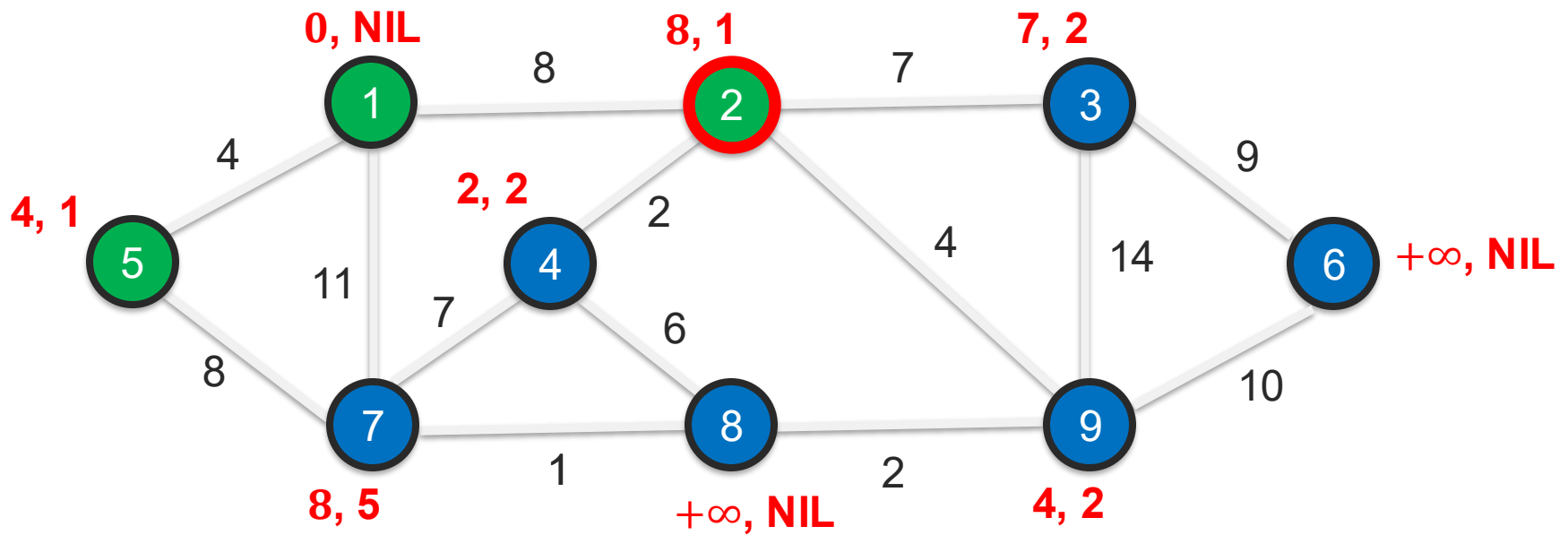
[Algoritmo di Prim]



Estrazione da Q del vertice 2

Adiacenti del vertice 2 $\rightarrow$ 1, 4, 9, 3

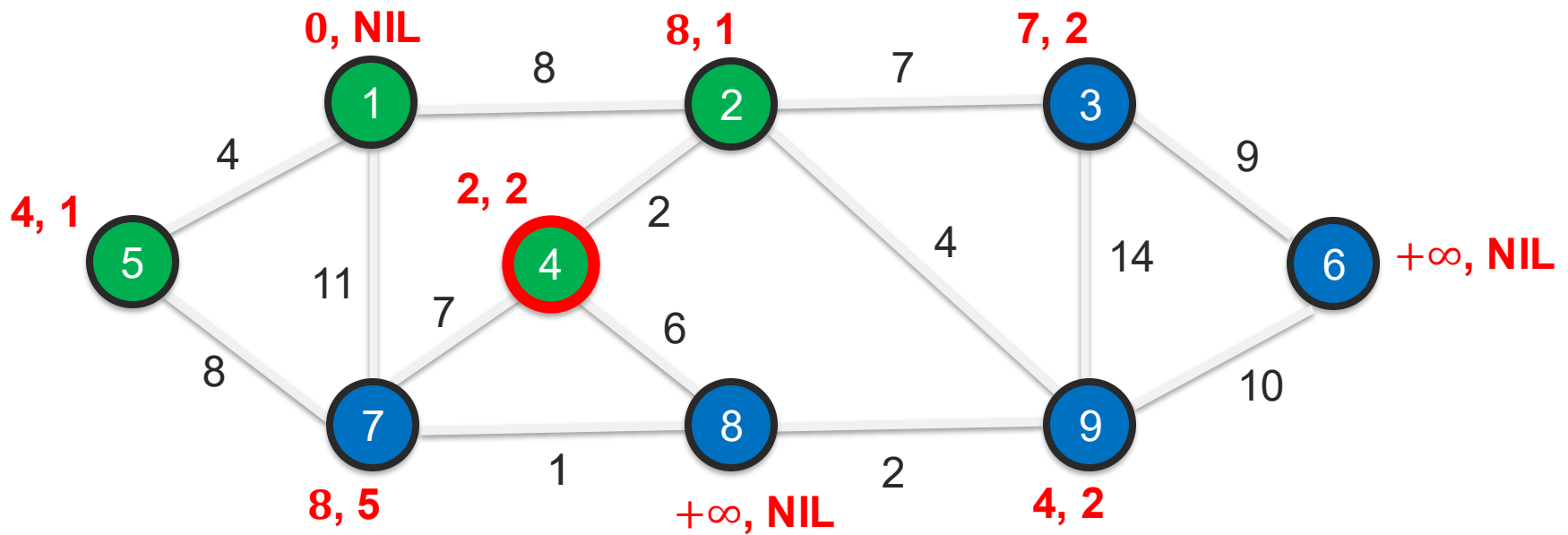
[Algoritmo di Prim]



Estrazione da Q del vertice 2

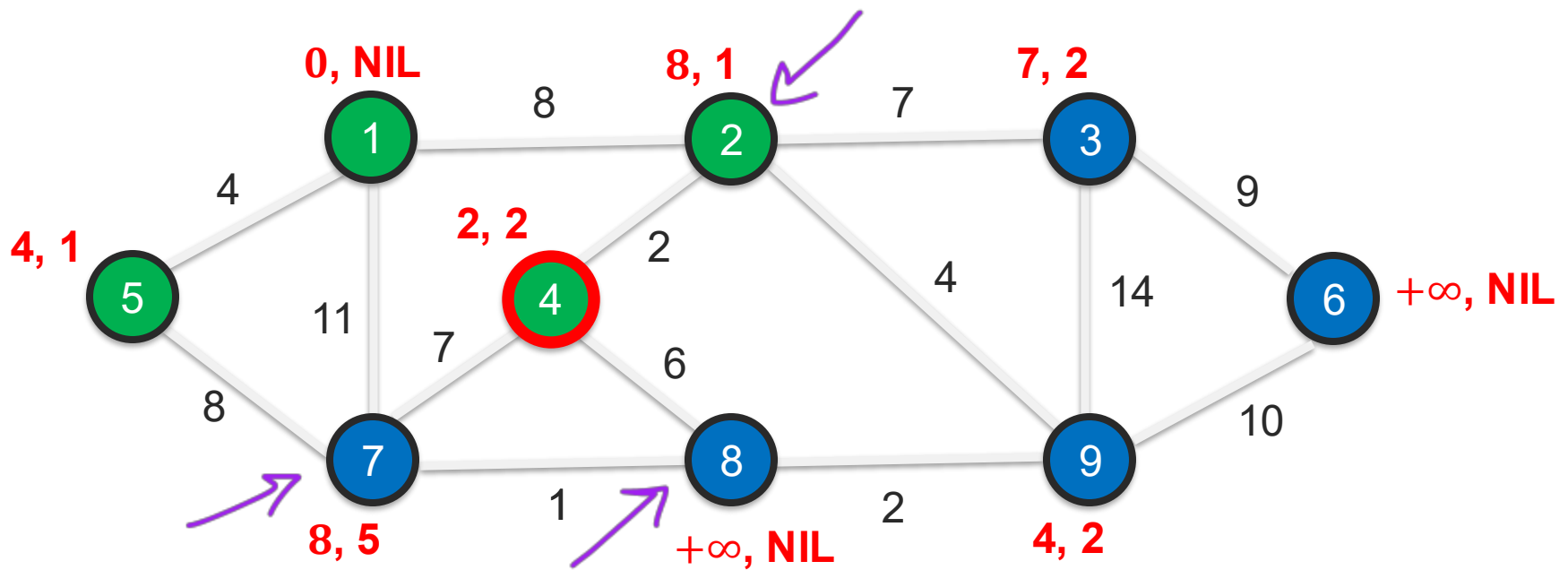
Adiacenti del vertice 2 $\rightarrow$ 1, 4, 9, 3

[Algoritmo di Prim]



Estrazione da Q del vertice 4

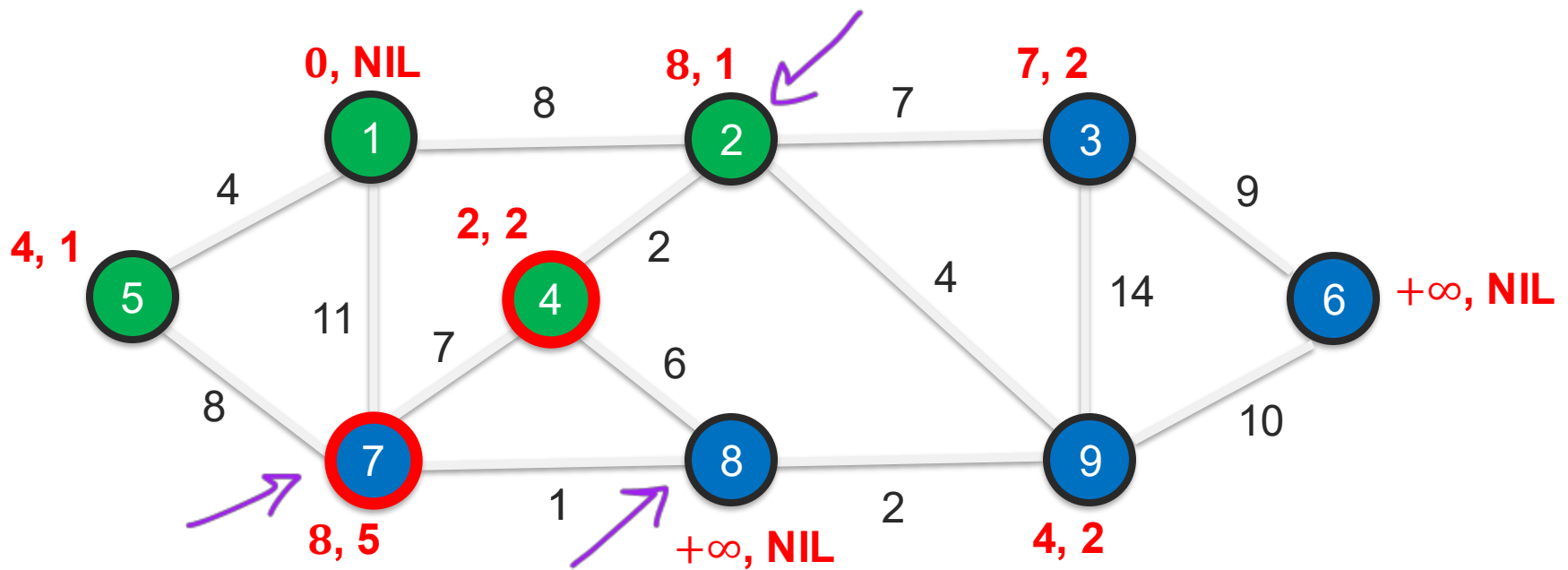
[Algoritmo di Prim]



Estrazione da Q del vertice 4

Adiacenti del vertice 4 $\rightarrow$ 2, 7, 8

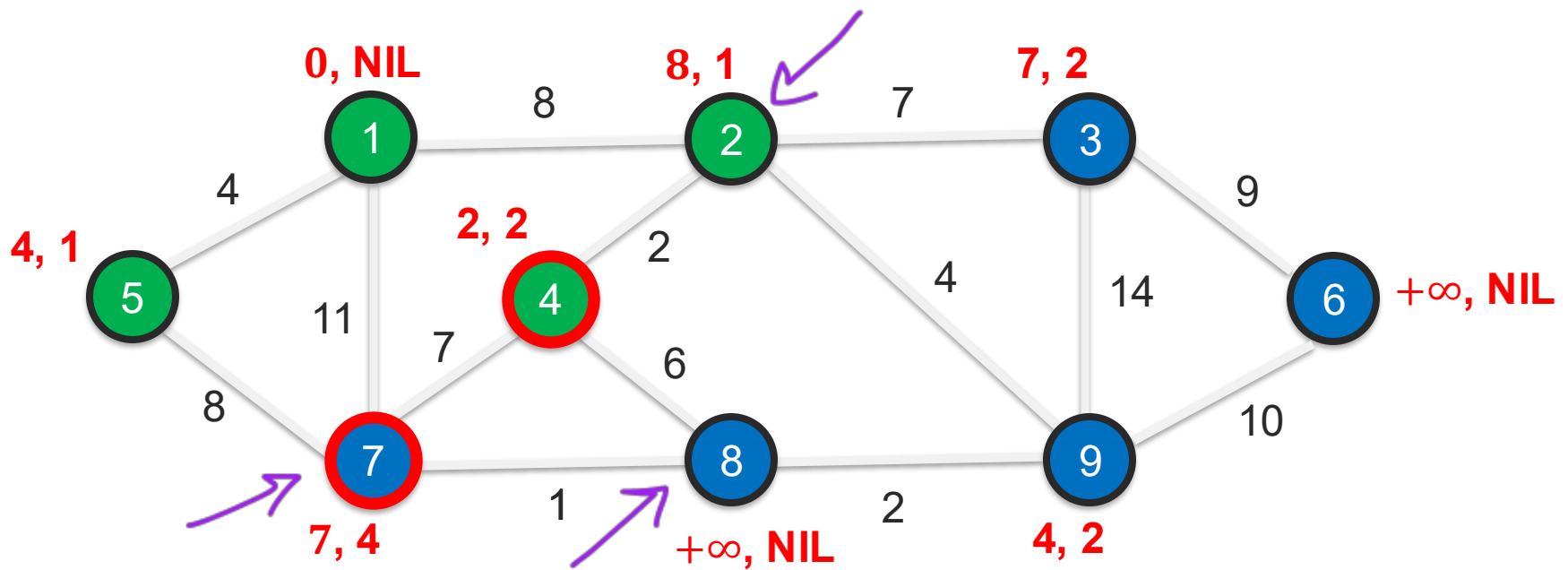
[Algoritmo di Prim]



Estrazione da Q del vertice 4

Adiacenti del vertice 4 $\rightarrow$ 2, 7, 8

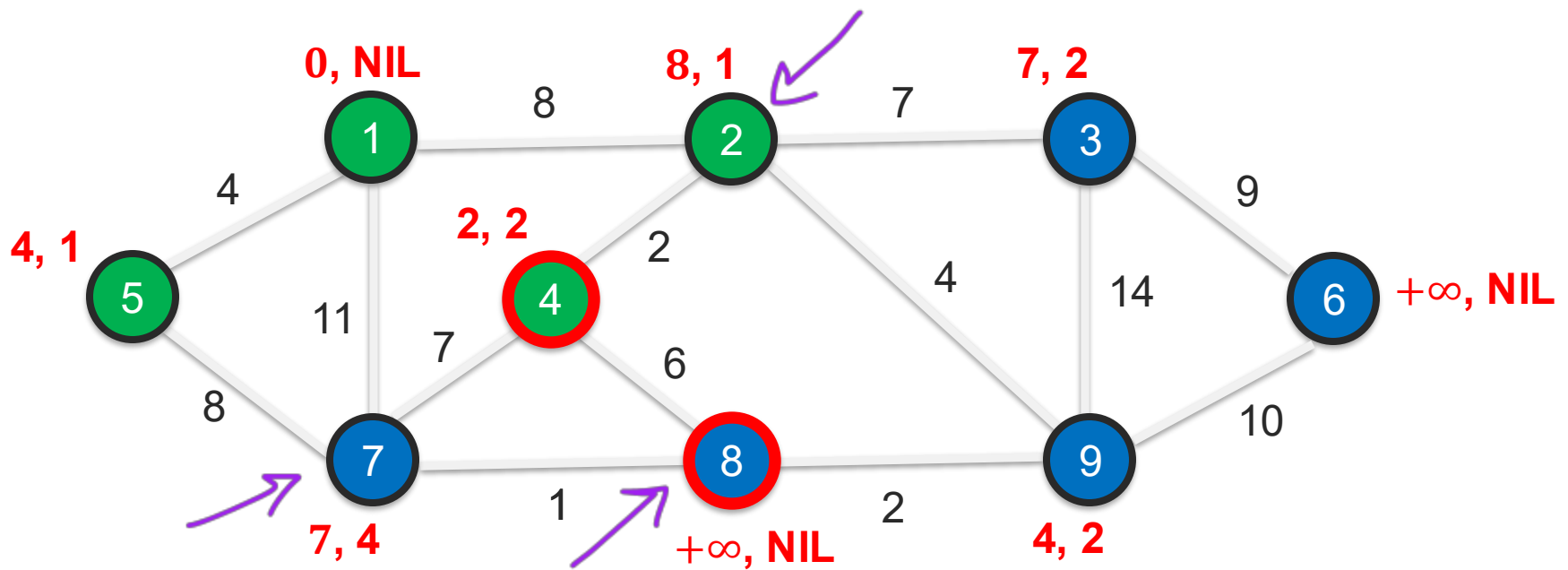
[Algoritmo di Prim]



Estrazione da Q del vertice 4

Adiacenti del vertice 4 $\rightarrow$ 2, 7, 8

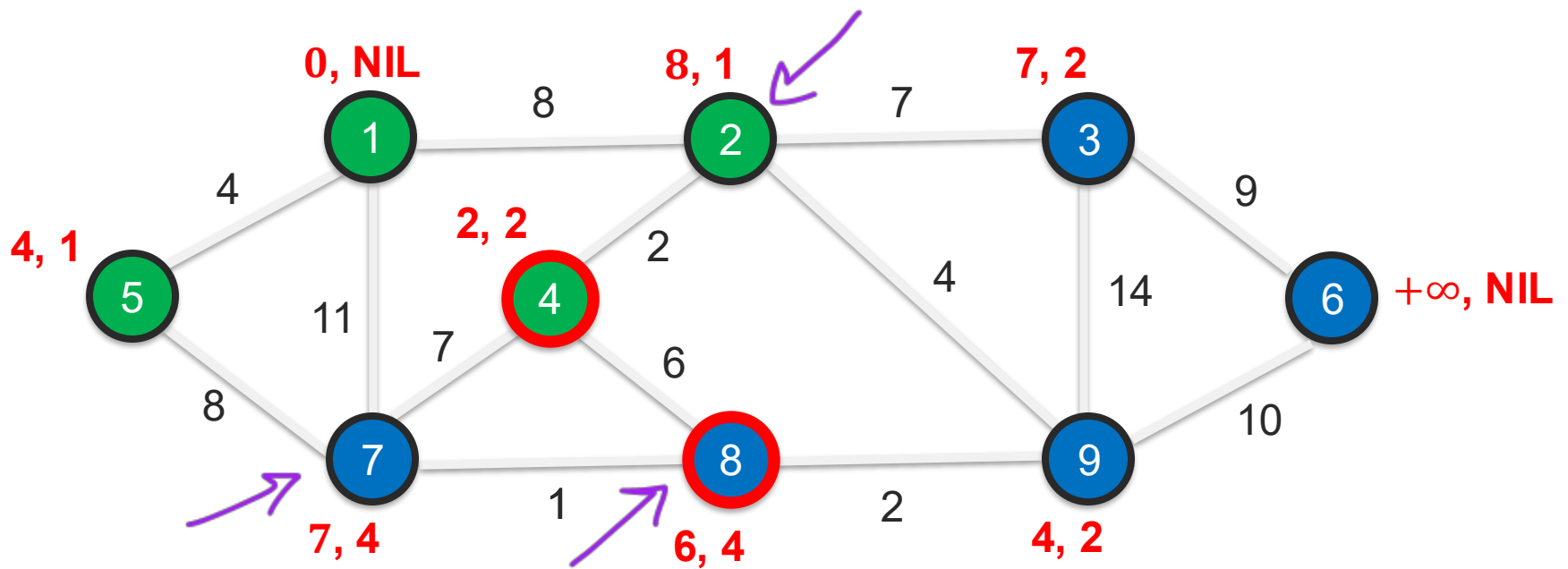
[Algoritmo di Prim]



Estrazione da Q del vertice 4

Adiacenti del vertice 4 $\rightarrow$ 2, 7, 8

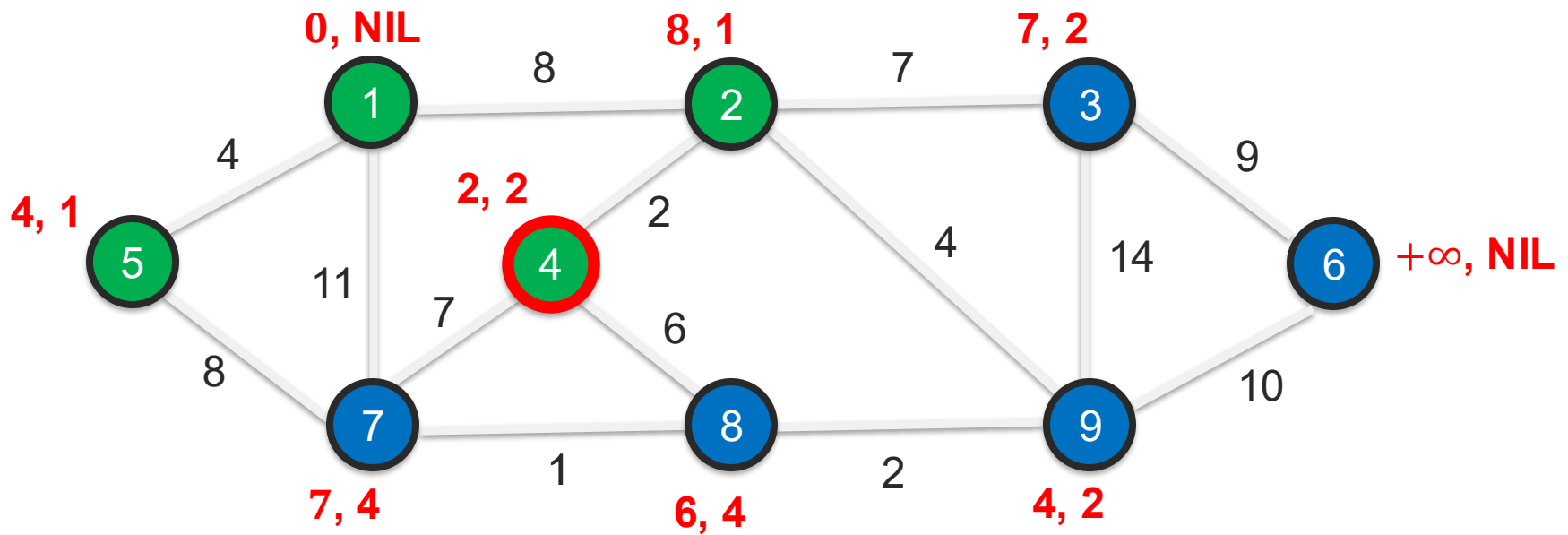
[Algoritmo di Prim]



Estrazione da Q del vertice 4

Adiacenti del vertice 4 $\rightarrow$ 2, 7, 8

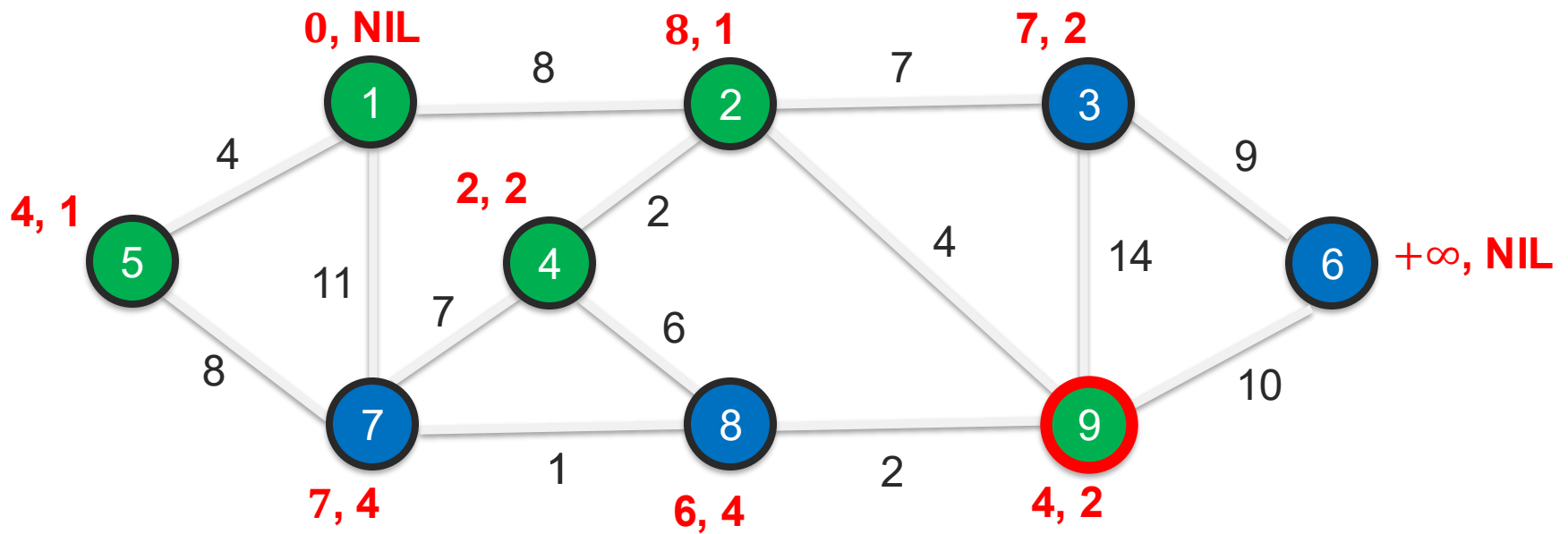
[Algoritmo di Prim]



Estrazione da Q del vertice 4

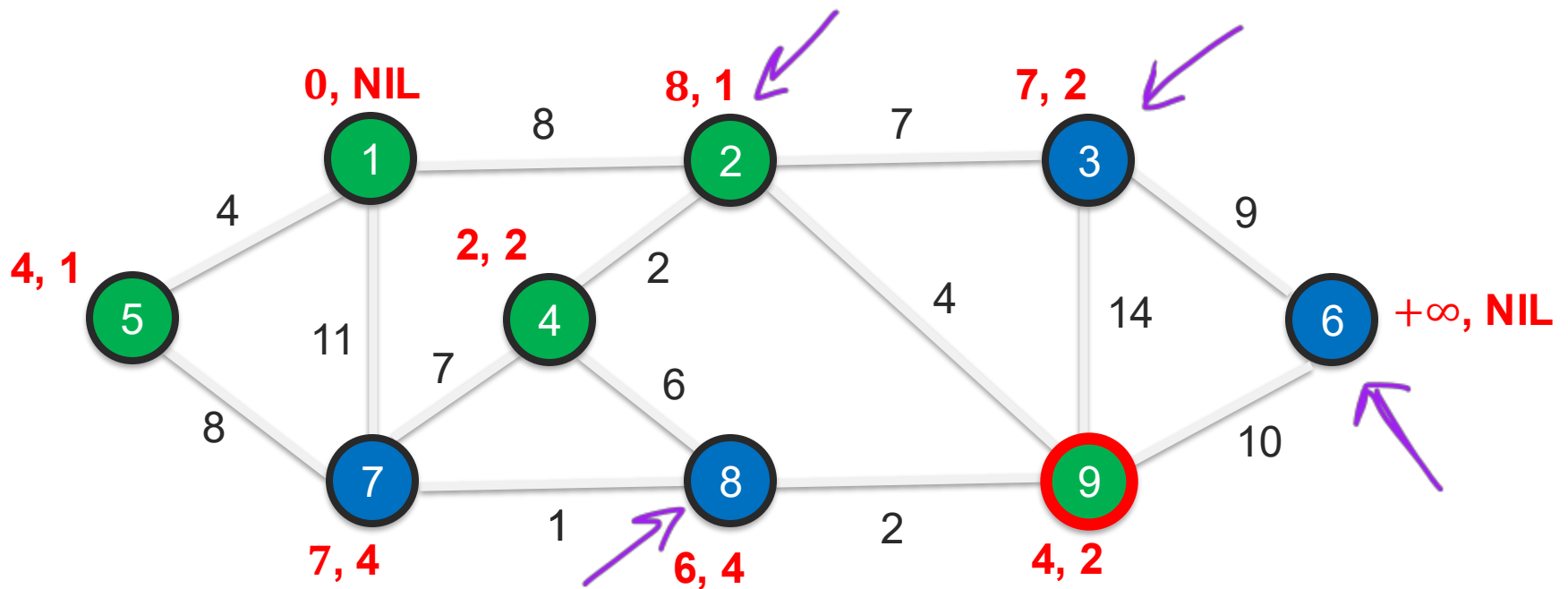
Adiacenti del vertice 4 $\rightarrow$ 2, 7, 8

[Algoritmo di Prim]



Estrazione da Q del vertice 9

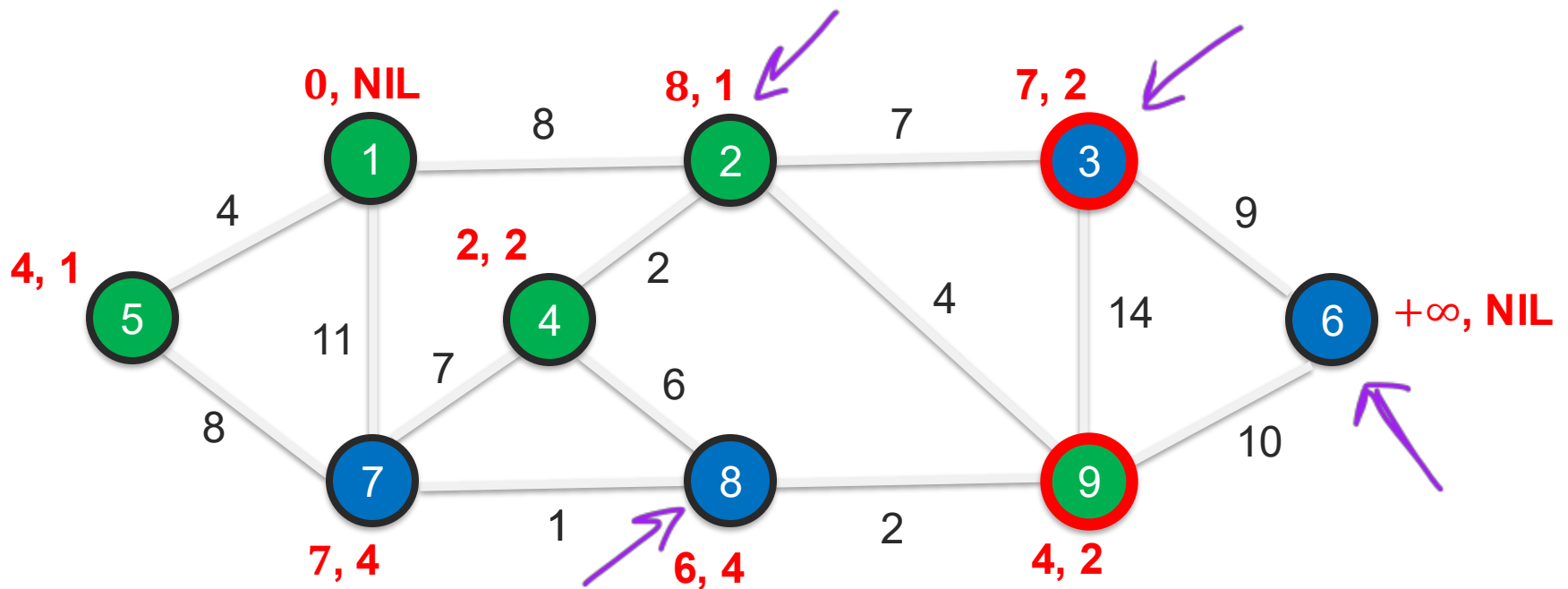
[Algoritmo di Prim]



Estrazione da Q del vertice 9

Adiacenti del vertice 9 $\rightarrow$ 2, 3, 6, 8

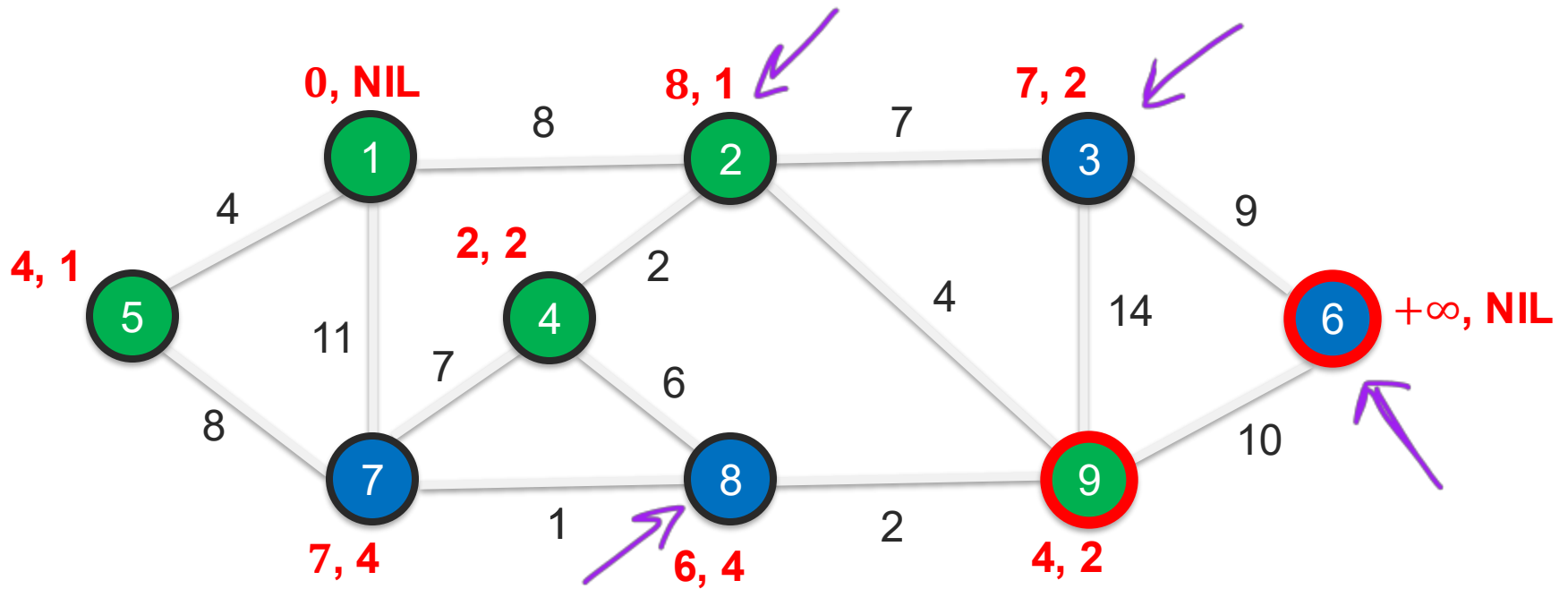
[Algoritmo di Prim]



Estrazione da Q del vertice 9

Adiacenti del vertice 9 $\rightarrow$ 2, 3, 6, 8

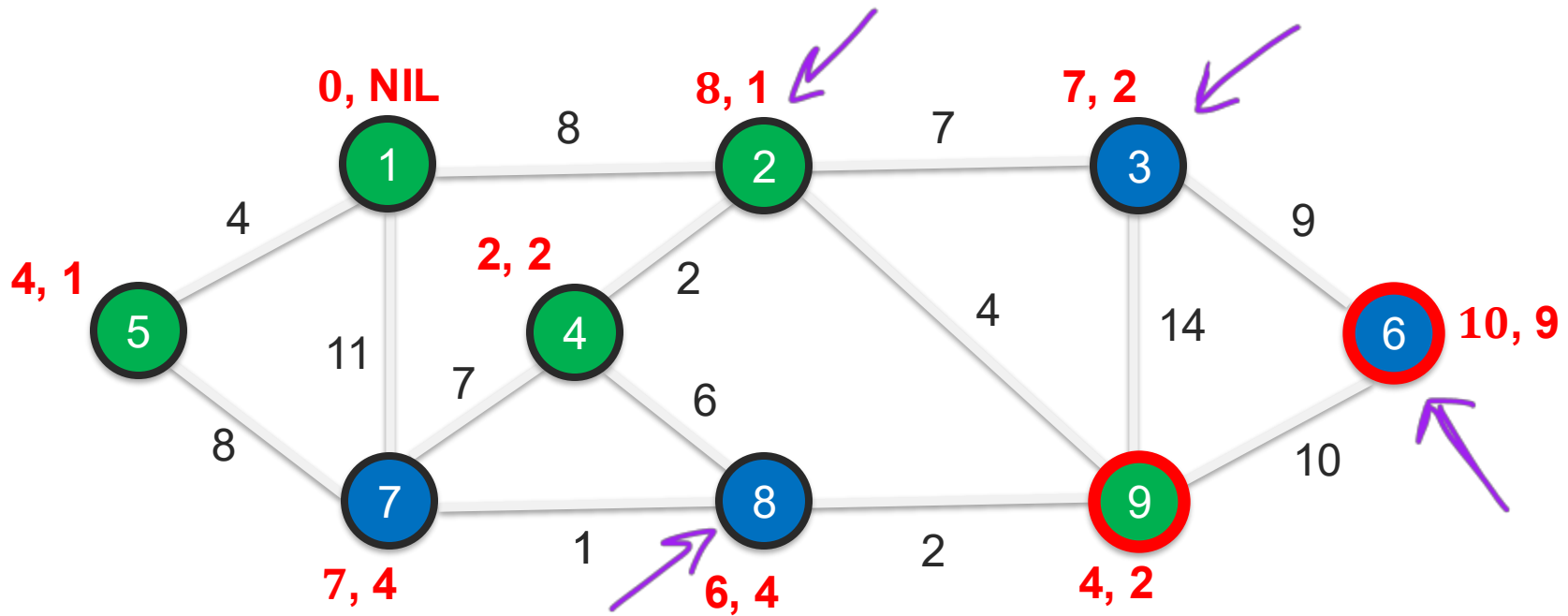
[Algoritmo di Prim]



Estrazione da Q del vertice 9

Adiacenti del vertice 9 $\rightarrow$ 2, 3, 6, 8

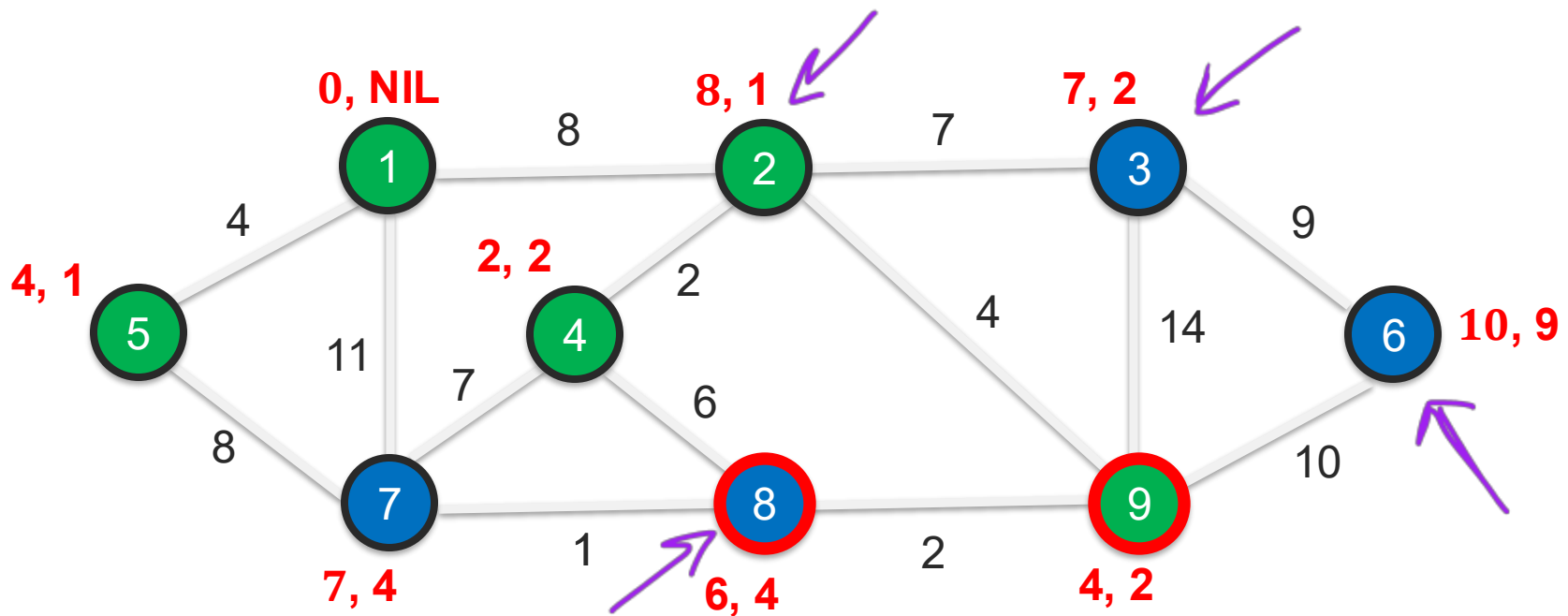
[Algoritmo di Prim]



Estrazione da Q del vertice 9

Adiacenti del vertice 9 $\rightarrow$ 2, 3, 6, 8

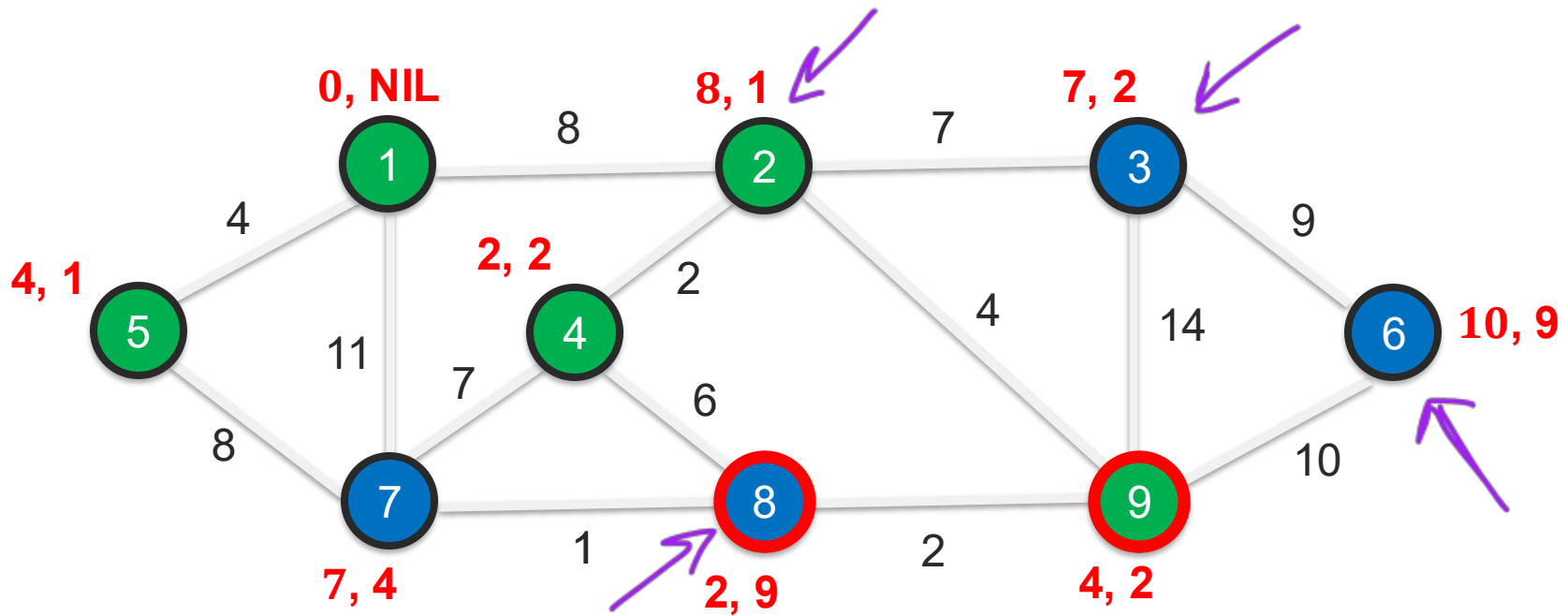
[Algoritmo di Prim]



Estrazione da Q del vertice 9

Adiacenti del vertice 9 $\rightarrow$ 2, 3, 6, 8

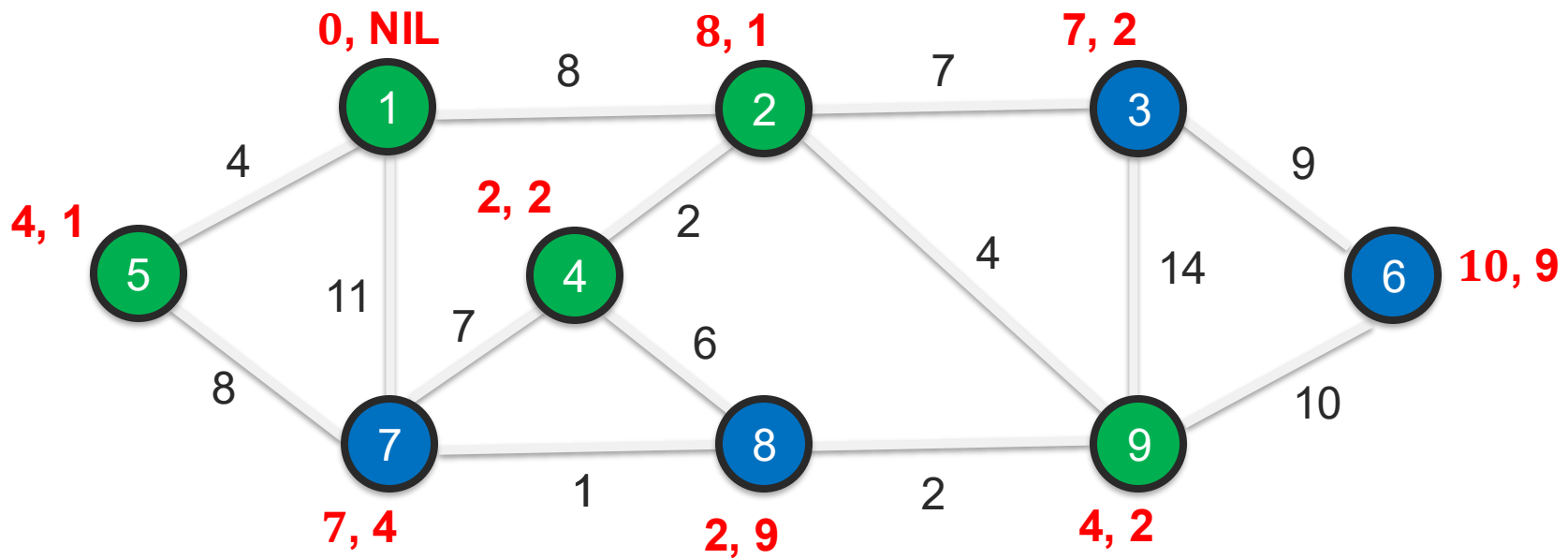
[Algoritmo di Prim]



Estrazione da Q del vertice 9

Adiacenti del vertice 9 $\rightarrow$ 2, 3, 6, 8

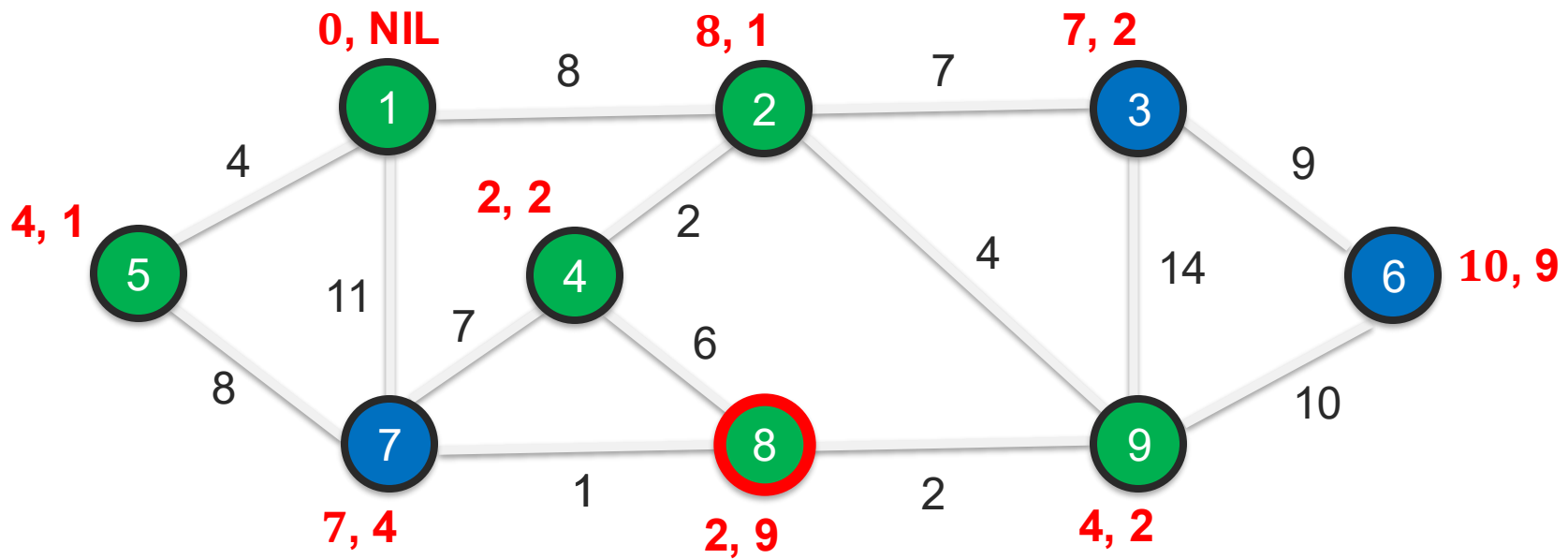
[Algoritmo di Prim]



Estrazione da Q del vertice 9

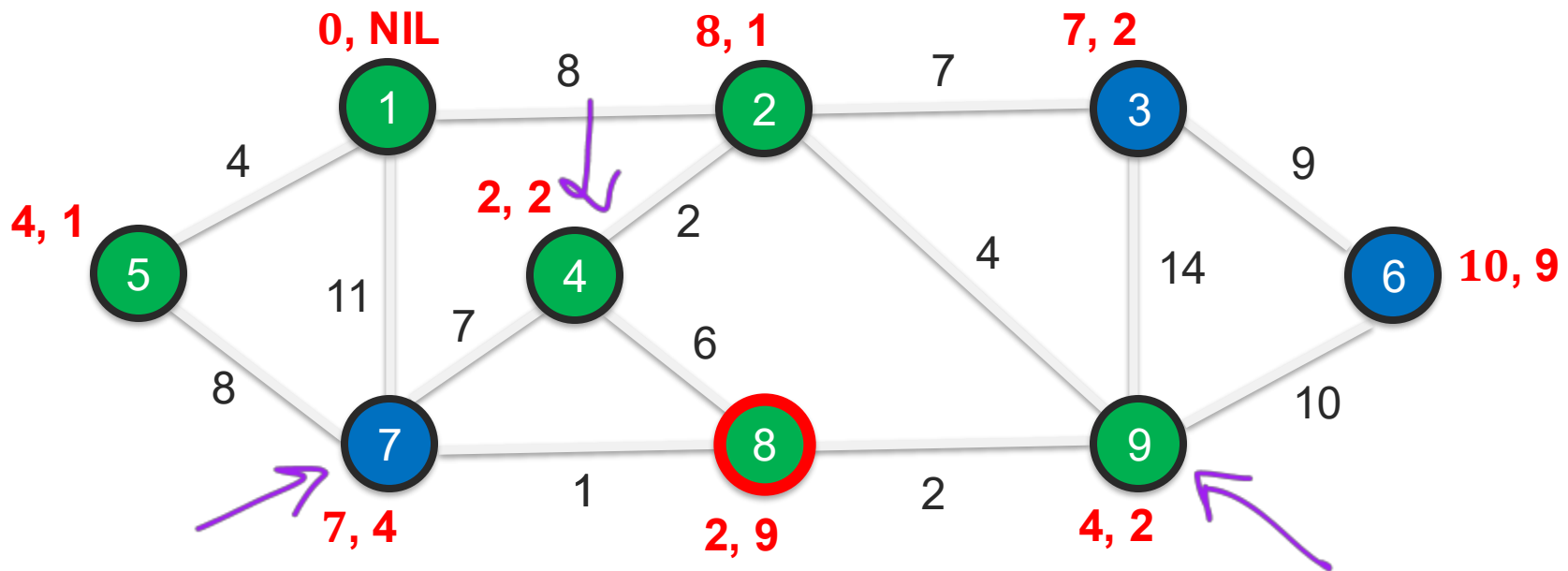
Adiacenti del vertice 9 $\rightarrow$ 2, 3, 6, 8

[Algoritmo di Prim]



Estrazione da Q del vertice 8

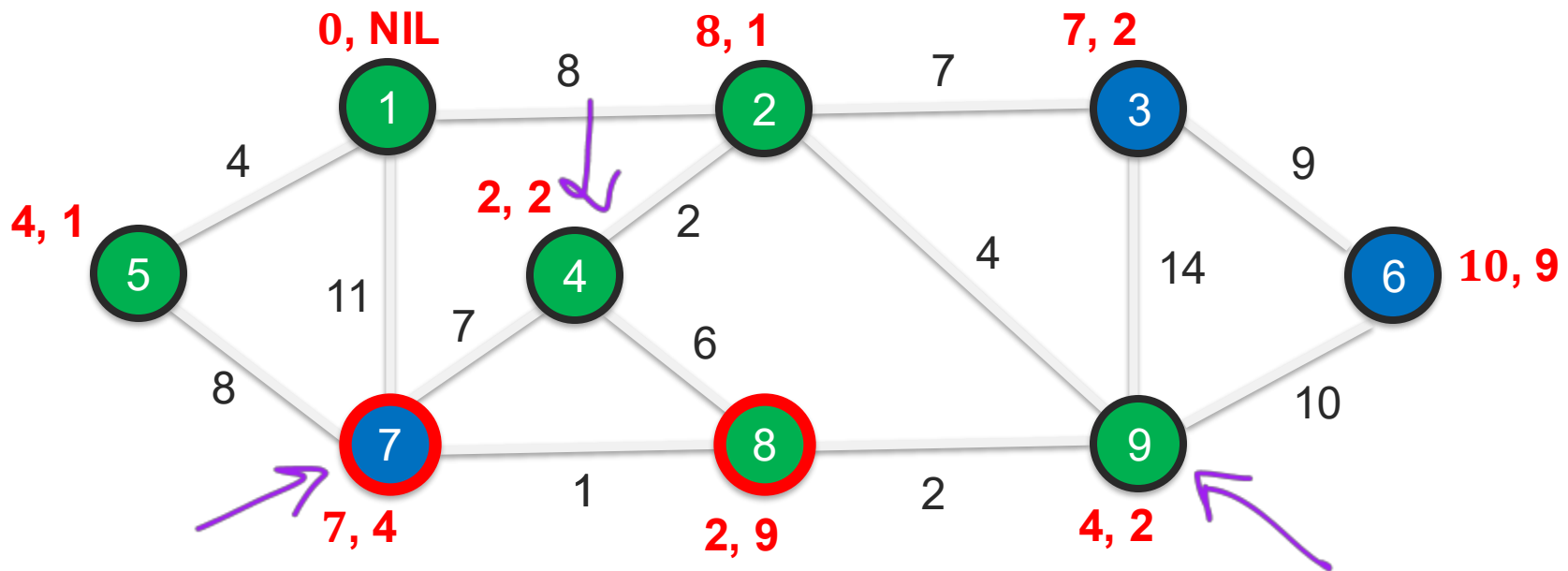
[Algoritmo di Prim]



Estrazione da Q del vertice 8

Adiacenti del vertice 8 $\rightarrow$ 4, 7, 9

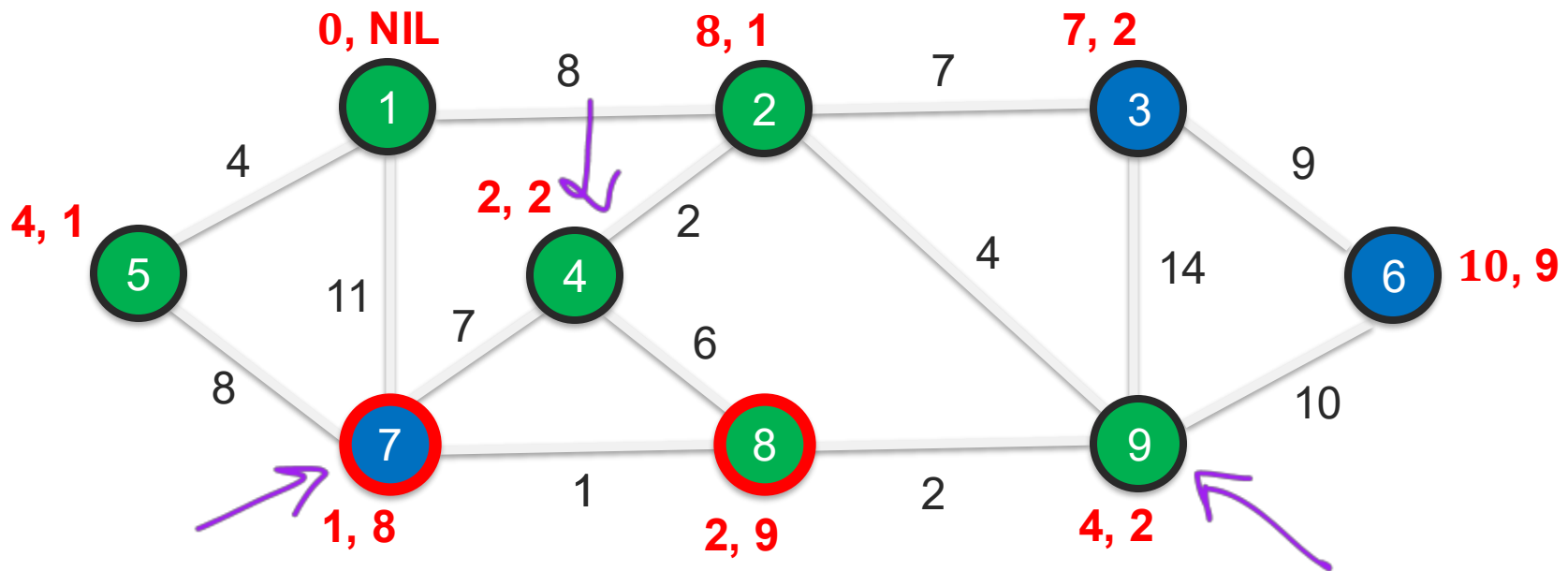
[Algoritmo di Prim]



Estrazione da Q del vertice 8

Adiacenti del vertice 8 $\rightarrow$ 4, 7, 9

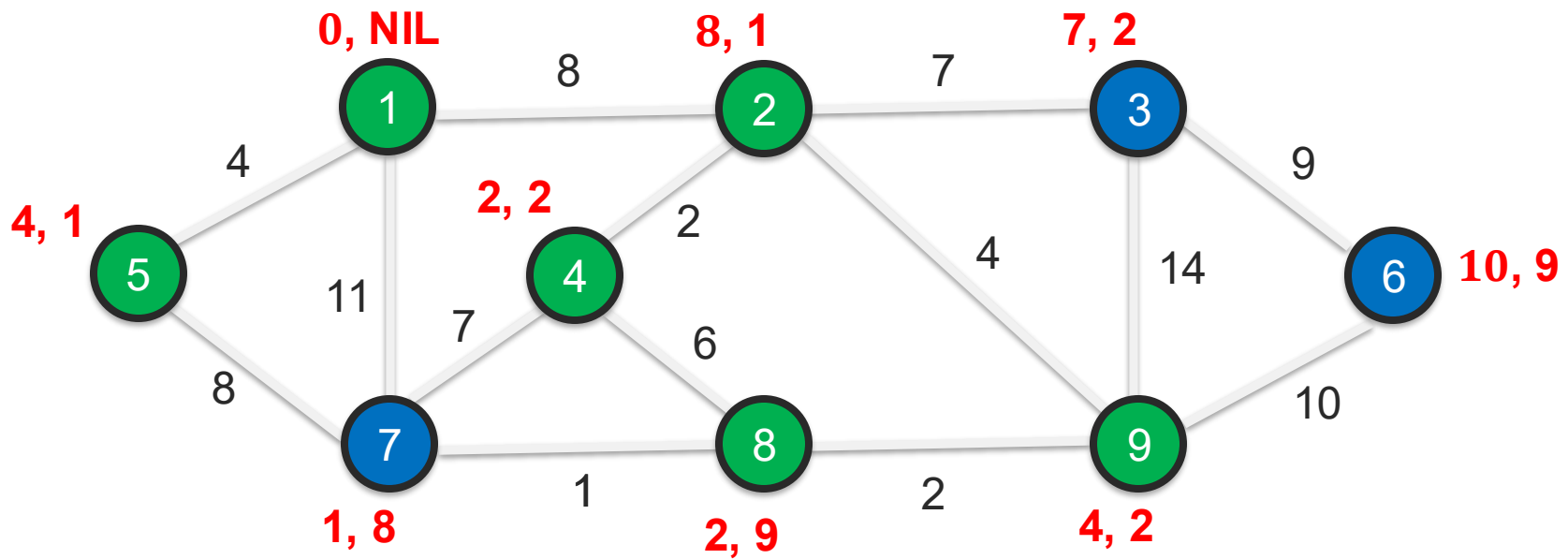
[Algoritmo di Prim]



Estrazione da Q del vertice 8

Adiacenti del vertice 8 $\rightarrow$ 4, 7, 9

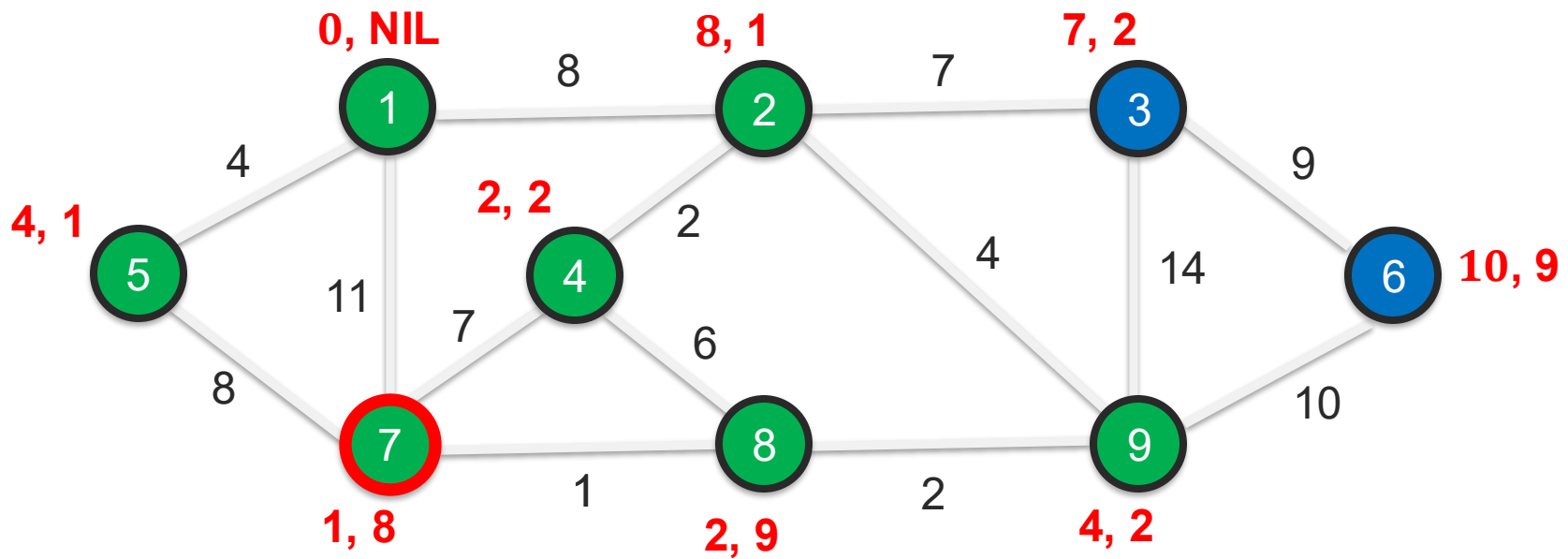
[Algoritmo di Prim]



Estrazione da Q del vertice 8

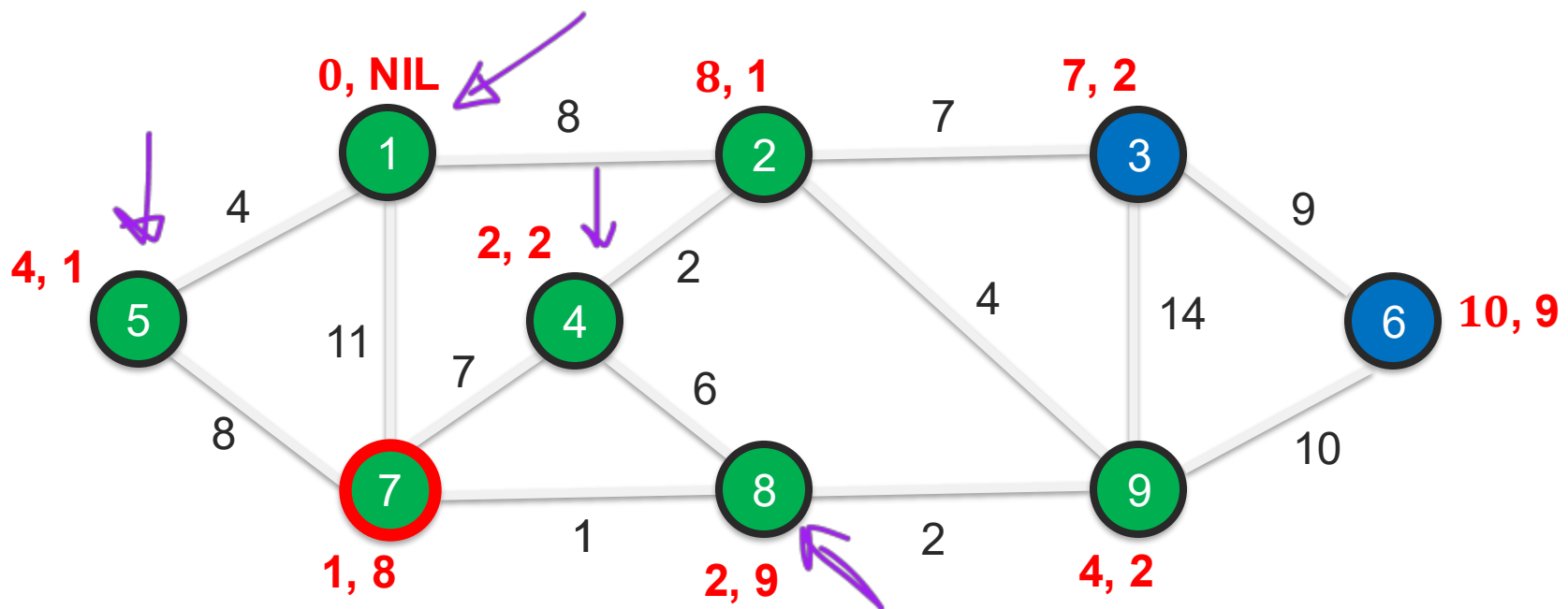
Adiacenti del vertice 8 $\rightarrow$ 4, 7, 9

[Algoritmo di Prim]



Estrazione da Q del vertice 7

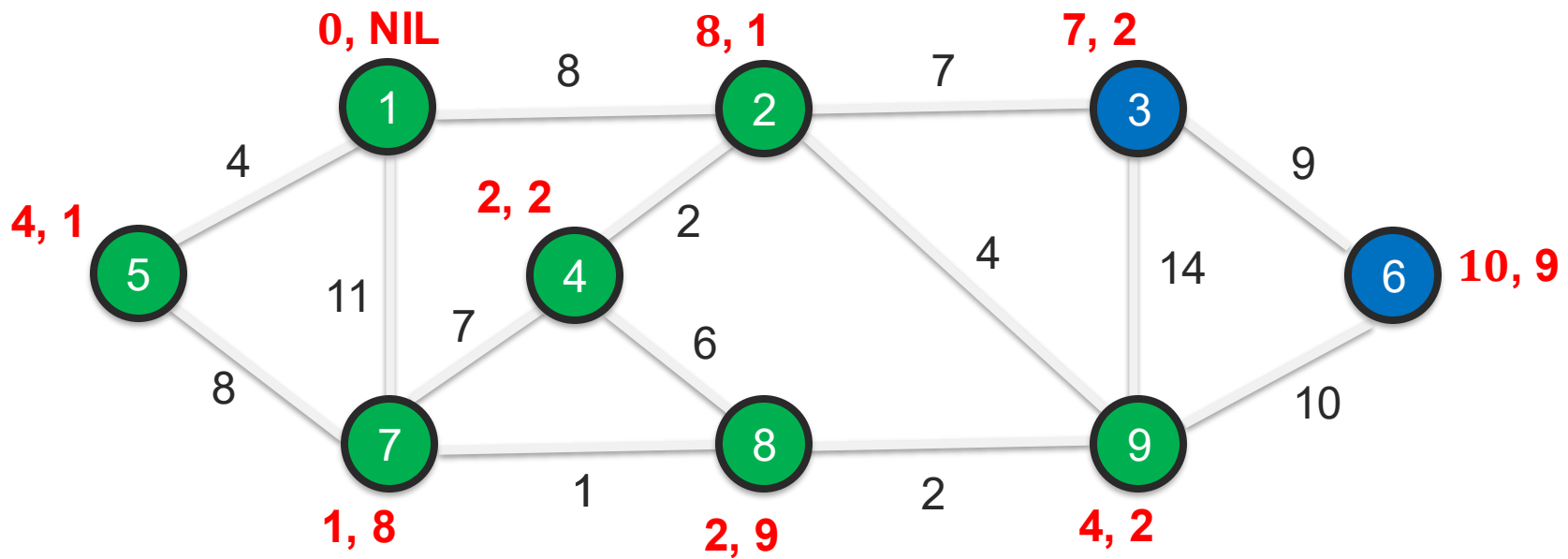
[Algoritmo di Prim]



Estrazione da Q del vertice 7

Adiacenti del vertice 7 $\rightarrow$ 1, 4, 5, 8

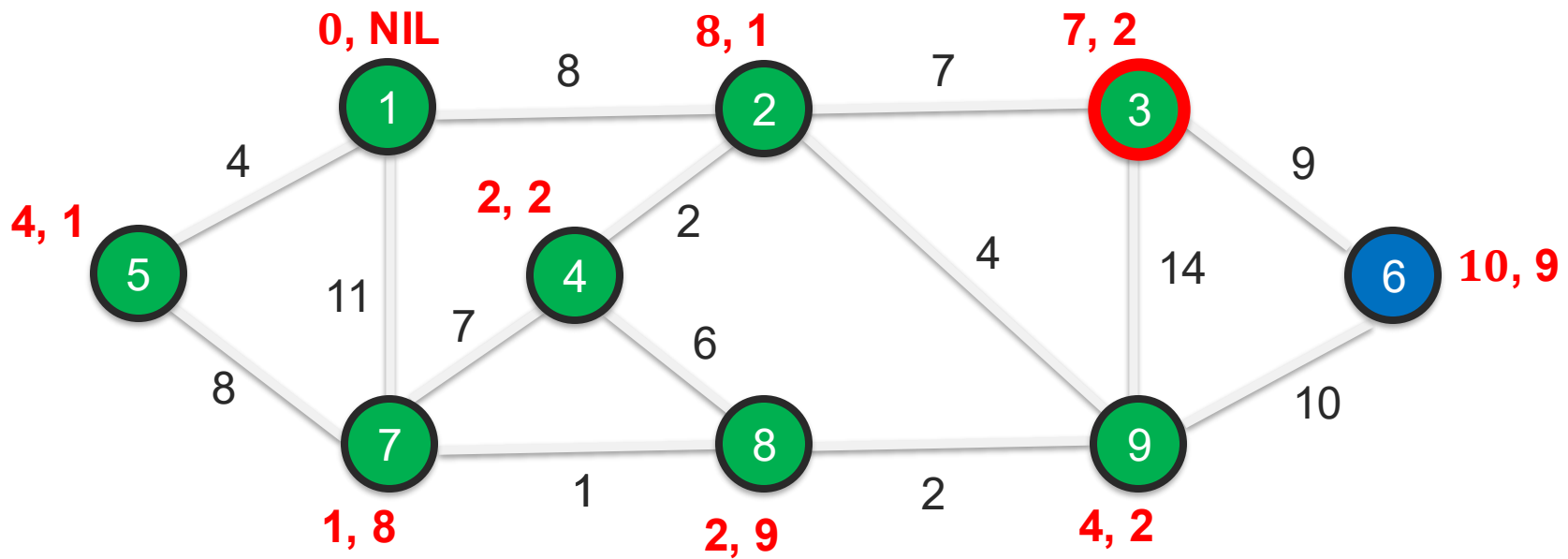
[Algoritmo di Prim]



Estrazione da Q del vertice 7

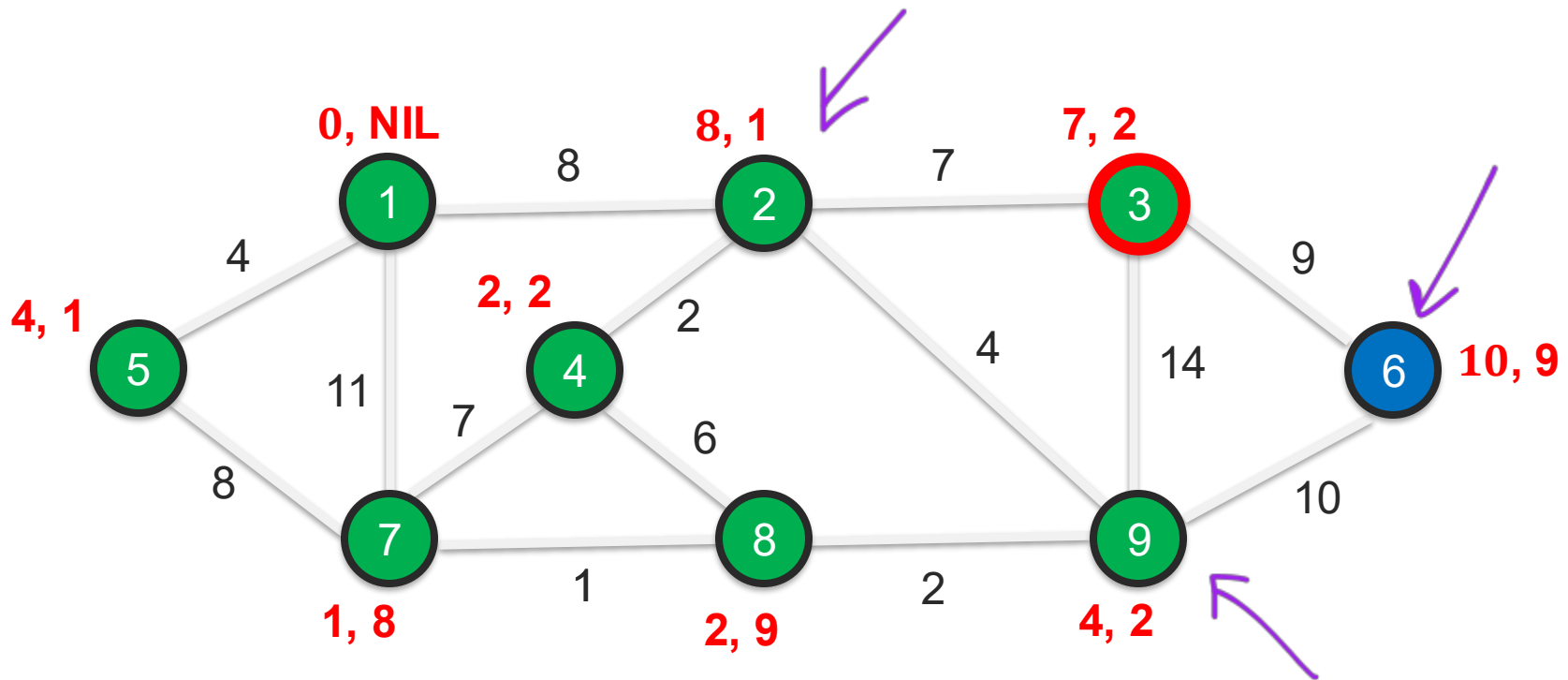
Adiacenti del vertice 7 $\rightarrow$ 1, 4, 5, 8

[Algoritmo di Prim]



Estrazione da Q del vertice 3

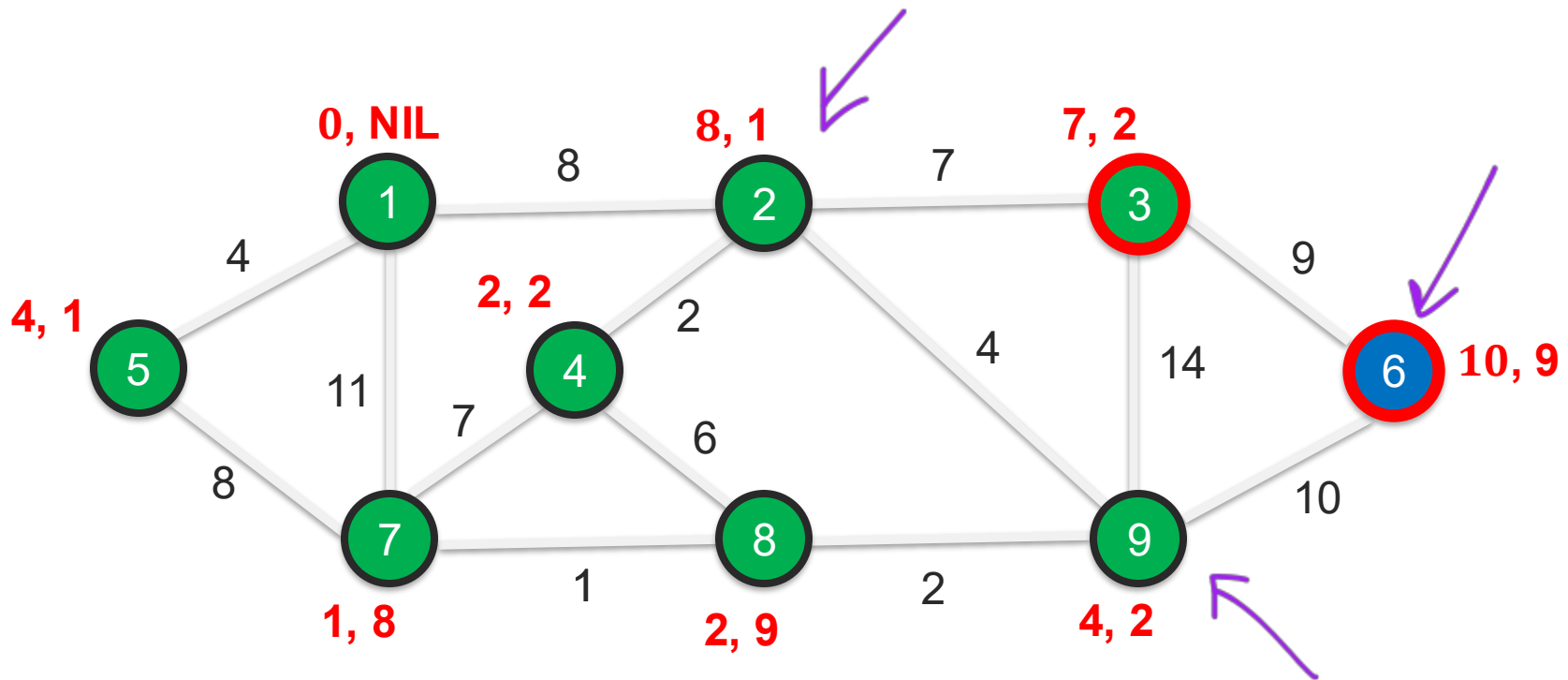
[Algoritmo di Prim]



Estrazione da Q del vertice 3

Adiacenti del vertice 3 $\rightarrow$ 2, 6, 9

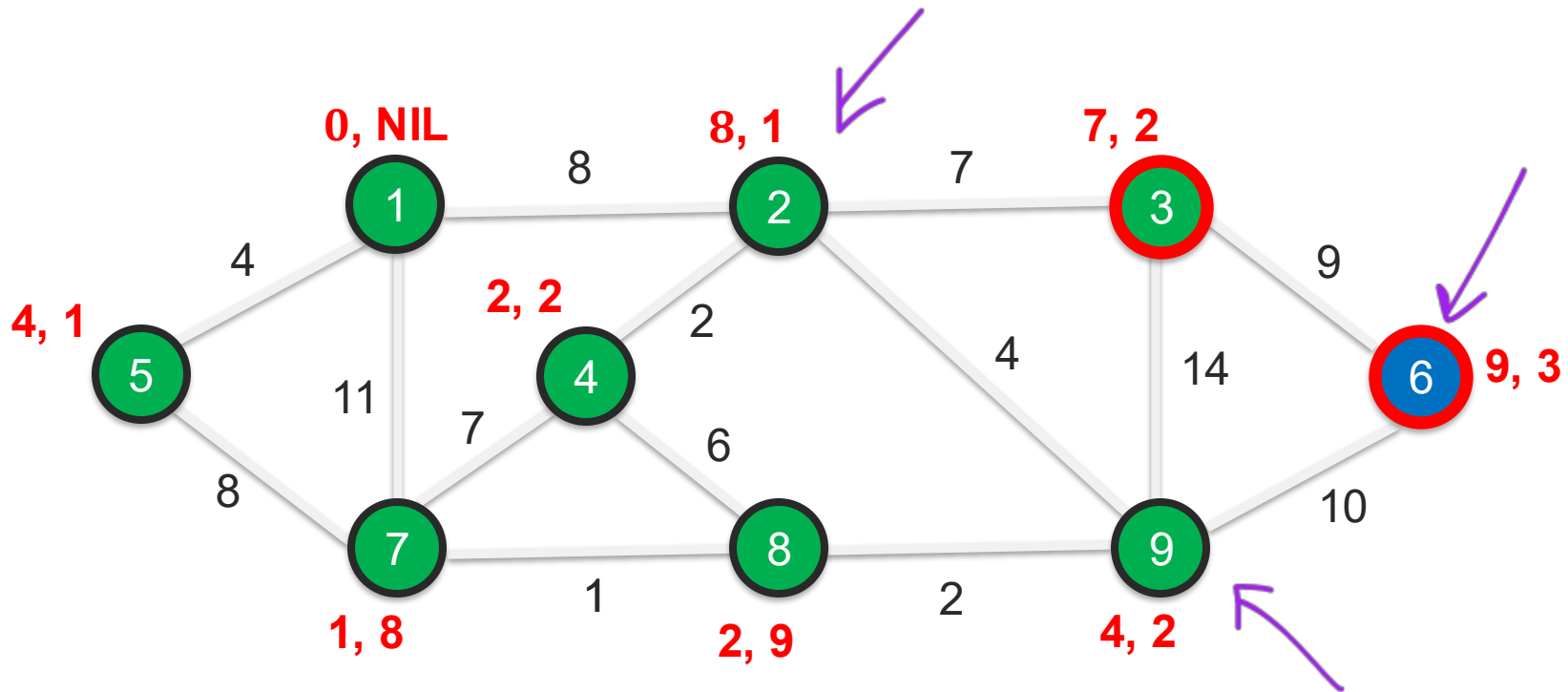
[Algoritmo di Prim]



Estrazione da Q del vertice 3

Adiacenti del vertice 3 $\rightarrow$ 2, 6, 9

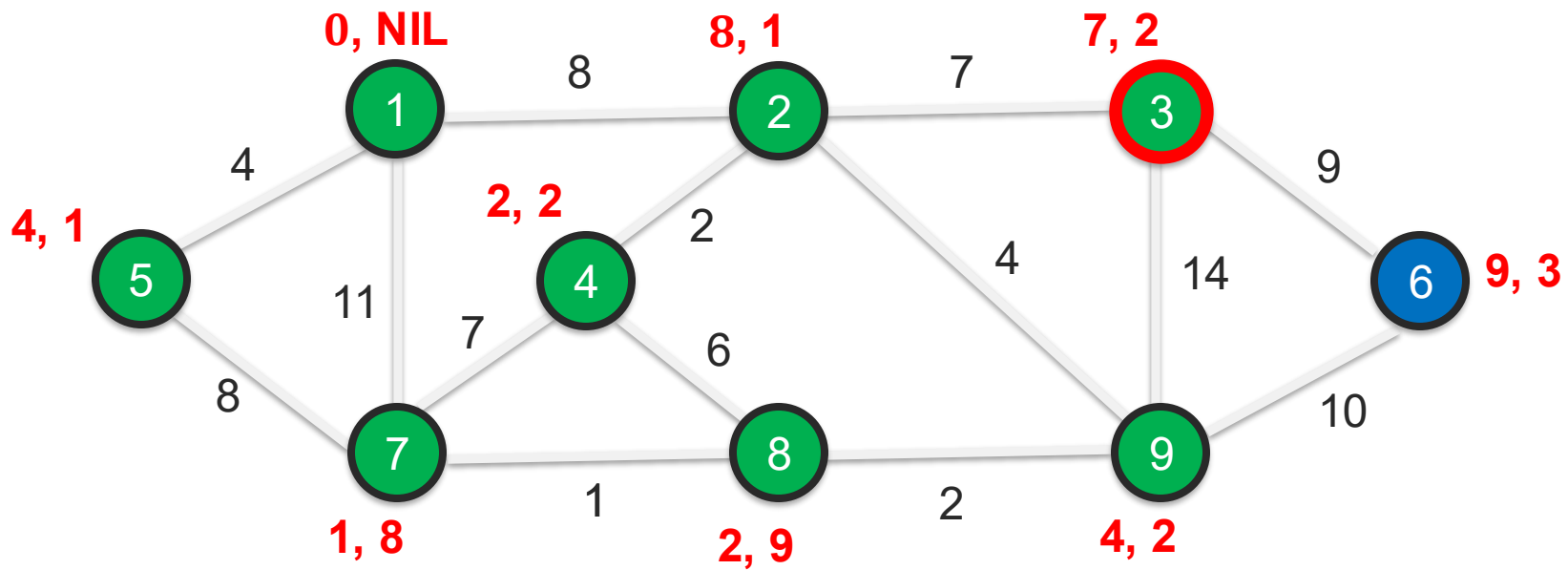
[Algoritmo di Prim]



Estrazione da Q del vertice 3

Adiacenti del vertice 3 $\rightarrow$ 2, 6, 9

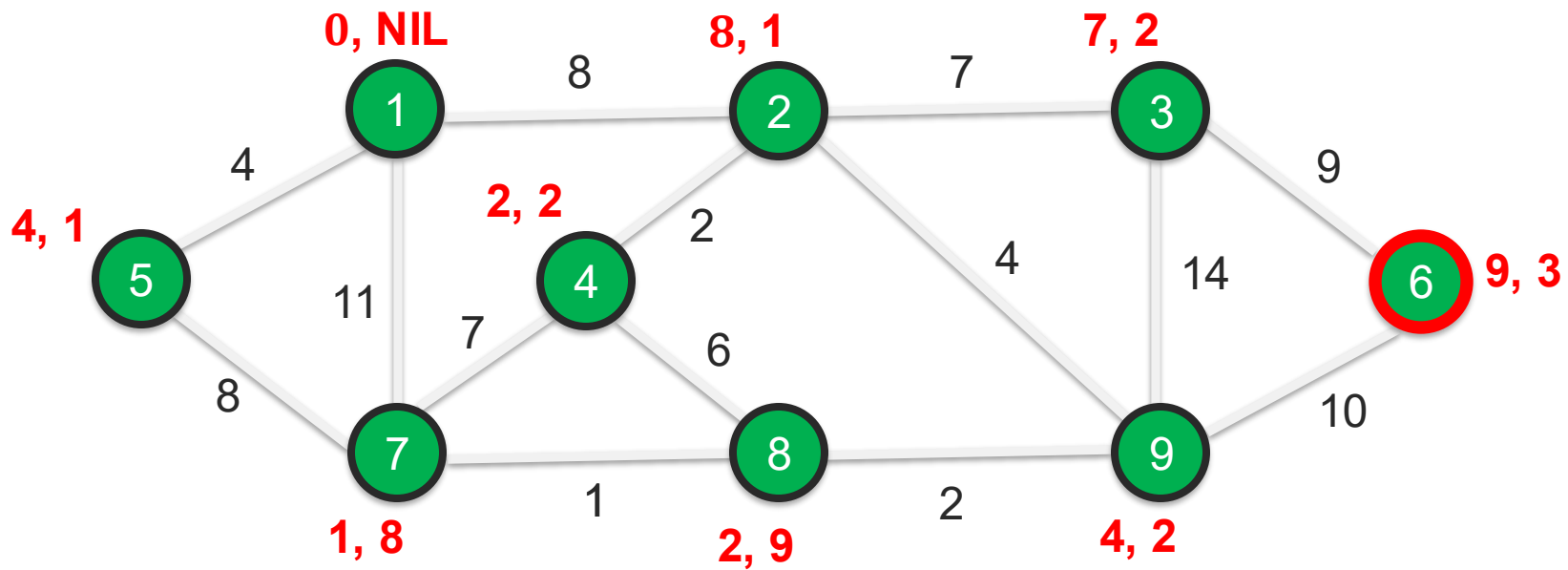
[Algoritmo di Prim]



Estrazione da Q del vertice 3

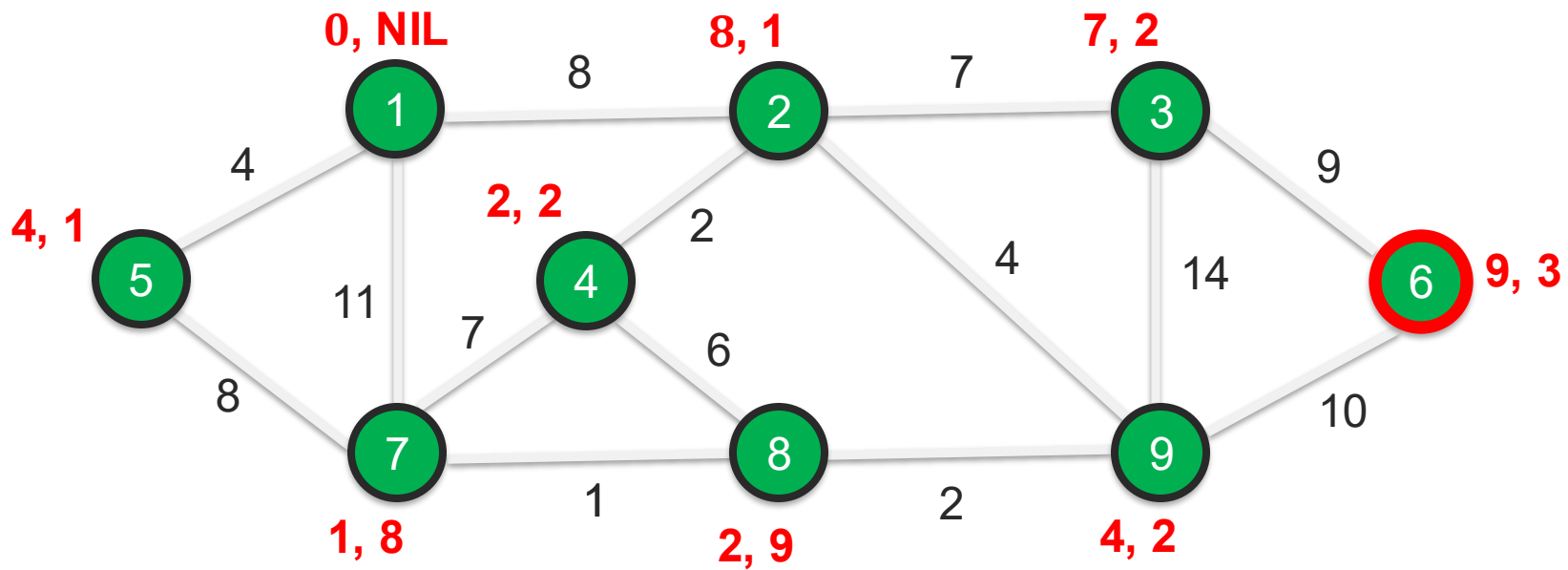
Adiacenti del vertice 3 $\rightarrow$ 2, 6, 9

[Algoritmo di Prim]



Estrazione da Q del vertice 6

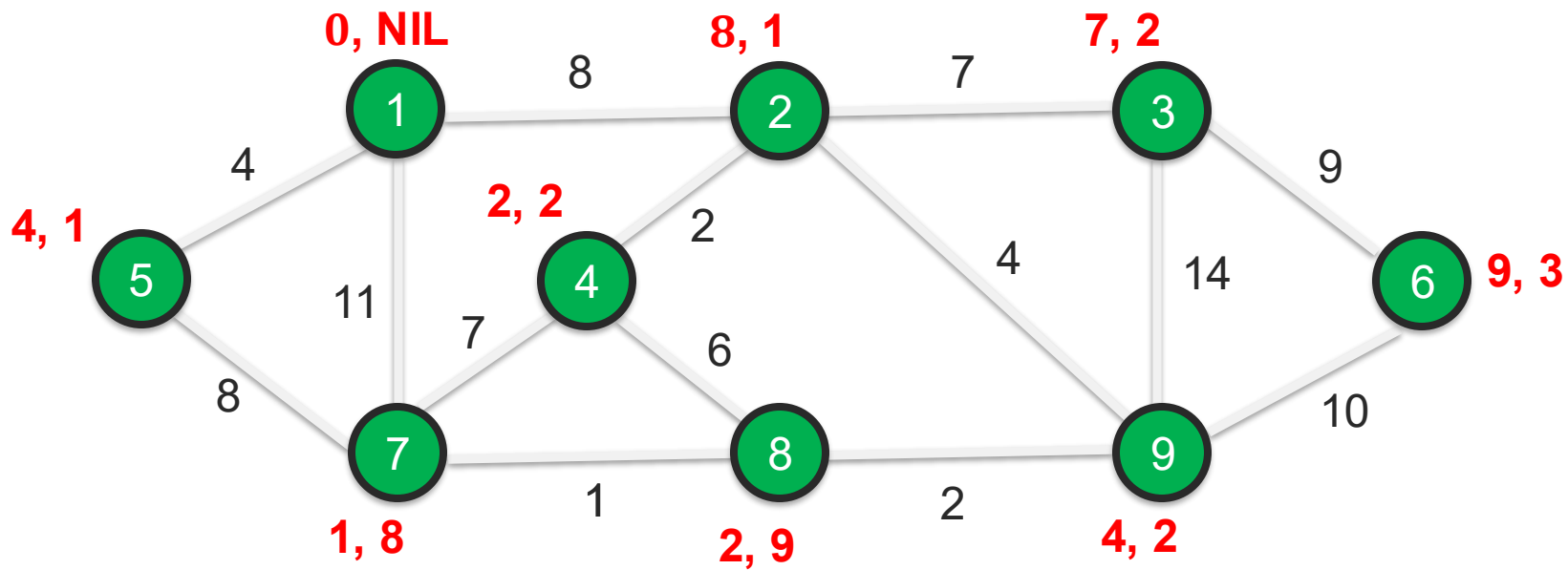
[Algoritmo di Prim]



Estrazione da Q del vertice 6

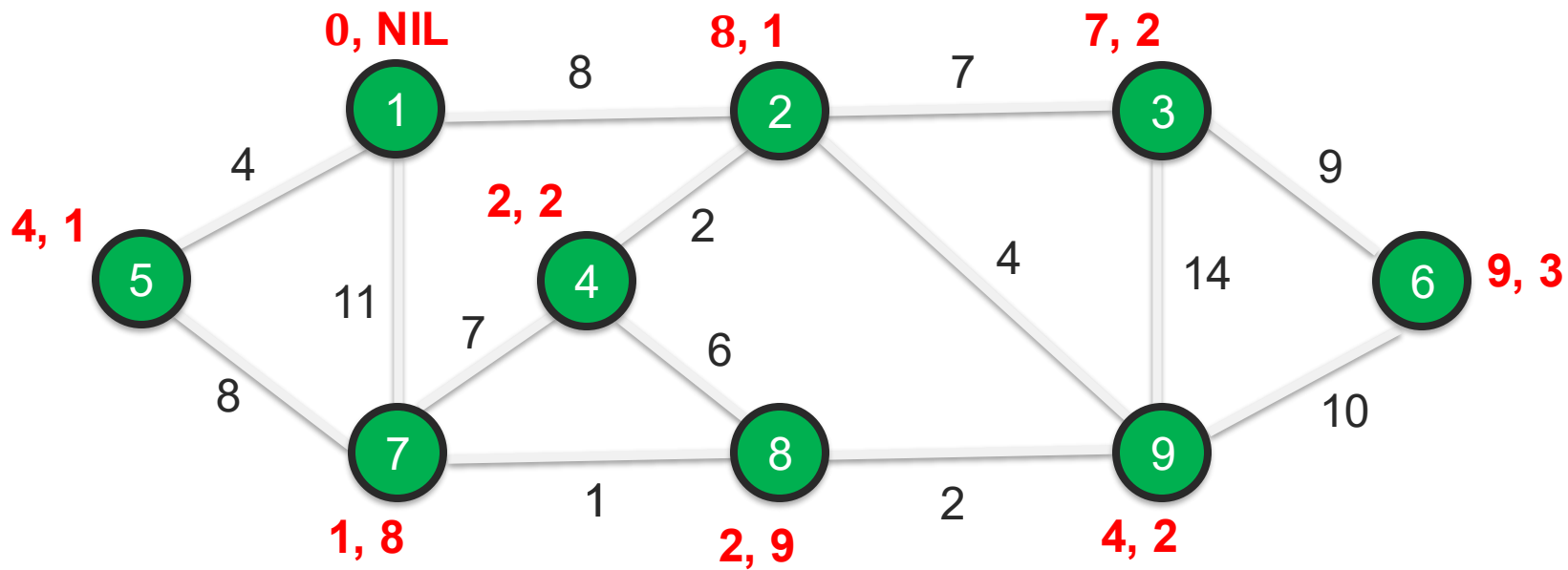
Adiacenti del vertice 6 $\rightarrow$ 3, 9

[Algoritmo di Prim]



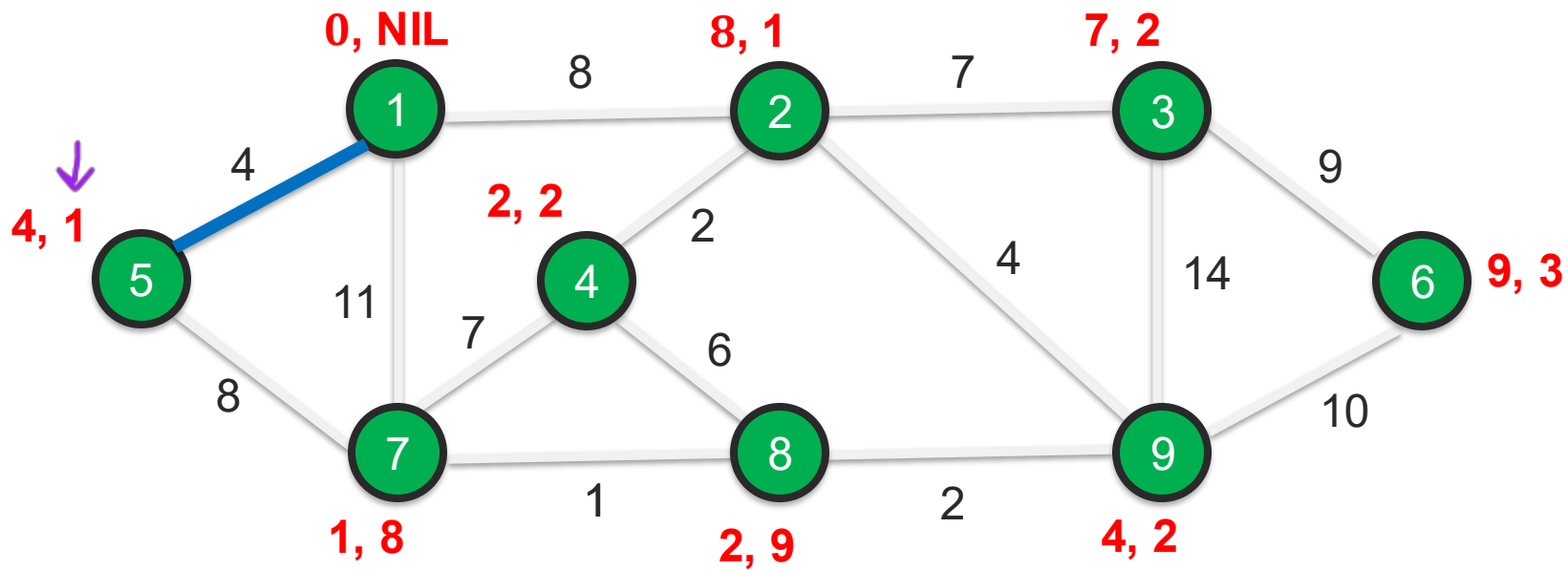
Q è vuota e l'algoritmo termina

[Algoritmo di Prim]



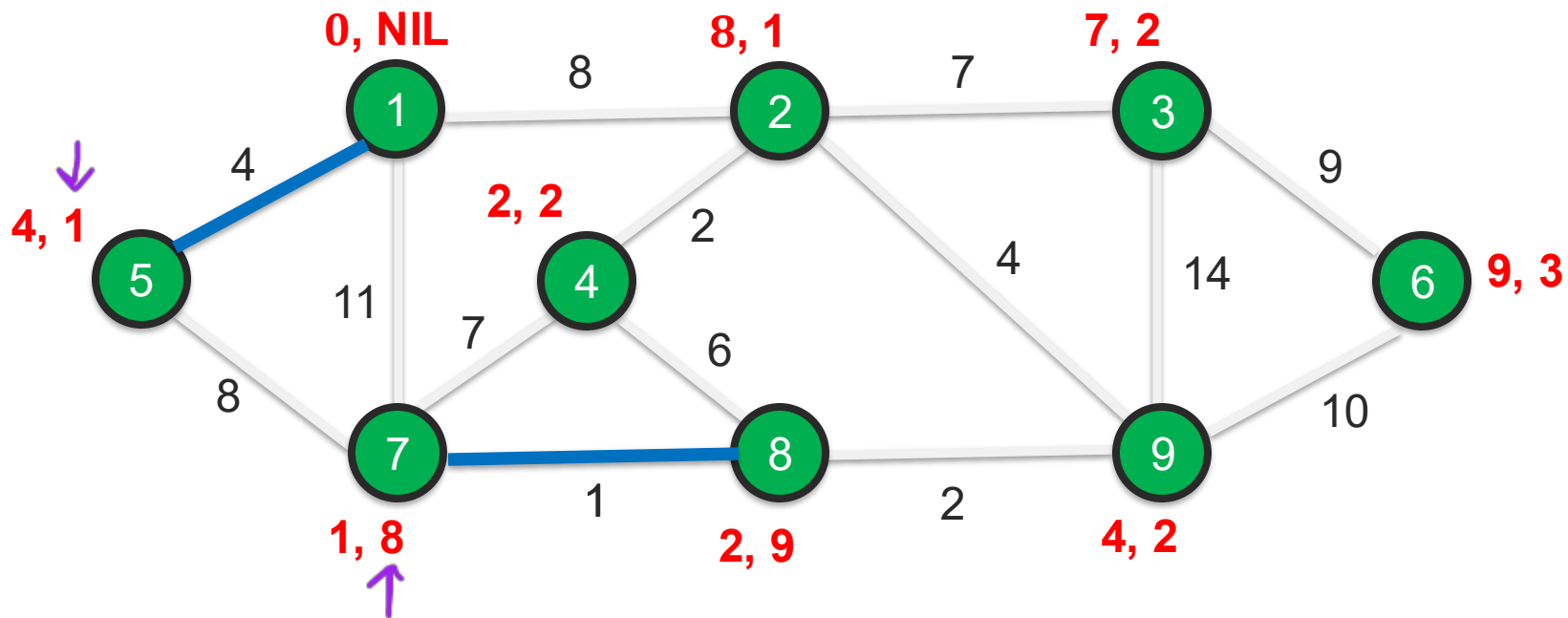
$$T = \{(u.\pi, u) \text{ t.c. } u \in V, u \neq r\}$$

[Algoritmo di Prim]



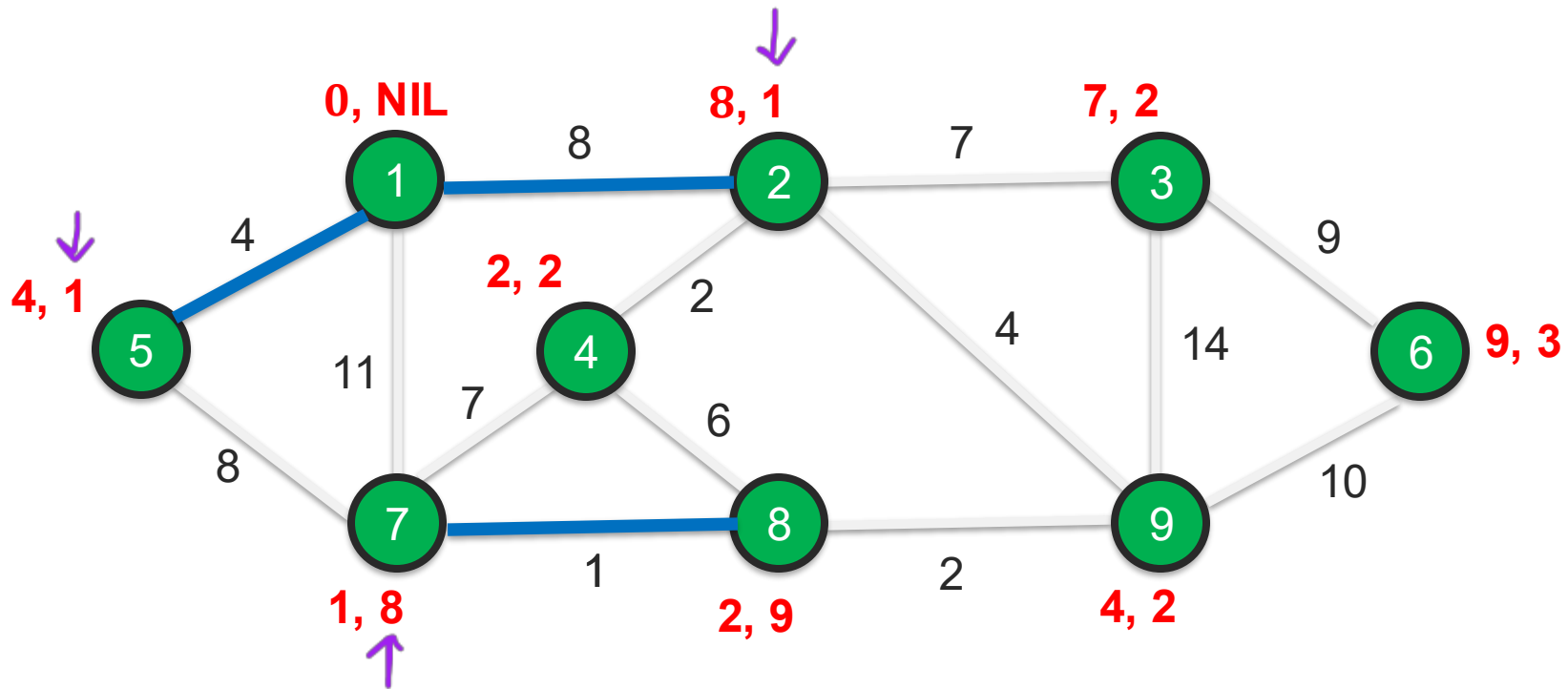
$$T = \{(u.\pi, u) \text{ t.c. } u \in V, u \neq r\}$$

[Algoritmo di Prim]



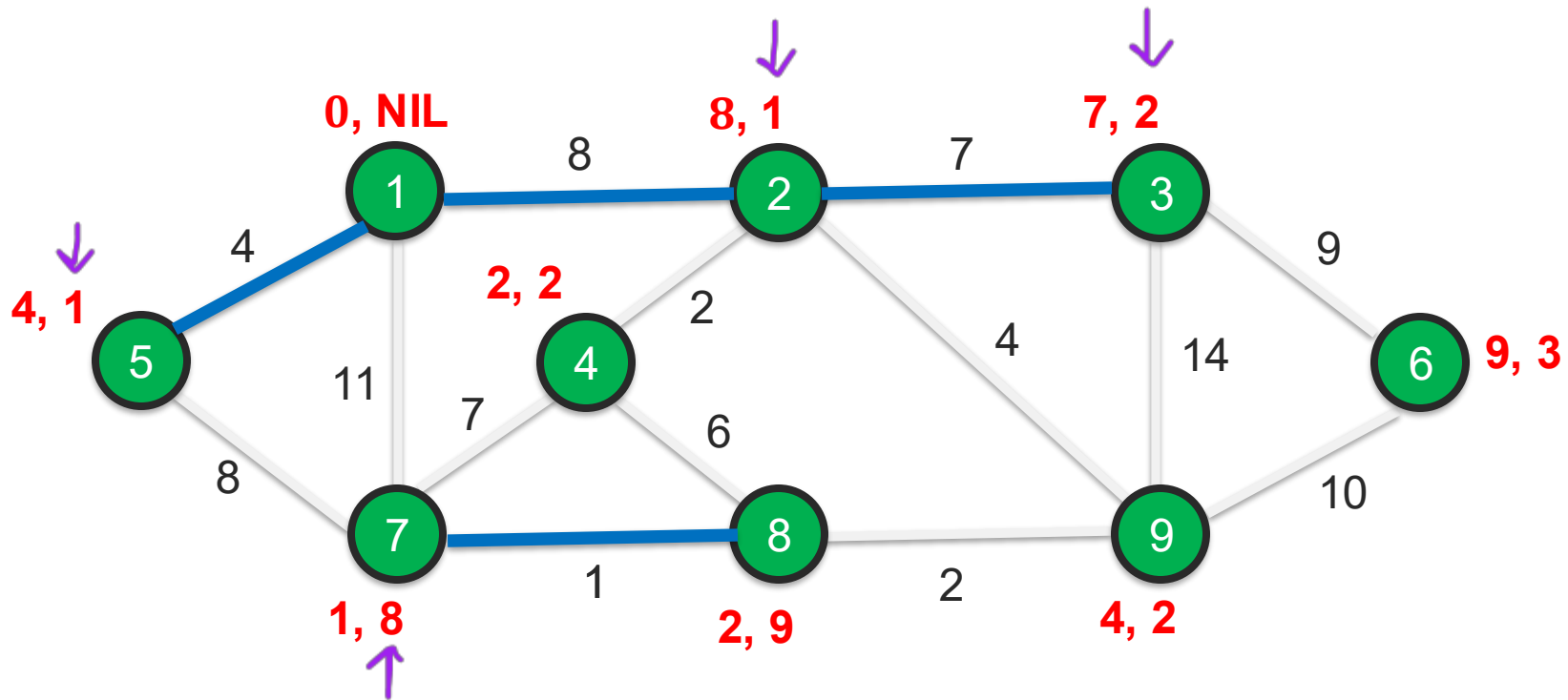
$$T = \{(u.\pi, u) \text{ t.c. } u \in V, u \neq r\}$$

[Algoritmo di Prim]



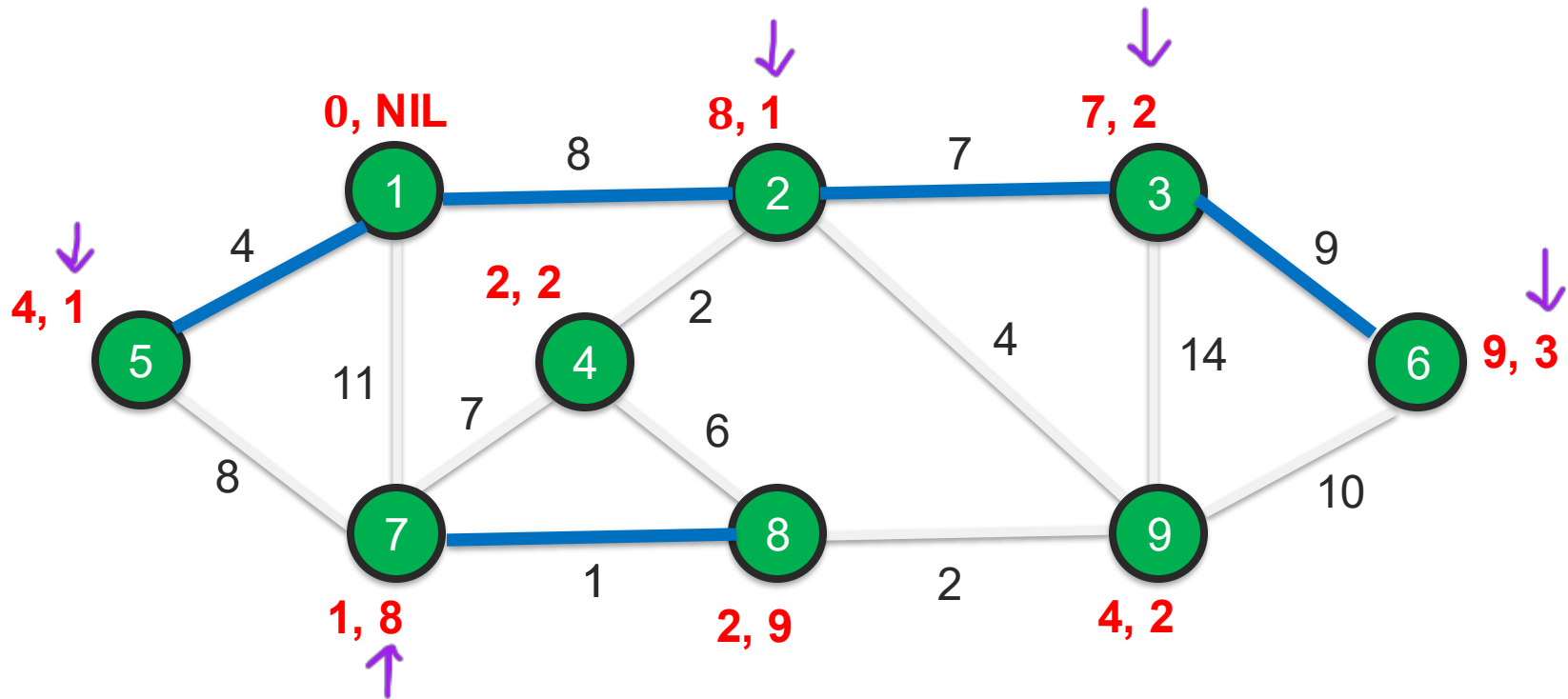
$$T = \{(u.\pi, u) \text{ t.c. } u \in V, u \neq r\}$$

[Algoritmo di Prim]



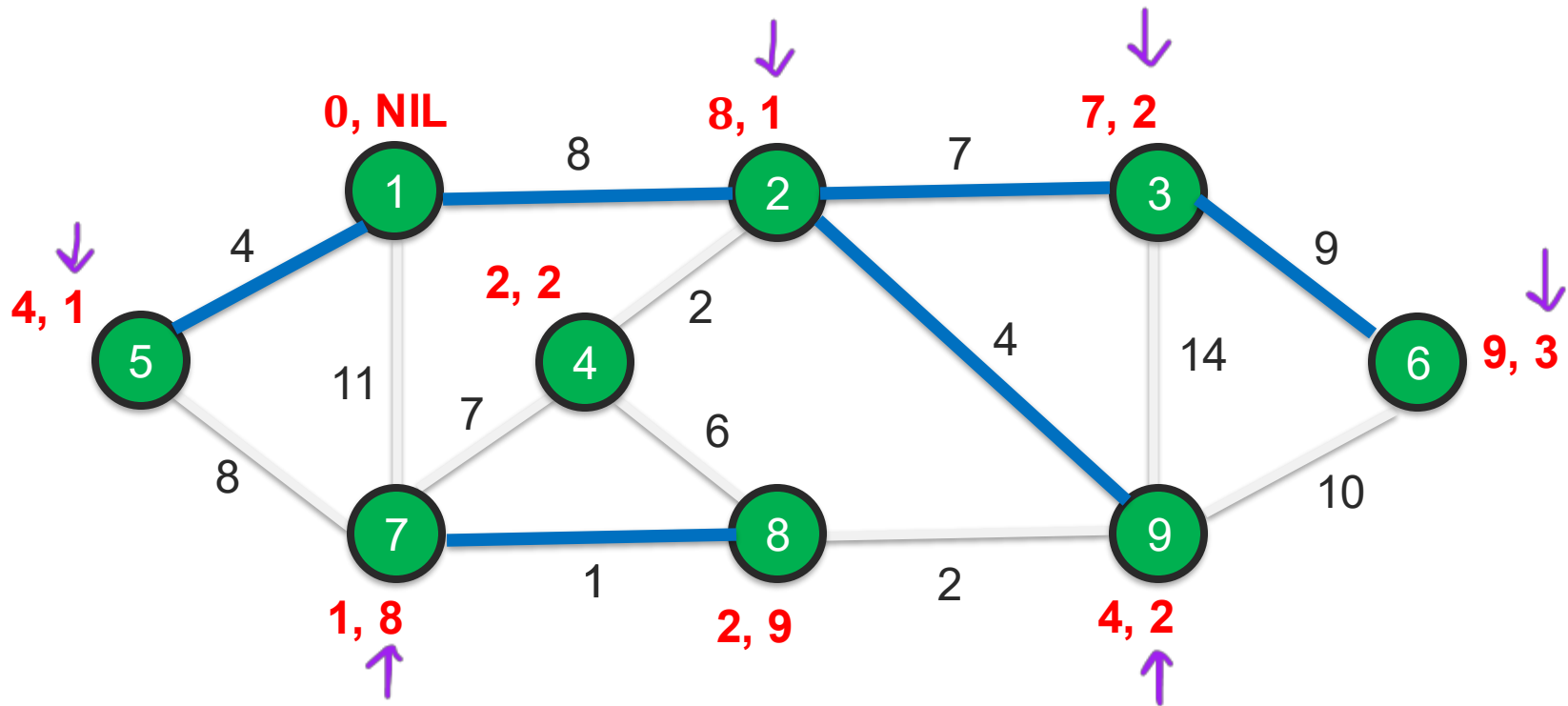
$$T = \{(u.\pi, u) \text{ t.c. } u \in V, u \neq r\}$$

[Algoritmo di Prim]



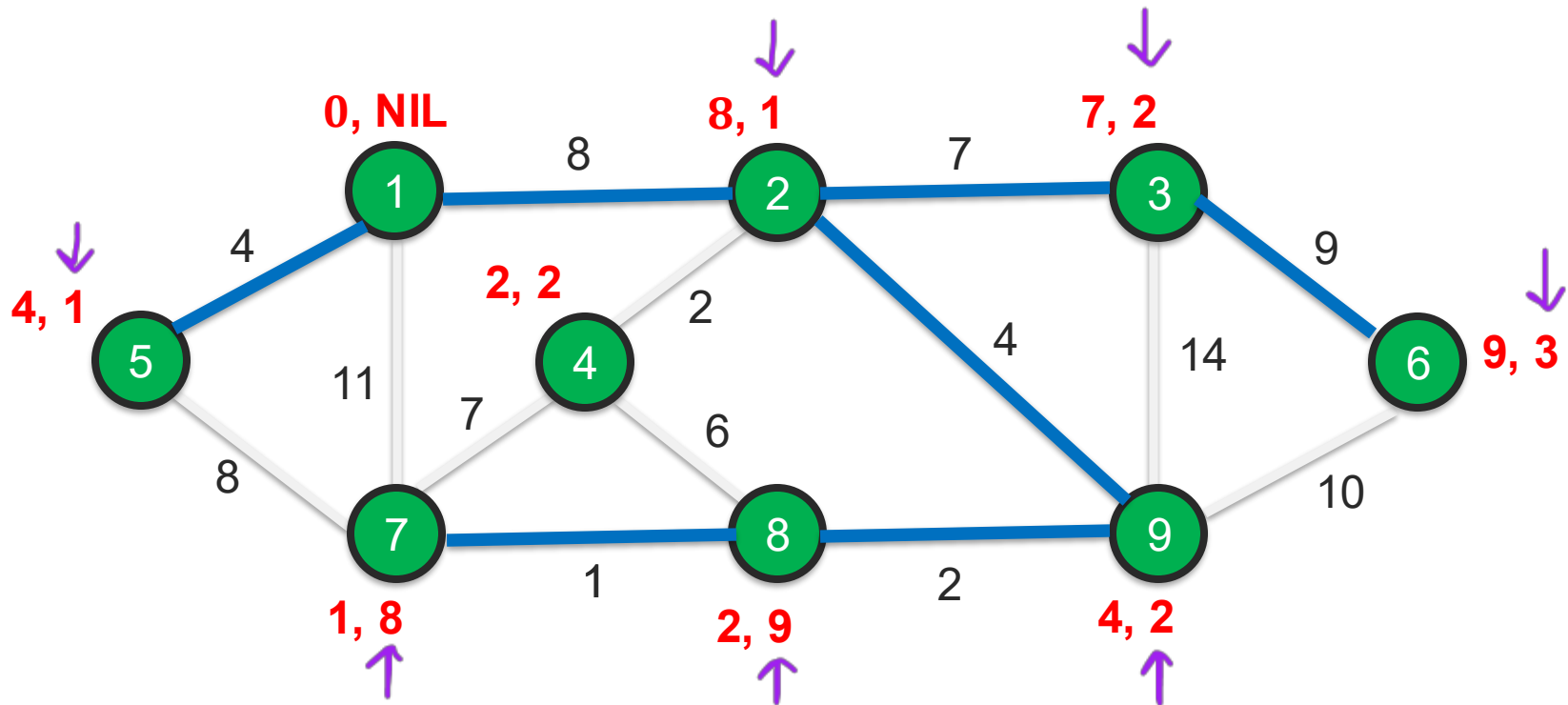
$$T = \{(u.\pi, u) \text{ t.c. } u \in V, u \neq r\}$$

[Algoritmo di Prim]



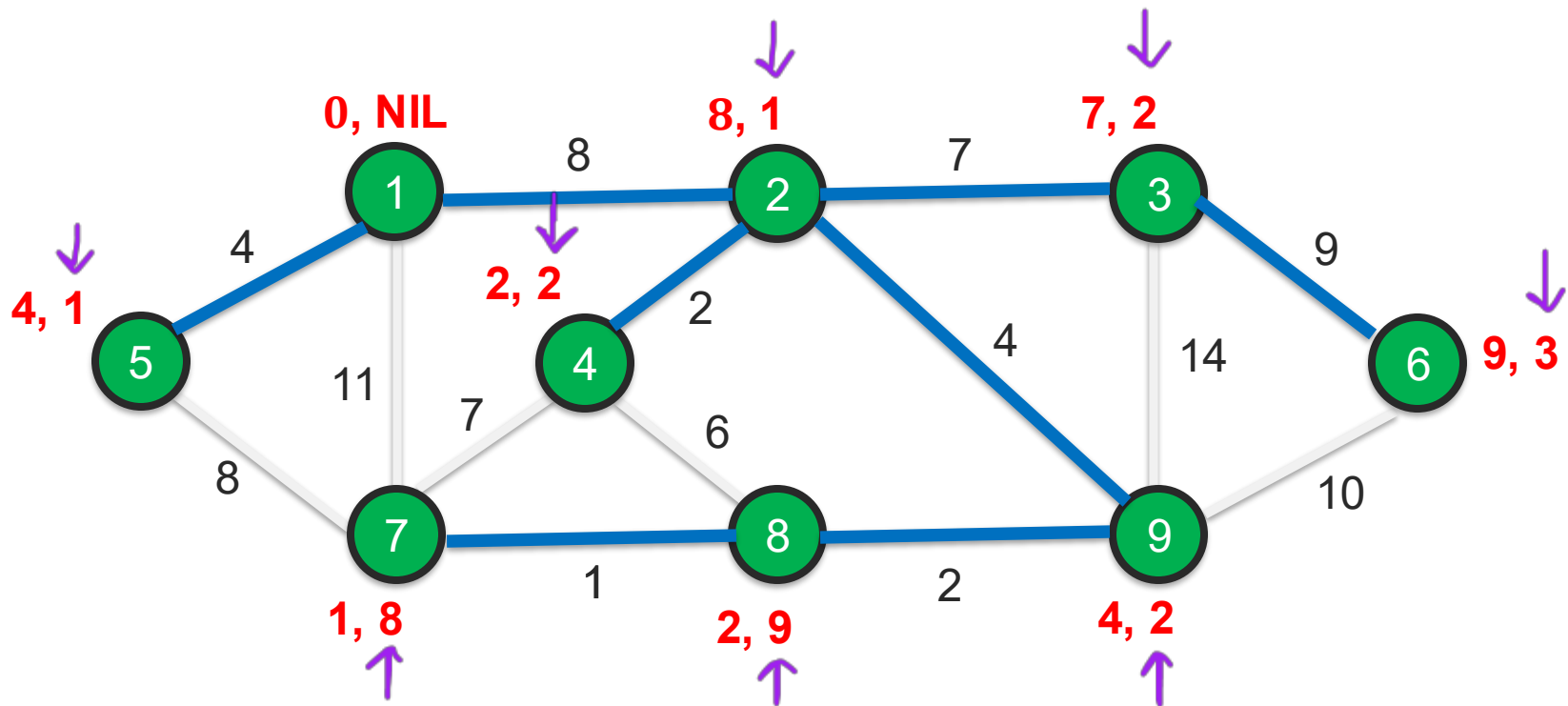
$$T = \{(u.\pi, u) \text{ t.c. } u \in V, u \neq r\}$$

[Algoritmo di Prim]



$$T = \{(u.\pi, u) \text{ t.c. } u \in V, u \neq r\}$$

[Algoritmo di Prim]



$$T = \{(u.\pi, u) \text{ t.c. } u \in V, u \neq r\}$$

[Algoritmo di Prim]

